Huawei Certification Training Courses

HCIP-Datacom-Advanced Routing & Switching Technology

Data Communication Senior Engineer Lab Guide

Issue: 1.0

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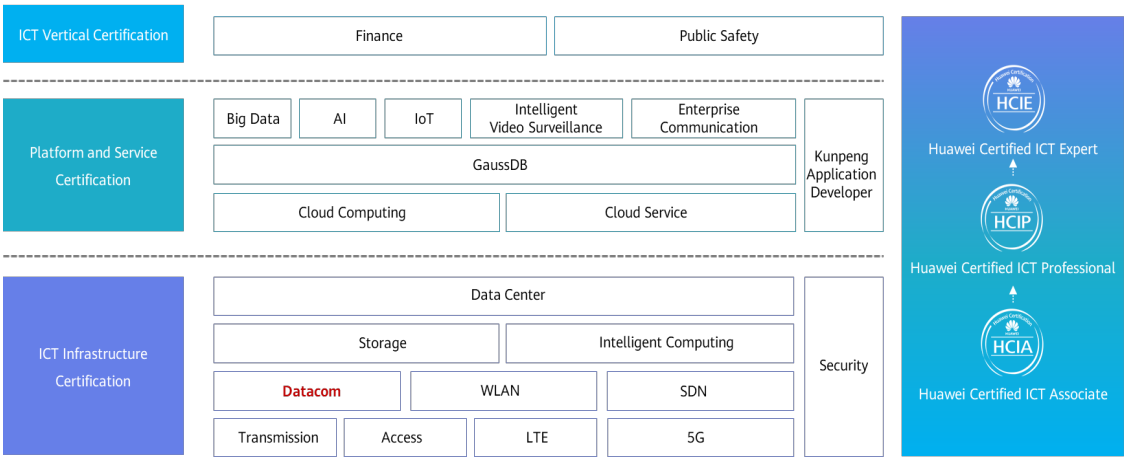
HCIP-Datacom-Advanced Routing & Switching Technology V1.0 certification covers advanced IGP features, advanced BGP features, IPv6 routing, advanced VLAN technologies, Ethernet switching security, MPLS principles, network O&M, network fault troubleshooting, and network migration. HCIP-Datacom-Advanced Routing & Switching Technology V1.0 certification is designed for medium- or large-sized enterprise network engineers that master characteristics and common technologies of medium-or large-sized enterprise networks. They have capabilities of planning and design, deployment and O&M, and fault location of medium- or large-sized enterprise networks using Huawei datacom devices, and can design high security, availability, reliability solutions.

The Huawei certification system introduces the industry, fosters innovation, and imparts cutting-edge datacom knowledge.

Huawei Certification



About This Document

Overview

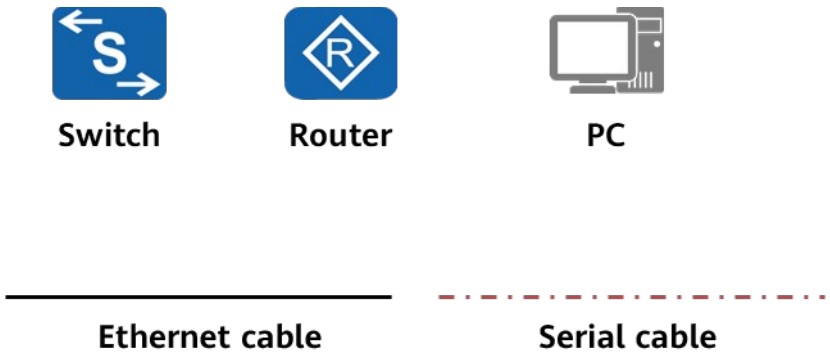
This document is an HCIP-Datacom-Advanced Routing & Switching Technology training course and is intended for trainees who are going to take the HCIP-Datacom-Advanced Routing & Switching Technology exam or readers who want to understand advanced Interior Gateway Protocol (IGP), advanced Border Gateway Protocol (BGP), IPv6 routing, advanced Virtual Local Area Network (VLAN) technologies, Ethernet switching security, Multiprotocol Label Switching (MPLS) principles, network O&M, network fault troubleshooting, and network migration.

Background Knowledge Required

This course is for Huawei's advanced certification. To better understand this course, familiarize yourself with the following requirements:

1. Have basic computer skills.
2. Have participated in HCIA-Datacom training.
3. Have passed the HCIA-Datacom exams.
4. Be familiar with the principles of the TCP/IP protocol stack.
5. Be familiar with the basic working principles of Ethernet switches and routers.

Symbol Conventions



Lab Environment

Network Description

This lab environment is intended for datacom engineers who are preparing for the HCIP-Datacom-Advanced Routing & Switching Technology exam. Each lab environment includes three switches (PoE not supported), two PoE switches, and five routers.

Device Requirements

To meet exercise requirements, the recommended configurations of the environment are as follows.

The following table lists required devices.

Device Name	Device Model	Software version
Switch	CloudEngine S5731-H24T4XC	V200R019C00 or later
PoE switch	CloudEngine S5731-H24P4XC	V200R019C00 or later
Router	NetEngine AR6120	V300R019 or later

NOTE

The port, output, and configuration information of devices in this document is provided based on the recommended topology. The actual information may vary according to the lab environment.

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1 Advanced IGP Features

1.1 Advanced IGP Features

1.1.1 Introduction

1.1.1.1 Objectives

Upon completion of this task, you will be able to:

- Deploy Fast Reroute (FRR) and Bidirectional Forwarding Detection (BFD) to speed up Open Shortest Path First (OSPF) convergence.
- Filter outgoing OSPF Link State Advertisements (LSAs).
- Deploy inter-area route filtering to reduce the OSPF link state database (LSDB) size.

1.1.1.1 Networking Topology

Networking of advanced IGP

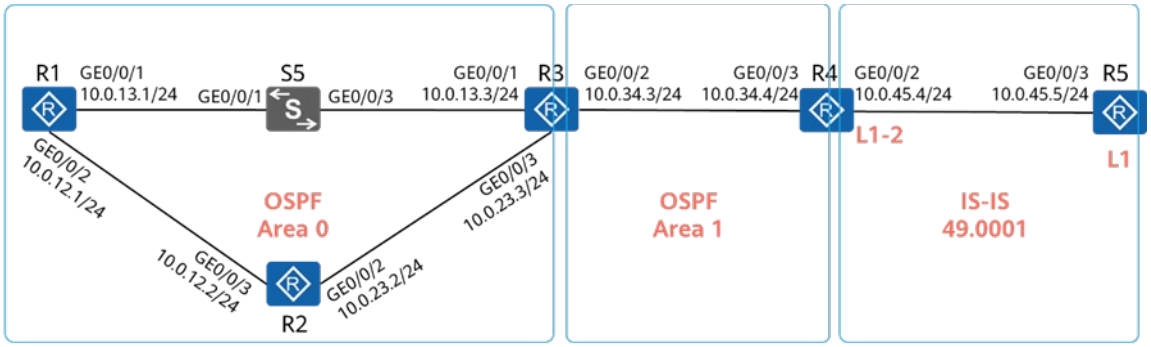


Figure 1-1 shows interconnected interfaces and their IP addresses. Loopback0 interfaces are created on all devices, and their IP addresses are 10.0.x.x/24, where x indicates the device number.

Interconnected interfaces of R1, R2, and R3 belong to OSPF area 0. Loopback0 interfaces of R1, R2, and R3 also belong to OSPF area 0, and interconnected interfaces of R3 and R4 and the Loopback0 interface of R4 belong to OSPF area 1.

R4 and R5 belong to Intermediate System-to-Intermediate System (IS-IS) area 49.0001. R5 is an IS-IS Level-1 router, and R4 is an IS-IS Level-1-2 router.

1.1.1.2 Background

There are four AR routers running OSPF on the intranet. To control the number of OSPF LSDBs, the four AR routers are assigned to different areas. IS-IS runs between R4 (ASBR) in OSPF area 1 and R5 in the branch.

To speed up OSPF convergence, the network administrator deploys OSPF IP FRR and association between OSPF and BFD.

To control the number of routes on the headquarters network, route filtering is configured on R3 (ABR) to limit the number of routes that can enter OSPF area 0.

The branch needs to access the headquarters network. The network administrator has delivered the default IS-IS routes to the branch instead of importing OSPF routes to the IS-IS routing table.

1.1.2 Lab Configuration

1.1.2.1 Configuration Roadmap

1. Configure IP addresses for devices.
2. Configure OSPF according to the planning.
3. Configure IS-IS according to the planning.
4. Enable OSPF IP FRR on R1 to generate a backup path to R4.
5. Enable BFD globally on all OSPF-enabled devices, enable BFD on interconnected interfaces, and shut down GE0/0/1 on R3. On R1, check whether the status of the OSPF neighbor relationship with R3 and route status can be rapidly switched.
6. Create Loopback2 interfaces with the same IP address on R1 and R2, activate OSPF on the interfaces, and check whether equal-cost routes exist in the OSPF routing table on R3. Then limit the number of equal-cost routes to 1. Create Loopback3 on R4 and activate OSPF. Configure inter-area route filtering on R3 (ABR) to prevent Loopback3 on R4 from advertising routes to OSPF area 0.
7. Advertise the default route in IS-IS process 1 of R4.

1.1.1.2 Configuration Procedure

2 Configure IP addresses for interconnection interfaces and loopback interfaces.

Name the devices.

The configuration details are not provided.

Disable the interfaces that are not used in this experiment.

The configuration details are not provided.

Configure IP addresses for GE0/0/1, GE0/0/2, and a loopback interface on R1.

```
[R1]interface GigabitEthernet0/0/1
[R1-GigabitEthernet0/0/1] ip address 10.0.13.1 255.255.255.0
[R1-GigabitEthernet0/0/1] quit
[R1]interface GigabitEthernet0/0/2
[R1-GigabitEthernet0/0/2] ip address 10.0.12.1 255.255.255.0
[R1-GigabitEthernet0/0/2] quit
[R1]interface LoopBack 0
[R1-LoopBack0] ip address 10.0.1.1 24
[R1-LoopBack0] quit
# Configure IP addresses for GE0/0/2, GE0/0/3, and a loopback interface on R2.
[R2]interface GigabitEthernet0/0/2
[R2-GigabitEthernet0/0/2] ip address 10.0.23.2 255.255.255.0
[R2-GigabitEthernet0/0/2] quit
[R2]interface GigabitEthernet0/0/3
[R2-GigabitEthernet0/0/3] ip address 10.0.12.2 255.255.255.0
[R2-GigabitEthernet0/0/3] quit
[R2]interface LoopBack0
[R2-LoopBack0] ip address 10.0.2.2 255.255.255.0
[R2-LoopBack0] quit
```

Configure IP addresses for GE0/0/1, GE0/0/2, GE0/0/3, and a loopback interface on R3.

```
[R3]interface GigabitEthernet0/0/1
[R3-GigabitEthernet0/0/1] ip address 10.0.13.3 255.255.255.0
[R3-GigabitEthernet0/0/1] quit
[R3]interface GigabitEthernet0/0/2
[R3-GigabitEthernet0/0/2] ip address 10.0.34.3 255.255.255.0
[R3-GigabitEthernet0/0/2]quit
[R3]interface GigabitEthernet0/0/3
[R3-GigabitEthernet0/0/3] ip address 10.0.23.3 255.255.255.0
[R3-GigabitEthernet0/0/3] quit
[R3]interface LoopBack0
[R3-LoopBack0] ip address 10.0.3.3 255.255.255.0
[R3-LoopBack0] quit
```

Configure IP addresses for GE0/0/2, GE0/0/3, and a loopback interface on R4.

```
[R4]interface GigabitEthernet0/0/2
[R4-GigabitEthernet0/0/2] ip address 10.0.45.4 255.255.255.0
[R4-GigabitEthernet0/0/2] quit
[R4]interface GigabitEthernet0/0/3
[R4-GigabitEthernet0/0/3] ip address 10.0.34.4 255.255.255.0
[R4-GigabitEthernet0/0/3] quit
[R4]interface LoopBack0
[R4-LoopBack0] ip address 10.0.4.4 255.255.255.0
[R4-LoopBack0] quit
```

Configure IP addresses for GE0/0/3 and a loopback interface on R5.

```
[R5]interface GigabitEthernet0/0/3
```

```
[R5-GigabitEthernet0/0/3] ip address 10.0.45.5 255.255.255.0
[R5-GigabitEthernet0/0/3] quit
[R5]interface LoopBack0
[R5-LoopBack0] ip address 10.0.5.5 255.255.255.0
[R5-LoopBack0] quit
```

Check the connectivity on R1, R2, and R4.

```
<R1>ping -c 1 10.12.2
Error: Unknown host 10.12.2.
<R1>ping -c 1 10.0.12.2
  PING 10.0.12.2: 56  data bytes, press CTRL_C to break
    Reply from 10.0.12.2: bytes=56 Sequence=1 ttl=255 time=60 ms

  --- 10.0.12.2 ping statistics ---
    1 packet(s) transmitted
    1 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 60/60/60 ms

<R1>ping -c 1 10.0.13.3
  PING 10.0.13.3: 56  data bytes, press CTRL_C to break
    Reply from 10.0.13.3: bytes=56 Sequence=1 ttl=255 time=50 ms

  --- 10.0.13.3 ping statistics ---
    1 packet(s) transmitted
    1 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 50/50/50 ms

<R2>ping -c 1 10.0.23.3
  PING 10.0.23.3: 56  data bytes, press CTRL_C to break
    Reply from 10.0.23.3: bytes=56 Sequence=1 ttl=255 time=60 ms

  --- 10.0.23.3 ping statistics ---
    1 packet(s) transmitted
    1 packet(s) received
    0.00% packet loss
round-trip min/avg/max = 60/60/60 ms

<R4>ping -c  1 10.0.34.3
  PING 10.0.34.3: 56  data bytes, press CTRL_C to break
    Reply from 10.0.34.3: bytes=56 Sequence=1 ttl=255 time=40 ms

  --- 10.0.34.3 ping statistics ---
    1 packet(s) transmitted
    1 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 40/40/40 ms

<R4>ping -c 1 10.0.45.5
  PING 10.0.45.5: 56  data bytes, press CTRL_C to break
    Reply from 10.0.45.5: bytes=56 Sequence=1 ttl=255 time=80 ms

  --- 10.0.45.5 ping statistics ---
    1 packet(s) transmitted
```

```
  1 packet(s) received
  0.00% packet loss
    round-trip min/avg/max = 80/80/80 ms
```

3 Configure OSPF

Configure OSPF on R1, R2, R3, and R4, use the IP address of Loopback0 as the router ID, set the OSPF process ID to 1, and activate OSPF on the corresponding interfaces.

Configure IP addresses of Loopback0 interfaces as OSPF router IDs of R1, R2, R3, and R4 and set the OSPF process ID to 1.

```
[R1]ospf 1 router-id 10.0.1.1
```

```
[R2]ospf 1 router-id 10.0.2.2
```

```
[R3]ospf 1 router-id 10.0.3.3
```

```
[R4]ospf 1 router-id 10.0.4.4
```

Configure OSPF on R1 and activate OSPF on GE0/0/1, GE0/0/2, and Loopback0.

```
[R1]ospf 1
[R1-ospf-1] area 0.0.0.0
[R1-ospf-1-area-0.0.0.0] network 10.0.13.1 0.0.0.0
[R1-ospf-1-area-0.0.0.0] network 10.0.12.1 0.0.0.0
[R1-ospf-1-area-0.0.0.0] network 10.0.1.1 0.0.0.0
[R1-ospf-1-area-0.0.0.0] quit
[R1-ospf-1] quit
```

Configure OSPF on R2 and activate OSPF on GE0/0/2, GE0/0/3, and Loopback0.

```
[R2]ospf 1
[R2-ospf-1] area 0.0.0.0
[R2-ospf-1-area-0.0.0.0] network 10.0.12.2 0.0.0.0
[R2-ospf-1-area-0.0.0.0] network 10.0.23.2 0.0.0.0
[R2-ospf-1-area-0.0.0.0] network 10.0.2.2 0.0.0.0
[R2-ospf-1-area-0.0.0.0] quit
[R2-ospf-1] quit
```

Configure OSPF on R3 and activate OSPF on GE0/0/1, GE0/0/2, and Loopback0.

```
[R3]ospf 1
[R3-ospf-1] area 0.0.0.0
[R3-ospf-1-area-0.0.0.0] network 10.0.13.3 0.0.0.0
[R3-ospf-1-area-0.0.0.0] network 10.0.23.3 0.0.0.0
```

```
[R3-ospf-1-area-0.0.0.0] network 10.0.3.3 0.0.0.0
[R3-ospf-1-area-0.0.0.0] area 0.0.0.1
[R3-ospf-1-area-0.0.0.1] network 10.0.34.3 0.0.0.0
[R3-ospf-1-area-0.0.0.1] quit
[R3-ospf-1] quit
```

Configure OSPF on R4 and activate OSPF on GE0/0/3 and Loopback0.

```
[R4]ospf 1
[R4-ospf-1] area 0.0.0.1
[R4-ospf-1-area-0.0.0.1] network 10.0.34.4 0.0.0.0
[R4-ospf-1-area-0.0.0.1] network 10.0.4.4 0.0.0.0
[R4-ospf-1-area-0.0.0.1] quit
[R4-ospf-1] quit
```

Check the OSPF neighbor relationship status on R2 and R3.

```
<R2>display ospf peer brief

 OSPF Process 1 with Router ID 10.0.2.2
   Peer Statistic Information
 ----------------------------------------------------------------------------
 Area Id          Interface                    Neighbor id      State
 0.0.0.0          GigabitEthernet0/0/2           10.0.3.3         Full
 0.0.0.0          GigabitEthernet0/0/3           10.0.1.1         Full
 ----------------------------------------------------------------------------

<R3>display ospf peer brief

 OSPF Process 1 with Router ID 10.0.3.3
   Peer Statistic Information
 ----------------------------------------------------------------------------
 Area Id          Interface                    Neighbor id      State
 0.0.0.0          GigabitEthernet0/0/1           10.0.1.1         Full
 0.0.0.0          GigabitEthernet0/0/3           10.0.2.2         Full
 0.0.0.1          GigabitEthernet0/0/2           10.0.4.4         Full
 ----------------------------------------------------------------------------
```

All OSPF neighbor relationships have been established.

Check the OSPF routing table on R1.

```
<R1>display ospf routing

  OSPF Process 1 with Router ID 10.0.1.1
    Routing Tables

 Routing for Network
 Destination        Cost   Type       NextHop        AdvRouter          Area
 10.0.1.1/32          0     Stub       10.0.1.1         10.0.1.1          0.0.0.0
 10.0.12.0/24         1     Transit    10.0.12.1        10.0.1.1          0.0.0.0
 10.0.13.0/24         1     Transit    10.0.13.1        10.0.1.1          0.0.0.0
 10.0.2.2/32          1     Stub       10.0.12.2        10.0.2.2          0.0.0.0
 10.0.3.3/32          1     Stub       10.0.13.3        10.0.3.3          0.0.0.0
 10.0.4.4/32          2     Inter-area   10.0.13.3        10.0.3.3          0.0.0.0
 10.0.23.0/24         2     Transit    10.0.13.3        10.0.2.2          0.0.0.0
 10.0.23.0/24         2     Transit    10.0.12.2        10.0.2.2          0.0.0.0
 10.0.34.0/24         2     Inter-area   10.0.13.3        10.0.3.3          0.0.0.0
```

```
 Total  Nets: 9
 Intra Area: 7   Inter Area: 2   ASE: 0   NSSA: 0
R1 has learned OSPF routes on the entire network.
```

4 Configure IS-IS

Configure IS-IS processes on routers one by one according to the topology design. Set the process ID to 1 and NET to the device ID. Here, the NET of R4 is 49.0001.0000.0000.0004.00.

Configure IS-IS on R4.

```
[R4]isis 1
[R4-isis-1] is-level level-1-2
[R4-isis-1] network-entity 49.0001.0000.0000.0004.00
[R4-isis-1] quit
[R4]interface GigabitEthernet0/0/2
[R4-GigabitEthernet0/0/2] isis enable 1
[R4-GigabitEthernet0/0/2] quit
```

Configure IS-IS on R5.

```
[R5]isis 1
[R5-isis-1] is-level level-1
[R5-isis-1] network-entity 49.0001.0000.0000.0005.00
[R5-isis-1] quit
[R5]interface LoopBack 0
[R5-LoopBack0] isis enable 1
[R5-LoopBack0] quit
[R5]interface GigabitEthernet0/0/3
[R5-GigabitEthernet0/0/3] isis enable 1
[R5-GigabitEthernet0/0/3] quit
```

Check the IS-IS neighbor relationship and IS-IS routing table on R4.

```
[R4]display isis peer

                    Peer information for ISIS(1)

 System Id     Interface          Circuit Id        State  HoldTime Type     PRI
--------------------------------------------------------------------------------
0000.0000.0005  GE0/0/2          0000.0000.0005.01  Up     8s       L1       64

Total Peer(s): 1
```

An IS-IS Level-1 neighbor relationship has been established between R4 and R5.

```
[R4]display isis route

                    Route information for ISIS(1)
                    -----------------------------

                   ISIS(1) Level-1 Forwarding Table
                   --------------------------------

IPV4 Destination      IntCost       ExtCost   ExitInterface      NextHop        Flags
```

```
-------------------------------------------------------------------------------
10.0.45.0/24          10          NULL     GE0/0/2          Direct           D/-/L/-
10.0.5.0/24           10          NULL     GE0/0/2          10.0.45.5        A/-/L/-
      Flags:D-Direct,A-Added to URT,L-Advertised in LSPs, S-IGP Shortcut,U-Up/Down Bit Set


                         ISIS(1) Level-2 Forwarding Table
                         --------------------------------

IPV4 Destination      IntCost       ExtCost   ExitInterface     NextHop        Flags
-------------------------------------------------------------------------------
10.0.45.0/24          10          NULL     GE0/0/2          Direct           D/-/L/-
      Flags:D-Direct,A-Added to URT,L-Advertised in LSPs, S-IGP Shortcut,U-Up/Down Bit Set
```

R4 has learned the route to the Loopback0 interface of R5.

5 Configure OSPF FRR

To back up the path from R1 to Loopback0 on R4, enable OSPF IP FRR on R1.

Check the OSPF route 10.0.4.4/32 on R1.

```
<R1>display ospf routing 10.0.4.4 24

  OSPF Process 1 with Router ID 10.0.1.1
<R1>display ospf routing 10.0.4.4 32

  OSPF Process 1 with Router ID 10.0.1.1

 Destination     : 10.0.4.4/32
 AdverRouter     : 10.0.3.3                    Area          : 0.0.0.0
 Cost            : 2                           Type          : Inter-area
 NextHop         : 10.0.13.3                   Interface     : GigabitEthernet0/0/1
 Priority        : Medium                      Age           : 00h35m44s
```

In this case, the next hop of the OSPF route 10.0.4.4/32 is 10.0.13.3.

Enable FRR on R1.

```
[R1]ospf 1
[R1-ospf-1] frr
[R1-ospf-1-frr] loop-free-alternate
[R1-ospf-1-frr] quit
[R1-ospf-1] quit
```

Check the OSPF route 10.0.4.4/32 on R1.

```
<R1>display ospf routing 10.0.4.4 32

  OSPF Process 1 with Router ID 10.0.1.1

 Destination     : 10.0.4.4/32
 AdverRouter     : 10.0.3.3                    Area          : 0.0.0.0
```

```
Cost          : 2                          Type          : Inter-area
NextHop       : 10.0.13.3                  Interface     : GigabitEthernet0/0/1
Priority      : Medium                     Age           : 00h01m00s
Backup Nexthop : 10.0.12.2                 Backup Interface: GigabitEthernet0/0/2
Backup Type    : LFA LINK
```

A backup route destined for R4's Loopback0 interface has been generated on R1, with the next hop address being 10.0.12.2 and outbound interface being GE0/0/2. The backup route is destined for R4's Loopback0 interface through R2.

6 Configure BFD for OSPF

To speed up OSPF convergence, enable BFD on all routers in the OSPF domain, configure BFD on interconnected interfaces, and disable GE0/0/1 on R3. Check whether R1 can quickly detect the BFD session termination and trigger OSPF route switching.

Enable BFD globally on R1, R2, R3, and R4.

```
[R1]bfd
[R1-bfd] quit
[R1]ospf 1
[R1-ospf-1] bfd all-interfaces enable
[R1-ospf-1] quit
```

```
[R2]bfd
[R2-bfd] quit
[R2]ospf 1
[R2-ospf-1] bfd all-interfaces enable
[R2-ospf-1] quit
```

```
[R3]bfd
[R3-bfd] quit
[R3]ospf 1
[R3-ospf-1] bfd all-interfaces enable
[R3-ospf-1] quit
```

```
[R4]bfd
[R4-bfd] quit
[R4]ospf 1
[R4-ospf-1] bfd all-interfaces enable
[R4-ospf-1] quit
```

Configure BFD on R1, set the minimum intervals for sending and receiving packets to 500 ms, and the local detection multiplier to 4.

```
[R1]interface GigabitEthernet0/0/1
[R1-GigabitEthernet0/0/1] ospf bfd enable
```

```
[R1-GigabitEthernet0/0/1] ospf bfd min-tx-interval 500 min-rx-interval 500 detect-multiplier 4
[R1-GigabitEthernet0/0/1] quit
[R1]interface GigabitEthernet0/0/2
[R1-GigabitEthernet0/0/2] ospf bfd enable
[R1-GigabitEthernet0/0/2] ospf bfd min-tx-interval 500 min-rx-interval 500 detect-multiplier 4
[R1-GigabitEthernet0/0/2] quit
```

Configure BFD on R2, set the minimum intervals for sending and receiving packets to 500 ms, and the local detection multiplier to 4.

```
[R2]interface GigabitEthernet0/0/2
[R2-GigabitEthernet0/0/2] ospf bfd enable
[R2-GigabitEthernet0/0/2] ospf bfd min-tx-interval 500 min-rx-interval 500 detect-multiplier 4
[R2-GigabitEthernet0/0/2] quit
[R2]interface GigabitEthernet0/0/3
[R2-GigabitEthernet0/0/3] ospf bfd enable
[R2-GigabitEthernet0/0/3] ospf bfd min-tx-interval 500 min-rx-interval 500 detect-multiplier 4
[R2-GigabitEthernet0/0/3] quit
```

Configure BFD on R3, set the minimum intervals for sending and receiving packets to 500 ms, and the local detection multiplier to 4.

```
[R3]interface GigabitEthernet0/0/1
[R3-GigabitEthernet0/0/1] ospf bfd enable
[R3-GigabitEthernet0/0/1] ospf bfd min-tx-interval 500 min-rx-interval 500 detect-multiplier 4
[R3-GigabitEthernet0/0/1] quit
[R3]interface GigabitEthernet0/0/2
[R3-GigabitEthernet0/0/2] ospf bfd enable
[R3-GigabitEthernet0/0/2] ospf bfd min-tx-interval 500 min-rx-interval 500 detect-multiplier 4
[R3-GigabitEthernet0/0/2] quit
[R3]interface GigabitEthernet0/0/3
[R3-GigabitEthernet0/0/3] ospf bfd enable
[R3-GigabitEthernet0/0/3] ospf bfd min-tx-interval 500 min-rx-interval 500 detect-multiplier 4
[R3-GigabitEthernet0/0/3] quit
```

Configure BFD on R4, set the minimum intervals for sending and receiving packets to 500 ms, and the local detection multiplier to 4.

```
[R4]interface GigabitEthernet0/0/3
[R4-GigabitEthernet0/0/3] ospf bfd enable
[R4-GigabitEthernet0/0/3] ospf bfd min-tx-interval 500 min-rx-interval 500 detect-multiplier 4
[R4-GigabitEthernet0/0/3] quit
```

Check the BFD session status on R1 and R3.

```
[R1]display ospf bfd session all

 OSPF Process 1 with Router ID 10.0.1.1
 Area 0.0.0.0 interface 10.0.13.1(GigabitEthernet0/0/1)'s BFD Sessions

NeighborId     :10.0.3.3        AreaId          :0.0.0.0              Interface  :GigabitEthernet0/0/1
BFDState       :up              rx              :500                  tx         :500
Multiplier     :4               BFD Local Dis   :8193                 LocalIpAdd:10.0.13.1
RemoteIpAdd:10.0.13.3           Diagnostic Info:No diagnostic information

 Area 0.0.0.0 interface 10.0.12.1(GigabitEthernet0/0/2)'s BFD Sessions

NeighborId     :10.0.2.2        AreaId          :0.0.0.0              Interface  :GigabitEthernet0/0/2
```

```
BFDState     :up              rx              :500              tx         :500
Multiplier   :4               BFD Local Dis   :8192              LocalIpAdd:10.0.12.1
RemoteIpAdd :10.0.12.2        Diagnostic Info:No diagnostic information
```

The BFD sessions between R1 and R2 and between R1 and R3 are up.

```
[R3]display ospf bfd session all

 OSPF Process 1 with Router ID 10.0.3.3
 Area 0.0.0.0 interface 10.0.13.3(GigabitEthernet0/0/1)'s BFD Sessions

NeighborId    :10.0.1.1        AreaId          :0.0.0.0          Interface :GigabitEthernet0/0/1
BFDState     :up              rx              :500              tx         :500
Multiplier   :4               BFD Local Dis   :8192              LocalIpAdd:10.0.13.3
RemoteIpAdd :10.0.13.1        Diagnostic Info:No diagnostic information

 Area 0.0.0.0 interface 10.0.23.3(GigabitEthernet0/0/3)'s BFD Sessions

NeighborId    :10.0.2.2        AreaId          :0.0.0.0          Interface :GigabitEthernet0/0/3
BFDState     :up              rx              :500              tx         :500
Multiplier   :4               BFD Local Dis   :8193              LocalIpAdd:10.0.23.3
RemoteIpAdd :10.0.23.2        Diagnostic Info:No diagnostic information

 Area 0.0.0.1 interface 10.0.34.3(GigabitEthernet0/0/2)'s BFD Sessions

NeighborId    :10.0.4.4        AreaId          :0.0.0.1          Interface :GigabitEthernet0/0/2
BFDState     :up              rx              :500              tx         :500
Multiplier   :4               BFD Local Dis   :8194              LocalIpAdd:10.0.34.3
RemoteIpAdd :10.0.34.4        Diagnostic Info:No diagnostic information
```

The BFD sessions between R3 and R1, R2, and R4 are Up.

Shut down GE0/0/1 of R3 and test the association between BFD and OSPF.

```
[R3]interface GigabitEthernet0/0/1
[R3-GigabitEthernet0/0/1] shutdown
[R3-GigabitEthernet0/0/1] quit
```

Check the status of GE0/0/1 on R1.

```
[R1]display interface GigabitEthernet 0/0/1
GigabitEthernet0/0/1 current state : UP
Line protocol current state : UP
```

R1 and R3 are connected through S5. After GE0/0/1 on R3 is shut down, GE0/0/1 on R1 is still Up and cannot detect the connectivity interruption between R1 and R3.

Check the BFD session status on GE0/0/1 of R1.

```
[R1]display ospf bfd session GigabitEthernet 0/0/1

 OSPF Process 1 with Router ID 10.0.1.1
```

There is no BFD session.

Check the OSPF neighbor relationship status on R1.

```
<R1>display ospf peer brief
```

```
 OSPF Process 1 with Router ID 10.0.1.1
   Peer Statistic Information
----------------------------------------------------------------------------
Area Id          Interface                    Neighbor id      State
0.0.0.0          GigabitEthernet0/0/2            10.0.2.2         Full
----------------------------------------------------------------------------
```

The OSPF neighbor relationship between R1 and R3 has been terminated, and the time of shutting down GE0/0/1 on R3 is smaller than the OSPF dead interval.

Run the display ospf peer last-nbr-down command on R1 to check the reason why the neighbor relationship is interrupted.

```
<R1>display ospf peer last-nbr-down

  OSPF Process 1 with Router ID 10.0.1.1

   Last Down OSPF Peer

 Neighbor IpAddress      : 10.0.13.3
 Neighbor Area   Id      : 0.0.0.0
 Neighbor Router Id      : 10.0.3.3
 Interface               : GigabitEthernet0/0/1
 Immediate Reason        : Neighbor Down Due to KillNeighbor
 Primary Reason          : BFD Session Down
 Down Time               : 2020-07-09 18:54:48-08:00
```

The command output shows that the neighbor relationship between R1 and R3 is interrupted because the BFD session is Down.

Check OSPF route 10.0.4.4/32 on R1.

```
[R1]display ospf routing 10.0.4.4 32

 OSPF Process 1 with Router ID 10.0.1.1

 Destination    : 10.0.4.4/32
 AdverRouter    : 10.0.3.3          Area        : 0.0.0.0
 Cost           : 3                 Type        : Inter-area
 NextHop        : 10.0.12.2         Interface   : GigabitEthernet0/0/2
 Priority       : Medium            Age         : 00h01m25s
```

The next hop of the route to R4's Loopback0 interface is changed to 10.0.12.2, and the outbound interface is GE0/0/2.

7 Configure OSPF route control

On R1 and R2, create Loopback2 interfaces on R1 and R2 and assign IP addresses on network segment 172.16.2.1/24 to Loopback2 interfaces. Activate OSPF on Loopback2 interfaces and set the number of equal-cost routes to 1 on R3.

On R4, create the loopback 3 interface with the IP address on network segment 172.16.3.1/24, change the OSPF interface type to broadcast, activate OSPF on the Loopback3 interface, and perform inter-area Type 3 LSA filtering on R3 (ABR) to prevent the OSPF inter-area route 172.16.3.0/24 from being transmitted to OSPF area 0.

Create Loopback2 interfaces on R1 and R2.

```
[R1]interface LoopBack2
[R1-LoopBack2] ip address 172.16.2.1 255.255.255.0
[R1-LoopBack2] quit

[R2]interface LoopBack2
[R2-LoopBack2] ip address 172.16.2.1 255.255.255.0
[R2-LoopBack2] quit
```

Activate OSPF on Loopback 2 interfaces of R1 and R2.

```
[R1]ospf 1
[R1-ospf-1]area 0.0.0.0
[R1-ospf-1-area-0.0.0.0] network 172.16.2.1 0.0.0.0
[R1-ospf-1-area-0.0.0.0] quit
[R1-ospf-1] quit

[R2]ospf 1
[R2-ospf-1]area 0.0.0.0
[R2-ospf-1-area-0.0.0.0] network 172.16.2.1 0.0.0.0
[R2-ospf-1-area-0.0.0.0] quit
[R2-ospf-1]quit
```

Enable GE0/0/1 of R3.

```
[R3]interface GigabitEthernet0/0/1
[R3-GigabitEthernet0/0/1] undo shutdown
```

After the OSPF neighbor relationship between R1 and R3 is reestablished, check the OSPF routing table of R3.

```
[R3-ospf-1]display ospf routing

  OSPF Process 1 with Router ID 10.0.3.3
    Routing Tables

 Routing for Network
Destination        Cost   Type      NextHop       AdvRouter       Area
10.0.3.3/32        0      Stub      10.0.3.3      10.0.3.3        0.0.0.0
10.0.13.0/24       1      Transit   10.0.13.3     10.0.3.3        0.0.0.0
10.0.23.0/24       1      Transit   10.0.23.3     10.0.3.3        0.0.0.0
10.0.34.0/24       1      Transit   10.0.34.3     10.0.3.3        0.0.0.1
10.0.1.1/32        1      Stub      10.0.13.1     10.0.1.1        0.0.0.0
10.0.2.2/32        1      Stub      10.0.23.2     10.0.2.2        0.0.0.0
10.0.4.4/32        1      Stub      10.0.34.4     10.0.4.4        0.0.0.1
10.0.12.0/24       2      Transit   10.0.23.2     10.0.2.2        0.0.0.0
10.0.12.0/24       2      Transit   10.0.13.1     10.0.2.2        0.0.0.0
172.16.2.1/32      1      Stub      10.0.23.2     10.0.2.2        0.0.0.0
172.16.2.1/32      1      Stub      10.0.13.1     10.0.1.1        0.0.0.0

 Total Nets: 11
```

```
Intra Area: 11  Inter Area: 0  ASE: 0  NSSA: 0
```

The OSPF route 172.16.2.1/32 on R3 is used for load balancing, with R1 and R2 as next hops.

Set the maximum number of equal-cost routes for load balancing to 1 on R3.

```
[R3]ospf 1
[R3-ospf-1] maximum load-balancing 1
```

Check the routing table on R3 again.

```
[R3]display ospf routing

 OSPF Process 1 with Router ID 10.0.3.3
   Routing Tables

 Routing for Network
 Destination        Cost   Type      NextHop       AdvRouter       Area
 10.0.3.3/32        0      Stub      10.0.3.3       10.0.3.3       0.0.0.0
 10.0.13.0/24       1      Transit   10.0.13.3      10.0.3.3       0.0.0.0
 10.0.23.0/24       1      Transit   10.0.23.3      10.0.3.3       0.0.0.0
 10.0.34.0/24       1      Transit   10.0.34.3      10.0.3.3       0.0.0.1
 10.0.1.1/32        1      Stub      10.0.13.1      10.0.1.1       0.0.0.0
 10.0.2.2/32        1      Stub      10.0.23.2      10.0.2.2       0.0.0.0
 10.0.4.4/32        1      Stub      10.0.34.4      10.0.4.4       0.0.0.1
 10.0.12.0/24       2      Transit   10.0.23.2      10.0.2.2       0.0.0.0
 172.16.2.1/32       1      Stub      10.0.23.2      10.0.2.2       0.0.0.0

 Total Nets: 9
 Intra Area: 9  Inter Area: 0  ASE: 0  NSSA: 0
```

There is only one OSPF route 172.16.2.1/32 on R3, and the next hop is R2.

When the number of equal-cost routes is greater than number specified in the **maximum load-balancing** command, valid routes are selected for load balancing based on the following criteria:

1. Route priority: Routes with the highest priority (lowest weight) are selected for load balancing.
2. Interface index: If routes have the same priority, the routes with the largest interface index are selected for load balancing.
3. Next-hop IP address: If routes have the same priority and interface index, the routes with the largest next-hop IP addresses are selected for load balancing.

The index of GE0/0/3 on R3 connected to R2 is greater than that of GE0/0/1 on R1, so the OSPF route 172.16.2.1/32 from R2 becomes valid.

Create Loopback3 on R4 and activate OSPF.

```
[R4]interface LoopBack 3
[R4-LoopBack3] ip address 172.16.3.1 24
[R4-LoopBack3] ospf network-type broadcast
[R4-LoopBack3] quit

[R4]ospf 1
[R4-ospf-1]area 1
```

```
[R4-ospf-1-area-0.0.0.1] network 172.16.3.1 0.0.0.0
[R4-ospf-1-area-0.0.0.1] quit
[R4-ospf-1] quit
```

Check OSPF routing information on R1.

```
<R1>display ospf routing

  OSPF Process 1 with Router ID 10.0.1.1
    Routing Tables

 Routing for Network
 Destination        Cost   Type       NextHop        AdvRouter        Area
 10.0.1.1/32        0      Stub       10.0.1.1       10.0.1.1         0.0.0.0
 10.0.12.0/24       1      Transit    10.0.12.1      10.0.1.1         0.0.0.0
 10.0.13.0/24       1      Transit    10.0.13.1      10.0.1.1         0.0.0.0
 172.16.2.1/32      0      Stub       172.16.2.1     10.0.1.1         0.0.0.0
 10.0.2.2/32        1      Stub       10.0.12.2      10.0.2.2         0.0.0.0
 10.0.3.3/32        1      Stub       10.0.13.3      10.0.3.3         0.0.0.0
 10.0.4.4/32        2      Inter-area 10.0.13.3      10.0.3.3         0.0.0.0
 10.0.23.0/24       2      Transit    10.0.12.2      10.0.3.3         0.0.0.0
 10.0.23.0/24       2      Transit    10.0.13.3      10.0.3.3         0.0.0.0
 10.0.34.0/24       2      Inter-area 10.0.13.3      10.0.3.3         0.0.0.0
 172.16.3.0/24      2      Inter-area 10.0.13.3      10.0.3.3         0.0.0.0

 Total Nets: 11
 Intra Area: 8  Inter Area: 3  ASE: 0  NSSA: 0
```

R1 has learned the route to the Loopback3 interface of R4.

Check the OSPF LSDB on R3.

```
[R3]display ospf lsdb
[R3]display ospf lsdb

  OSPF Process 1 with Router ID 10.0.3.3
   Link State Database

             Area: 0.0.0.0
 Type       LinkState ID    AdvRouter          Age  Len   Sequence   Metric
 Router     10.0.3.3        10.0.3.3           599  60    8000001E        1
 Router     10.0.2.2        10.0.2.2           761  72    80000014        1
 Router     10.0.1.1        10.0.1.1           602  72    80000014        1
 Network    10.0.23.3       10.0.3.3           1693  32   80000002        0
 Network    10.0.13.1       10.0.1.1           602  32    80000002        0
 Network    10.0.12.2       10.0.2.2           373  32    80000008        0
 Sum-Net    10.0.34.0       10.0.3.3           535  28    80000007        1
 Sum-Net    172.16.3.0      10.0.3.3           455  28    80000001        1
 Sum-Net    10.0.4.4        10.0.3.3           39   28    80000007        1

             Area: 0.0.0.1
 Type       LinkState ID    AdvRouter          Age  Len   Sequence   Metric
 Router     10.0.3.3        10.0.3.3           337  36    8000000B        1
 Router     10.0.4.4        10.0.4.4           418  60    80000010        1
 Network    10.0.34.3       10.0.3.3           337  32    80000008        0
 Sum-Net    10.0.13.0       10.0.3.3           640  28    80000003        1
 Sum-Net    10.0.12.0       10.0.3.3           1693  28   80000002        2
```

```
Sum-Net   172.16.2.1     10.0.3.3        1398   28    80000001        1
Sum-Net   10.0.3.3       10.0.3.3        1680   28    80000006        0
Sum-Net   10.0.2.2       10.0.3.3        1693   28    80000002        1
Sum-Net   10.0.1.1       10.0.3.3         599  28    80000003        1
Sum-Net   10.0.23.0      10.0.3.3        1739   28    80000002        1
```

In the LSDB of OSPF area 0 on R3 (ABR), you can view the Type 3 LSA 172.16.3.0.

Configure Type 3 LSA filtering on R3 (ABR) to prevent the OSPF inter-area route 172.16.3.0/24 from being transmitted to OSPF area 0.

```
[R3]ip ip-prefix 1 index 10 deny 172.16.3.0 24 greater-equal 24 less-equal 24
[R3]ip ip-prefix 1 index 20 permit 0.0.0.0 0 less-equal 32

[R3]ospf 1
[R3-ospf-1] area 1
[R3-ospf-1-area-0.0.0.1] filter ip-prefix 1 export
```

Check the OSPF LSDB on R3 again.

```
[R3]display ospf lsdb

  OSPF Process 1 with Router ID 10.0.3.3
   Link State Database

             Area: 0.0.0.0
Type       LinkState ID     AdvRouter          Age  Len   Sequence   Metric
Router      10.0.3.3         10.0.3.3           966  60    8000001E        1
Router      10.0.2.2         10.0.2.2          1128   72    80000014        1
Router      10.0.1.1         10.0.1.1           969  72    80000014        1
Network      10.0.23.3        10.0.3.3           259  32    80000003        0
Network      10.0.13.1        10.0.1.1           969  32    80000002        0
Network      10.0.12.2        10.0.2.2           740  32    80000008        0
Sum-Net    10.0.34.0        10.0.3.3            13  28    80000001        1
Sum-Net    10.0.4.4         10.0.3.3            13  28    80000001        1

             Area: 0.0.0.1
Type       LinkState ID     AdvRouter          Age  Len   Sequence   Metric
Router      10.0.3.3         10.0.3.3           704  36    8000000B        1
Router      10.0.4.4         10.0.4.4           785  60    80000010        1
Network      10.0.34.3        10.0.3.3           704  32    80000008        0
Sum-Net    10.0.13.0        10.0.3.3          1007   28    80000003        1
Sum-Net    10.0.12.0        10.0.3.3           259  28    80000003        2
Sum-Net    172.16.2.1       10.0.3.3          1765   28    80000001        1
Sum-Net    10.0.3.3         10.0.3.3           246  28    80000007        0
Sum-Net    10.0.2.2         10.0.3.3           259  28    80000003        1
Sum-Net    10.0.1.1         10.0.3.3           966  28    80000003        1
Sum-Net    10.0.23.0        10.0.3.3           307  28    80000003        1
```

The Type 3 LSA 172.16.3.0 cannot be found in the LSDB of OSPF area 0 on R3.

Check the OSPF route 172.6.3.1/24 on R1.

```
<R1>display ospf routing 172.16.3.1 24

 OSPF Process 1 with Router ID 10.0.1.1
```

The OSPF route 172.6.3.1/24 does not exist.

8

Configure default IS-IS route control.

Advertise the default route on R4 (IS-IS Level-1-2) and check whether the default route is generated on R5 (IS-IS Level-1).

Advertise a default route on R4.

```
[R4]isis 1
[R4-isis-1] default-route-advertise always level-1-2
```

Check IS-IS routes on R5.

```
<R5>display isis route

                    Route information for ISIS(1)
                    -----------------------------

                   ISIS(1) Level-1 Forwarding Table
                   --------------------------------

IPV4 Destination      IntCost      ExtCost   ExitInterface     NextHop          Flags
--------------------------------------------------------------------------------
0.0.0.0/0             10           NULL      GE0/0/3           10.0.45.4        A/-/-/-
10.0.45.0/24          10           NULL      GE0/0/3           Direct           D/-/L/-
10.0.5.0/24           0            NULL      Loop0             Direct           D/-/L/-
      Flags: D-Direct, A-Added to URT, L-Advertised in LSPs, S-IGP Shortcut, U-Up/Down Bit Set
```

R5 has learned the default route advertised by R4.

Check the IS-IS LSDB on R5.

```
<R5>display isis lsdb verbose

                Database information for ISIS(1)
                --------------------------------

                  Level-1 Link State Database

LSPID             Seq Num      Checksum     Holdtime     Length  ATT/P/OL
--------------------------------------------------------------------------
0000.0000.0004.00-00  0x0000000f   0xc0d4       723          70      0/0/0
 SOURCE       0000.0000.0004.00
 NLPID        IPV4
 AREA ADDR    49.0001
 INTF ADDR    10.0.45.4
 NBR  ID      0000.0000.0005.01  COST: 10
 IP-Internal  10.0.45.0       255.255.255.0    COST: 10

0000.0000.0004.00-01  0x00000001   0x9df        974          41      0/0/0
 SOURCE       0000.0000.0004.00
```

```
 IP-Internal  0.0.0.0        0.0.0.0         COST: 0

0000.0000.0005.00-00* 0x00000012   0x9d40       472         86      0/0/0
 SOURCE       0000.0000.0005.00
 NLPID        IPV4
 AREA ADDR    49.0001
 INTF ADDR    10.0.5.5
 INTF ADDR    10.0.45.5
 NBR  ID      0000.0000.0005.01  COST: 10
 IP-Internal  10.0.5.0        255.255.255.0    COST: 0
 IP-Internal  10.0.45.0       255.255.255.0    COST: 10

0000.0000.0005.01-00* 0x0000000e   0xf77e       472         55      0/0/0
 SOURCE       0000.0000.0005.01
 NLPID        IPV4
 NBR  ID      0000.0000.0005.00  COST: 0
 NBR  ID      0000.0000.0004.00  COST: 0

Total LSP(s): 4
   *(In TLV)-Leaking Route, *(By LSPID)-Self LSP, +-Self LSP(Extended),
         ATT-Attached, P-Partition, OL-Overload
```

The LSP 0000.0000.0004.00-01 from R4 carries the default route.

1.1.3 Quiz

Why can the **filter ip-prefix** *ip-prefix-name* **export** command be used on an ABR to filter OSPF inter-area routes but cannot be used on a router in an OSPF area to filter OSPF intra-area routes?

1.1.4 Configuration Reference

Configuration on R1

```
#
sysname R1
#
bfd
#
interface GigabitEthernet0/0/1
 ip address 10.0.13.1 255.255.255.0
 ospf bfd enable
 ospf bfd min-tx-interval 500 min-rx-interval 500 detect-multiplier 4
#
interface GigabitEthernet0/0/2
 ip address 10.0.12.1 255.255.255.0
 ospf bfd enable
 ospf bfd min-tx-interval 500 min-rx-interval 500 detect-multiplier 4
#
interface LoopBack1
 ip address 10.0.1.1 255.255.255.0
#
interface LoopBack2
 ip address 172.16.2.1 255.255.255.0
#
ospf 1 router-id 10.0.1.1
```

```
 bfd all-interfaces enable
 frr
  loop-free-alternate
 area 0.0.0.0
  network 10.0.13.1 0.0.0.0
  network 10.0.12.1 0.0.0.0
  network 10.0.1.1 0.0.0.0
  network 172.16.2.1 0.0.0.0
#
return
```

Configuration on R2

```
#
sysname R2
#
bfd
#
interface GigabitEthernet0/0/2
 ip address 10.0.23.2 255.255.255.0
 ospf bfd enable
 ospf bfd min-tx-interval 500 min-rx-interval 500 detect-multiplier 4
#
interface GigabitEthernet0/0/3
 ip address 10.0.12.2 255.255.255.0
 ospf bfd enable
 ospf bfd min-tx-interval 500 min-rx-interval 500 detect-multiplier 4
#
interface LoopBack0
 ip address 10.0.2.2 255.255.255.0
#
interface LoopBack2
 ip address 172.16.2.1 255.255.255.0
#
ospf 1 router-id 10.0.2.2
 bfd all-interfaces enable
 area 0.0.0.0
  network 10.0.12.2 0.0.0.0
  network 10.0.23.2 0.0.0.0
  network 10.0.2.2 0.0.0.0
  network 172.16.2.1 0.0.0.0
#
return
```

Configuration on R3

```
#
sysname R3
#
bfd
#
interface GigabitEthernet0/0/1
 ip address 10.0.13.3 255.255.255.0
```

```
 ospf bfd enable
 ospf bfd min-tx-interval 500 min-rx-interval 500 detect-multiplier 4
#
interface GigabitEthernet0/0/2
 ip address 10.0.34.3 255.255.255.0
 ospf bfd enable
 ospf bfd min-tx-interval 500 min-rx-interval 500 detect-multiplier 4
#
interface GigabitEthernet0/0/3
 ip address 10.0.23.3 255.255.255.0
 ospf bfd enable
 ospf bfd min-tx-interval 500 min-rx-interval 500 detect-multiplier 4
#
interface LoopBack0
 ip address 10.0.3.3 255.255.255.0
#
ospf 1 router-id 10.0.3.3
 bfd all-interfaces enable
 maximum load-balancing 1
 area 0.0.0.0
  network 10.0.13.3 0.0.0.0
  network 10.0.23.3 0.0.0.0
  network 10.0.3.3 0.0.0.0
 area 0.0.0.1
  filter ip-prefix 1 export
  network 10.0.34.3 0.0.0.0
#
ip ip-prefix 1 index 10 deny 172.16.3.0 24 greater-equal 24 less-equal 24
ip ip-prefix 1 index 20 permit 0.0.0.0 0 less-equal 32
#
return
```

Configuration on R4

```
#
sysname R4
#
bfd
#
isis 1
 network-entity 49.0001.0000.0000.0004.00
 default-route-advertise always level-1-2
#
interface GigabitEthernet0/0/2
 ip address 10.0.45.4 255.255.255.0
 isis enable 1
#
interface GigabitEthernet0/0/3
 ip address 10.0.34.4 255.255.255.0
 ospf bfd enable
 ospf bfd min-tx-interval 500 min-rx-interval 500 detect-multiplier 4
#
interface LoopBack0
 ip address 10.0.4.4 255.255.255.0
```

```
#
interface LoopBack3
 ip address 172.16.3.1 255.255.255.0
 ospf network-type broadcast
#
ospf 1 router-id 10.0.4.4
 bfd all-interfaces enable
 area 0.0.0.1
  network 10.0.34.4 0.0.0.0
  network 10.0.4.4 0.0.0.0
  network 172.16.3.1 0.0.0.0
#
return
```

Configuration on R5

```
#
sysname R5
#
isis 1
 is-level level-1
 network-entity 49.0001.0000.0000.0005.00
#
interface GigabitEthernet0/0/3
 ip address 10.0.45.5 255.255.255.0
 isis enable 1
#
interface LoopBack0
 ip address 10.0.5.5 255.255.255.0
 isis enable 1
#
return
```

2 Advanced BGP Features

2.1 Advanced BGP Features

2.1.1 Introduction

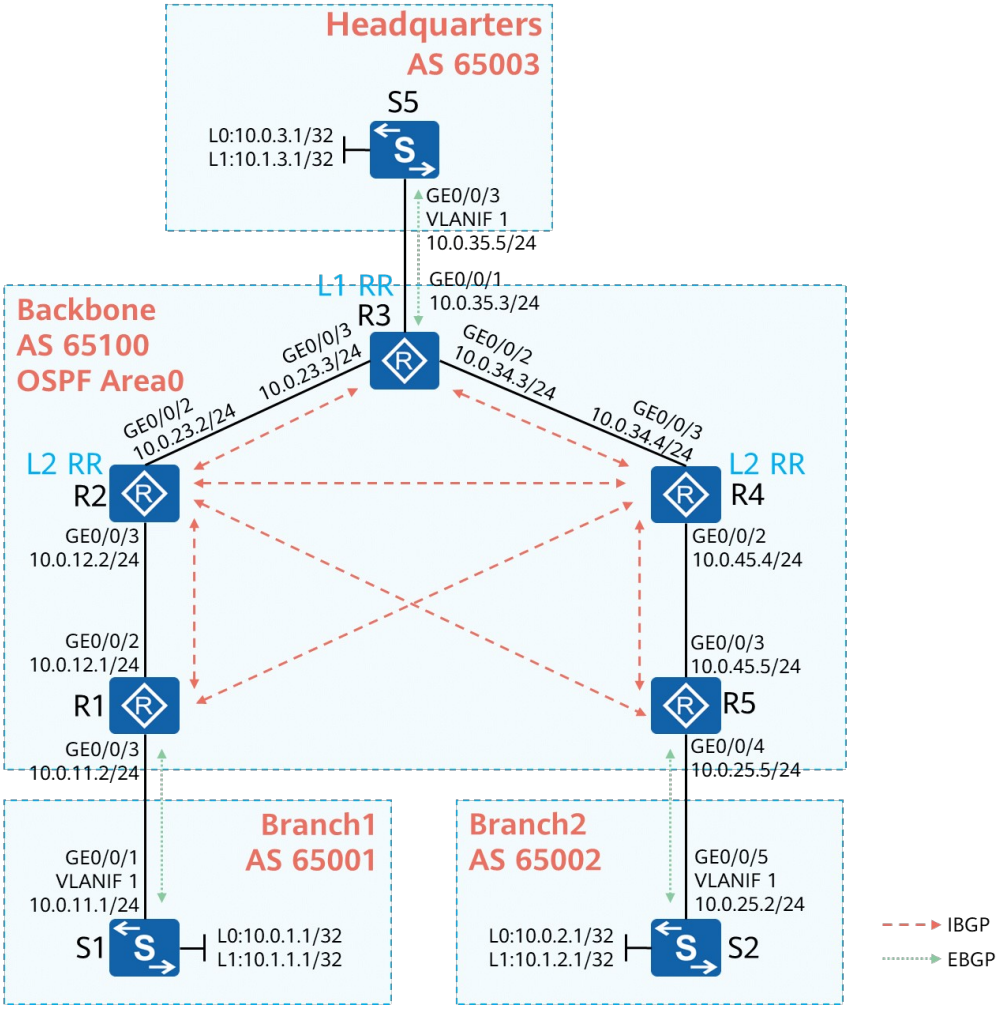
2.1.1.1 Objectives

Upon completion of this task, you will be able to:

- Establish BGP peer relationships based on peer groups.
- Configure BGP routing policies.
- Configure BGP security features.

2.1.1.2 Networking Topology

Networking of advanced BGP



Ошибка: источник перекрёстной ссылки не найден shows IP addresses of interconnection interfaces, BGP autonomous system (AS) numbers, and BGP peer relationships. R2 and R4 are level-2 route reflectors (RRs), and R1 and R5 are clients of level-2 RRs. R3 is a level-1 RR and receives routes from level-2 RRs. Loopback interfaces on S1, S2, and S5 are used to simulate users.

2.1.1.3 Lab Background

An enterprise has two branches and one headquarters. The enterprise has the following services:

OA: IP addresses of Loopback0 interfaces on S1, S2, and S5 are on the OA service network segment. OA data can be transmitted between branches and between branches and the headquarters. The routes related to OA services must be marked as originating ASs.

Finance: IP addresses of Loopback1 interfaces of S1, S2, and S5 belong to the financial service network segment. Confidential financial data can be transmitted only between branches and the headquarters.

The network administrator needs to build a secure network that meets these requirements.

2.1.2 Lab Configuration

2.1.2.1 Configuration Roadmap

1. Configure IP addresses for devices.
2. Configure OSPF in the backbone area to build the underlying network.
3. Deploy Generalized TTL Security Mechanism (GTSM) and BGP authentication between branches and the backbone network to ensure BGP network security.
4. Configure R1, R3, and R5 to establish Internal BGP (IBGP) peer relationships with R2 and R4, and configure R1, R3, and R5 as RR clients of R2 and R4.
5. Configure R3 as a level-1 RR, establish IBGP peer relationships with R2 and R4, and configure R2 and R4 as RR clients of R3.
6. Configure R1, R2, and R3 to add the community attribute to the routes of Loopback0 interfaces to mark the originating AS of the OA service.
7. Configure a routing policy on R1, R3, and R5 and use the AS-Path Filter tool to filter routes of Loopback1 interfaces.

2.1.2.2 Configuration Procedure

Step 1 Configure IP addresses for interconnection interfaces and loopback interfaces.

Name the devices.

The configuration details are not provided.

Disable the interfaces that are not used in this experiment.

The configuration details are not provided.

Configure IP addresses for GE0/0/2, GE0/0/3, and Loopback0 on R1.

```
[R1]interface LoopBack0
[R1-LoopBack0] ip address 10.10.10.1 255.255.255.255
[R1-LoopBack0] quit
[R1]interface GigabitEthernet0/0/2
[R1-GigabitEthernet0/0/2] ip address 10.0.12.1 255.255.255.0
[R1-GigabitEthernet0/0/2] quit
[R1]interface GigabitEthernet0/0/3
[R1-GigabitEthernet0/0/3] ip address 10.0.11.2 255.255.255.0
[R1-GigabitEthernet0/0/3] quit
```

Configure IP addresses for GE0/0/2, GE0/0/3, and Loopback0 on R2.

```
[R2]interface LoopBack0
[R2-LoopBack0] ip address 10.10.10.2 255.255.255.255
[R2-LoopBack0] quit
```

```
[R2]interface GigabitEthernet0/0/2
[R2-GigabitEthernet0/0/2] ip address 10.0.23.2 255.255.255.0
[R2-GigabitEthernet0/0/2] quit
[R2]interface GigabitEthernet0/0/3
[R2-GigabitEthernet0/0/3] ip address 10.0.12.2 255.255.255.0
[R2-GigabitEthernet0/0/3] quit
```

Configure IP addresses for GE0/0/1, GE0/0/2, GE0/0/3, and Loopback0 on R3.

```
[R3]interface LoopBack0
[R3-LoopBack0] ip address 10.10.10.3 255.255.255.255
[R3-LoopBack0] quit
[R3]interface GigabitEthernet0/0/1
[R3-GigabitEthernet0/0/1] ip address 10.0.35.3 255.255.255.0
[R3-GigabitEthernet0/0/1] quit
[R3]interface GigabitEthernet0/0/2
[R3-GigabitEthernet0/0/2] ip address 10.0.34.3 255.255.255.0
[R3-GigabitEthernet0/0/2] quit
[R3]interface GigabitEthernet0/0/3
[R3-GigabitEthernet0/0/3] ip address 10.0.23.3 255.255.255.0
[R3-GigabitEthernet0/0/3] quit
```

Configure IP addresses for GE0/0/2, GE0/0/3, GE0/0/5, and Loopback0 on R4.

```
[R4]interface LoopBack0
[R4-LoopBack0] ip address 10.10.10.4 255.255.255.255
[R4-LoopBack0] quit
[R4]interface GigabitEthernet0/0/2
[R4-GigabitEthernet0/0/2] ip address 10.0.45.4 255.255.255.0
[R4-GigabitEthernet0/0/2] quit
[R4]interface GigabitEthernet0/0/3
[R4-GigabitEthernet0/0/3] ip address 10.0.34.4 255.255.255.0
[R4-GigabitEthernet0/0/3] quit
```

Configure IP addresses for GE0/0/3, GE0/0/4, and Loopback0 on R5.

```
[R5]interface LoopBack0
[R5-LoopBack0] ip address 10.10.10.5 255.255.255.255
[R5-LoopBack0] quit
[R5]interface GigabitEthernet0/0/3
[R5-GigabitEthernet0/0/3] ip address 10.0.45.5 255.255.255.0
[R5-GigabitEthernet0/0/3] quit
[R5]interface GigabitEthernet0/0/4
[R5-GigabitEthernet0/0/4] ip address 10.0.25.5 255.255.255.0
[R5-GigabitEthernet0/0/4] quit
```

Configure the VLAN to which GE0/0/1 of S1 belongs and IP addresses for VLANIF 1, Loopback0, and Loopback1.

```
[S1]interface LoopBack0
[S1-LoopBack0] ip address 10.0.1.1 255.255.255.255
[S1-LoopBack0] quit
[S1]interface LoopBack1
[S1-LoopBack1] ip address 10.1.1.1 255.255.255.255
[S1-LoopBack1] quit
[S1]interface GigabitEthernet0/0/1
[S1-GigabitEthernet0/0/1] port link-type access
[S1-GigabitEthernet0/0/1] port default vlan 1
```

```
[S1-GigabitEthernet0/0/1] quit
[S1]interface Vlanif 1
[S1-Vlanif1] ip address 10.1.11.1 24
[S1-Vlanif1] quit
```

Configure the VLAN to which GE0/0/5 of S2 belongs and IP addresses for VLANIF 1, Loopback0, and Loopback1.

```
[S2]interface LoopBack0
[S2-LoopBack0] ip address 10.0.2.1 255.255.255.255
[S2-LoopBack0] quit
[S2]interface LoopBack1
[S2-LoopBack1] ip address 10.1.2.1 255.255.255.255
[S2-LoopBack1] quit
[S2]interface GigabitEthernet0/0/5
[S2-GigabitEthernet0/0/5] port link-type access
[S2-GigabitEthernet0/0/5] port default vlan 1
[S2-GigabitEthernet0/0/5] quit
[S2]interface Vlanif 1
[S2-Vlanif1] ip address 10.1.25.2 24
[S2-Vlanif1] quit
```

Configure the VLAN to which GE0/0/3 of S5 belongs and IP addresses for VLANIF 1, Loopback0, and Loopback1.

```
[S5]interface LoopBack0
[S5-LoopBack0] ip address 10.0.2.1 255.255.255.255
[S5-LoopBack0] quit
[S5]interface LoopBack1
[S5-LoopBack1] ip address 10.1.2.1 255.255.255.255
[S5-LoopBack1] quit
[S5]interface GigabitEthernet0/0/3
[S5-GigabitEthernet0/0/3] port link-type access
[S5-GigabitEthernet0/0/3] port default vlan 1
[S5-GigabitEthernet0/0/3] quit
[S5]interface Vlanif 1
[S5-Vlanif1] ip address 10.1.35.5 24
[S5-Vlanif1] quit
```

Check the connectivity of IP addresses of interconnection interfaces on R1, R3, and R5.

```
<R1>ping -c 1 10.0.11.1
  PING 10.0.11.1: 56  data bytes, press CTRL_C to break
    Reply from 10.0.11.1: bytes=56 Sequence=1 ttl=255 time=90 ms

  --- 10.0.11.1ping statistics ---
    1 packet(s) transmitted
    1 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 90/90/90 ms

<R1>ping -c 1 10.0.12.2
  PING 10.0.12.2: 56  data bytes, press CTRL_C to break
    Reply from 10.0.12.2: bytes=56 Sequence=1 ttl=255 time=140 ms

  --- 10.0.12.2 ping statistics ---
```

```
    1 packet(s) transmitted
    1 packet(s) received
    0.00% packet loss
round-trip min/avg/max = 140/140/140 ms

<R3>ping -c 1 10.0.23.2
  PING 10.0.23.2: 56  data bytes, press CTRL_C to break
    Reply from 10.0.23.2: bytes=56 Sequence=1 ttl=255 time=70 ms

  --- 10.0.23.2ping statistics ---
    1 packet(s) transmitted
    1 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 70/70/70 ms

<R3>ping -c 1 10.0.34.4
  PING 10.0.34.4: 56  data bytes, press CTRL_C to break
    Reply from 10.0.34.4: bytes=56 Sequence=1 ttl=255 time=60 ms

  --- 10.0.34.4ping statistics ---
    1 packet(s) transmitted
    1 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 60/60/60 ms

<R3>ping -c 1 10.0.35.5
  PING 10.0.35.5: 56  data bytes, press CTRL_C to break
    Reply from 10.0.35.5: bytes=56 Sequence=1 ttl=255 time=50 ms

  --- 10.0.35.5 ping statistics ---
    1 packet(s) transmitted
    1 packet(s) received
    0.00% packet loss
round-trip min/avg/max = 50/50/50 ms

<R5>ping -c 1 10.0.45.4
  PING 10.0.45.4: 56  data bytes, press CTRL_C to break
    Reply from 10.0.45.4: bytes=56 Sequence=1 ttl=255 time=60 ms

  --- 10.0.45.4 ping statistics ---
    1 packet(s) transmitted
    1 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 60/60/60 ms

<R5>ping -c 1 10.0.25.2
  PING 10.0.25.2: 56  data bytes, press CTRL_C to break
    Reply from 10.0.25.2: bytes=56 Sequence=1 ttl=255 time=50 ms

  --- 10.0.25.2 ping statistics ---
    1 packet(s) transmitted
    1 packet(s) received
    0.00% packet loss
round-trip min/avg/max = 50/50/50 ms
```

9 Configure OSPF on the backbone network to set up an underlying network.

Configure an OSPF process on each router according to the topology design. Set the process ID to 1, area ID to 0, and router ID to the loopback interface's IP address of the router. R1 with the router ID of 10.10.10.1 is used as an example.

Configure routes of OSPF interfaces in AS 65100 to be accurately advertised.

Configure R1.

```
[R1]router id 10.10.10.1
[R1]ospf 1
[R1-ospf-1] area 0
[R1-ospf-1-area-0.0.0.0] network 10.10.10.1 0.0.0.0
[R1-ospf-1-area-0.0.0.0] network 10. 0.12.1 0.0.0.0
```

Configure R2.

```
[R2]router id 10.10.10.2
[R2]ospf 1
[R2-ospf-1] area 0
[R2-ospf-1-area-0.0.0.0] network 10.10.10.2 0.0.0.0
[R2-ospf-1-area-0.0.0.0] network 10.0.12.2 0.0.0.0
[R2-ospf-1-area-0.0.0.0] network 10.0.23.2 0.0.0.0
```

Configure R3.

```
[R3]router id 10.10.10.3
[R3]ospf 1
[R3-ospf-1] area 0
[R3-ospf-1-area-0.0.0.0] network 10.10.10.3 0.0.0.0
[R3-ospf-1-area-0.0.0.0] network 10.0.23.3 0.0.0.0
[R3-ospf-1-area-0.0.0.0] network 10.0.34.3 0.0.0.0
[R3-ospf-1-area-0.0.0.0] network 10.0.35.3 0.0.0.0
```

Configure R4.

```
[R4]router id 10.10.10.4
[R4]ospf 1
[R4-ospf-1] area 0
[R4-ospf-1-area-0.0.0.0] network 10.10.10.4 0.0.0.0
[R4-ospf-1-area-0.0.0.0] network 10.0.34.4 0.0.0.0
[R4-ospf-1-area-0.0.0.0] network 10.0.45.4 0.0.0.0
```

Configure R5.

```
[R5]router id 10.10.10.5
[R5]ospf 1
[R5-ospf-1] area 0
[R5-ospf-1-area-0.0.0.0] network 10.10.10.5 0.0.0.0
[R5-ospf-1-area-0.0.0.0] network 10.0.45.5 0.0.0.0
```

Check the OSPF configuration.

Check OSPF neighbor relationship information on R2 and R4.

```
[R2]display ospf peer brief

 OSPF Process 1 with Router ID 10.10.10.2
   Peer Statistic Information
 ----------------------------------------------------------------------------
 Area Id          Interface                    Neighbor id      State
 0.0.0.0          GigabitEthernet0/0/3             10.10.10.1       Full
 0.0.0.0          GigabitEthernet0/0/2             10.10.10.3       Full
 ----------------------------------------------------------------------------

[R4]display ospf peer brief

 OSPF Process 1 with Router ID 10.10.10.4
   Peer Statistic Information
 ----------------------------------------------------------------------------
 Area Id          Interface                    Neighbor id      State
 0.0.0.0          GigabitEthernet0/0/3             10.10.10.3       Full
 0.0.0.0          GigabitEthernet0/0/2             10.10.10.5       Full
 ----------------------------------------------------------------------------
```

Check the OSPF routing table and IP routing table on R2.

```
[R2]display ospf routing

 OSPF Process 1 with Router ID 10.10.10.2
   Routing Tables

 Routing for Network
 Destination        Cost  Type       NextHop        AdvRouter       Area
 10.0.12.0/24        1    Transit    10.0.12.2        10.10.10.2      0.0.0.0
 10.0.23.0/24        1    Transit    10.0.23.1        10.10.10.2      0.0.0.0
 10.10.10.2/32       0     Stub      10.10.10.2       10.10.10.2      0.0.0.0
 10.0.34.0/24        2    Transit    10.0.23.2        10.10.10.4      0.0.0.0
 10.0.35.0/24        2    Stub      10.0.23.2        10.10.10.3      0.0.0.0
 10.0.45.0/24        3    Transit    10.0.23.2        10.10.10.5      0.0.0.0
 10.10.10.1/32       1     Stub      10.0.12.1        10.10.10.1      0.0.0.0
 10.10.10.3/32       1     Stub      10.0.23.2       10.10.10.3      0.0.0.0
 10.10.10.4/32       2     Stub      10.0.23.2       10.10.10.4      0.0.0.0
 10.10.10.5/32       3     Stub      10.0.23.2       10.10.10.5      0.0.0.0

 Total Nets: 10
 Intra Area: 10  Inter Area: 0  ASE: 0  NSSA: 0

[R2]display ip routing-table
Route Flags: R - relay, D - download to fib
------------------------------------------------------------------------------
Routing Tables: Public
         Destinations : 18       Routes : 18

Destination/Mask    Proto   Pre  Cost      Flags NextHop         Interface

     10.0.12.0/24  Direct  0    0           D   10.0.12.2       GigabitEthernet0/0/3
     10.0.12.2/32  Direct  0    0           D   127.0.0.1       GigabitEthernet0/0/3
   10.0.12.255/32  Direct  0    0           D   127.0.0.1       GigabitEthernet0/0/3
```

```
     10.0.23.0/24  Direct  0    0         D   10.0.23.1       GigabitEthernet0/0/2
     10.0.23.2/32  Direct  0    0         D   127.0.0.1       GigabitEthernet0/0/2
   10.0.23.255/32  Direct  0    0         D   127.0.0.1       GigabitEthernet0/0/2
     10.0.34.0/24  OSPF  10   2         D   10.0.23.2       GigabitEthernet0/0/2
     10.0.35.0/24  OSPF  10   2         D   10.0.23.2       GigabitEthernet0/0/2
     10.0.45.0/24  OSPF  10   3         D   10.0.23.2       GigabitEthernet0/0/2
    10.10.10.1/32  OSPF  10   1         D   10.0.12.1       GigabitEthernet0/0/3
    10.10.10.2/32  Direct  0    0         D   127.0.0.1       LoopBack0
    10.10.10.3/32  OSPF  10   1         D   10.0.23.2       GigabitEthernet0/0/2
    10.10.10.4/32  OSPF  10   2         D   10.0.23.2       GigabitEthernet0/0/2
    10.10.10.5/32  OSPF  10   3         D   10.0.23.2       GigabitEthernet0/0/2
     127.0.0.0/8   Direct  0    0         D   127.0.0.1       InLoopBack0
     127.0.0.1/32  Direct  0    0         D   127.0.0.1       InLoopBack0
127.255.255.255/32  Direct  0    0         D   127.0.0.1       InLoopBack0
255.255.255.255/32  Direct  0    0         D   127.0.0.1       InLoopBack0
```

The preceding command output shows that the OSPF process of R2 has learned all routes in AS 65100 and the routes are preferentially selected in the IP routing table.

10 Configure EBGP peers and deploy GTSM and BGP authentication between EBGP peers.

Establish EBGP peer relationships between S1 and R1, between S5 and R3, and between S2 and R5. Configure BGP authentication, set the authentication password to Huawei@123, and set the TTL in GTSM to 255.

In this experiment, S1 belongs to AS 65001, S2 belongs to AS 65002, S5 belongs to AS 65003, R1, R3, and R5 belongs to AS 65100.

Configure EBGP peers on S1 and R1, and configure BGP authentication and GTSM. GTSM only needs to be configured on the backbone network side.

```
[R1]bgp 65100
[R1-bgp] peer 10.0.11.1 as-number 65001
[R1-bgp] peer 10.0.11.1 password cipher Huawei@123
[R1-bgp] peer 10.0.11.1 valid-ttl-hops 255

[S1]bgp 65001
[S1-bgp] peer 10.0.11.2 as-number 65100
[S1-bgp] peer 10.0.11.2 password cipher Huawei@123
```

Configure EBGP peers on S5 and R3, and configure BGP authentication and GTSM. GTSM only needs to be configured on the backbone network side.

```
[R3]bgp 65100
[R3-bgp] peer 10.0.35.5 as-number 65003
[R3-bgp] peer 10.0.35.5 password cipher Huawei@123
[R3-bgp] peer 10.0.35.5 valid-ttl-hops 255

[S5]bgp 65003
[S5-bgp] peer 10.0.35.3 as-number 65100
```

```
[S5-bgp] peer 10.0.35.3 password cipher Huawei@123
```

Configure EBGP peers on S2 and R5, and configure BGP authentication and GTSM. GTSM only needs to be configured on the backbone network side.

```
[R5]bgp 65100
[R5-bgp] peer 10.0.25.2 as-number 65002
[R5-bgp] peer 10.0.25.2 password cipher Huawei@123
[R5-bgp] peer 10.0.25.2 valid-ttl-hops 255

[S2]bgp 65002
[S2-bgp] peer 10.0.25.5 as-number 65001
[S2-bgp] peer 10.0.25.5 password cipher Huawei@123
```

Check the EBGP configuration.

Check the BGP peer relationship status on R1, R3, and R5.

```
[R1]display bgp peer

 BGP local router ID : 10.10.10.1
 Local AS number : 65100
 Total number of peers : 1    Peers in established state : 1

  Peer            V          AS  MsgRcvd  MsgSent  OutQ  Up/Down       State  PrefRcv

  10.0.11.1       4       65001      12       15     0   00:10:16  Established     0

[R3]display bgp peer

 BGP local router ID : 10.10.10.3
 Local AS number : 65100
 Total number of peers : 1    Peers in established state : 1

  Peer            V          AS  MsgRcvd  MsgSent  OutQ  Up/Down       State  PrefRcv

  10.0.35.5       4       65003      14       12     0   00:10:48  Established     0

[R5]display bgp peer

 BGP local router ID : 10.10.10.5
 Local AS number : 65100
 Total number of peers : 1    Peers in established state : 1

  Peer            V          AS  MsgRcvd  MsgSent  OutQ  Up/Down       State PrefRcv

  10.0.25.2       4       65002      13       13     0   00:11:34  Established    0
```

The neighbor relationships between R1, R3, and R5 and their peers are in Established state.

11 Configure IBGP peers and hierarchical RRs.

R2 and R4 are level-2 RRs, and R1, R3, and R5 are clients of R2 and R4. To prevent route transmission between RRs in the same cluster, you need to change the cluster ID to 24.24.24.24.

R3 is a level-1 RR, and R2 and R4 are clients of R3.

Deploy IBGP peers based on the topology and establish IBGP peer relationships based on loopback interface addresses. Because there are a large number of IBGP peers, configure a peer group.

Configure R1.

```
[R1]bgp 65100
[R1-bgp] group IBGP internal
[R1-bgp] peer IBGP connect-interface LoopBack0
[R1-bgp] peer IBGP next-hop-local
[R1-bgp] peer 10.10.10.2 group IBGP
[R1-bgp] peer 10.10.10.4 group IBGP
```

Configure R2.

```
[R2]bgp 65100
[R2-bgp] group IBGP internal
[R2-bgp] peer IBGP connect-interface LoopBack0
[R2-bgp] peer 10.10.10.1 group IBGP
[R2-bgp] peer 10.10.10.3 group IBGP
[R2-bgp] peer 10.10.10.4 group IBGP
[R2-bgp] peer 10.10.10.5 group IBGP
```

Configure R3.

```
[R3]bgp 65100
[R3-bgp] group IBGP internal
[R3-bgp] peer IBGP connect-interface LoopBack0
[R3-bgp] peer IBGP next-hop-local
[R3-bgp] peer 10.10.10.2 group IBGP
[R3-bgp] peer 10.10.10.4 group IBGP
```

Configure R4.

```
[R4]bgp 65100
[R4-bgp] roup IBGP internal
[R4-bgp] peer IBGP connect-interface LoopBack0
[R4-bgp] peer 10.10.10.1 group IBGP
[R4-bgp] peer 10.10.10.2 group IBGP
[R4-bgp] peer 10.10.10.3 group IBGP
[R4-bgp] peer 10.10.10.5 group IBGP
```

Configure R5.

```
[R5]bgp 65100
[R5-bgp] group IBGP internal
[R5-bgp] peer IBGP connect-interface LoopBack0
[R5-bgp] peer IBGP next-hop-local
[R5-bgp] peer 10.10.10.2 group IBGP
[R5-bgp] peer 10.10.10.4 group IBGP
```

Check the IBGP configuration.

Check the BGP peer relationship status on R2 and R4.

```
[R2]display bgp peer

 BGP local router ID : 10.10.10.2
 Local AS number : 65100
 Total number of peers : 4    Peers in established state : 4

  Peer            V          AS  MsgRcvd  MsgSent  OutQ  Up/Down       State  PrefRcv

  10.10.10.1      4       65100      25       25     0   00:23:16  Established      0
  10.10.10.3      4       65100      23       24     0   00:21:52  Established      0
  10.10.10.4      4       65100      18       19     0   00:16:58  Established      0
  10.10.10.5      4       65100      16       17     0   00:14:50  Established      0

[R4]display bgp peer

 BGP local router ID : 10.10.10.4
 Local AS number : 65100
 Total number of peers : 4    Peers in established state : 4

  Peer            V          AS  MsgRcvd  MsgSent  OutQ  Up/Down       State  PrefRcv

  10.10.10.1      4       65100      20       20     0   00:18:33  Established      0
  10.10.10.2      4       65100      20       20     0   00:18:25  Established      0
  10.10.10.3      4       65100      20       20     0   00:18:16  Established      0
  10.10.10.5      4       65100      18       19     0   00:16:06  Established      0
```

Based on peer tables of R2 and R4, you can find that IBGP peer relationships have been established between routers in AS 65100.

Deploy hierarchical RRs.

R2 and R4 are level-2 RRs, and R1, R3, and R5 are clients of R2 and R4. The same cluster ID is configured for R2 and R4.

R3 is a level-1 RR, and R2 and R4 are clients of R3.

Configure R2.

```
[R2]bgp 65100
[R2-bgp] peer IBGP reflect-client
[R2-bgp] reflector cluster-id 24.24.24.24
```

Configure R4.

```
[R4]bgp 65100
[R4-bgp] peer IBGP reflect-client
[R4-bgp] reflector cluster-id 24.24.24.24
```

Configure R3.

```
[R3]bgp 65100
[R3-bgp] peer IBGP reflect-client
```

Check the RR configuration.

Check the RR configuration on R2, R3, and R4.

```
[R2]display bgp group IBGP

 BGP peer-group: IBGP
```

```
 Remote AS: 65100
 Authentication type configured: None
 Type : internal
 Configured hold timer value: 180
 Keepalive timer value: 60
 Connect-retry timer value: 32
 Minimum route advertisement interval is 15 seconds
 Connect-interface has been configured
 PeerSession Members:
   10.10.10.1        10.10.10.3        10.10.10.4        10.10.10.5


 It's route-reflector-client
 Peer Preferred Value: 0
 No routing policy is configured
 Peer Members:
  Peer          V        AS  MsgRcvd  MsgSent  OutQ  Up/Down      State  PrefRcv

  10.10.10.1     4       65100     46      47     0  00:44:30  Established     0
  10.10.10.3     4       65100     45      47     0  00:43:06  Established     0
  10.10.10.4     4       65100     41      42     0  00:38:12  Established     0
  10.10.10.5     4       65100     38      40     0  00:36:04  Established     0

[R4]display bgp group IBGP

 BGP peer-group: IBGP
 Remote AS: 65100
 Authentication type configured: None
 Type : internal
 Configured hold timer value: 180
 Keepalive timer value: 60
 Connect-retry timer value: 32
 Minimum route advertisement interval is 15 seconds
 Connect-interface has been configured
 PeerSession Members:
   10.10.10.1        10.10.10.2        10.10.10.3        10.10.10.5


 It's route-reflector-client
 Peer Preferred Value: 0
 No routing policy is configured
 Peer Members:
  Peer          V        AS  MsgRcvd  MsgSent  OutQ  Up/Down      State  PrefRcv

  10.10.10.1     4       65100     50      51     0  00:48:18  Established     0
  10.10.10.2     4       65100     51      51     0  00:48:10  Established     0
  10.10.10.3     4       65100     50      51     0  00:48:01  Established     0
  10.10.10.5     4       65100     47      49     0  00:45:51  Established     0

[R3]display bgp group IBGP

 BGP peer-group: IBGP
 Remote AS: 65100
 Authentication type configured: None
 Type : internal
```

```
Configured hold timer value: 180
Keepalive timer value: 60
Connect-retry timer value: 32
Minimum route advertisement interval is 15 seconds
Connect-interface has been configured
PeerSession Members:
  10.10.10.2        10.10.10.4

Nexthop self has been configured
It's route-reflector-client
Peer Preferred Value: 0
No routing policy is configured
Peer Members:
 Peer          V        AS  MsgRcvd  MsgSent  OutQ  Up/Down      State  PrefRcv

 10.10.10.2     4      65100      57      56     0  00:54:36  Established     0
 10.10.10.4     4      65100      52      52     0  00:49:34  Established     0
```

Clients of the RR are correctly configured.

12 Advertise BGP routes.

The routes of Loopback0, Loopback1, and Loopback2 on S1, S2, and S5 need to be advertised to BGP. After the routes are advertised, the backbone network controls route advertisement through routing policies.

BGP routes can be advertised using the **import-route** or **network** command. In this experiment, there are only a few network segments. Therefore, you can use the **network** command.

Configure S1.

```
[R1]bgp 65001
[R1-bgp] network 10.0.1.1 32
[R1-bgp] network 10.1.1.1 32
```

Configure S2.

```
[R2]bgp 65002
[R2-bgp] network 10.0.2.1 32
[R2-bgp] network 10.1.2.1 32
```

Configure S5.

```
[R5]bgp 65003
[R5-bgp] network 10.0.3.1 32
[R5-bgp] network 10.1.3.1 32
```

Check route advertisement results.

Check route advertisement results on S1, S2, and S5. S1 is used as an example. The configurations of S2 and S5 are similar.

```
[S1]display bgp routing-table
```

```
BGP Local router ID is 10.1.11.1
Status codes: * - valid, > - best, d - damped,
              h - history,  i - internal, s - suppressed, S - Stale
              Origin : i - IGP, e - EGP, ? - incomplete


Total Number of Routes: 9
     Network          NextHop       MED       LocPrf    PrefVal Path/Ogn

 *>  10.0.1.1/32        0.0.0.0         0                 0      i
 *>  10.0.2.1/32        10.0.11.2                          0      65100 65002i
 *>  10.0.3.1/32        10.0.11.2                          0      65100 65003i
 *>  10.1.1.1/32        0.0.0.0         0                 0      i
 *>  10.1.2.1/32        10.0.11.2                          0      65100 65002i
 *>  10.1.3.1/32        10.0.11.2                          0      65100 65003i
```

The command output shows that S1, S2, and S5 have learned routes from each other.

13 Configure a routing policy to label the OA service.

The community attribute can be used to label a route so that the network administrator can know the AS from which the route comes.

In this experiment, you only need to add attributes to the routes of Loopback0 interfaces on S1, S2, and S5.

By default, the community attribute is not sent to peers. You need to manually configure devices to advertise the community attribute.

Generally, the community attribute is in the format of AS:NN. In this experiment, the community attribute is defined as AS:01.

Configure R1, R2, R3, R4, and R5 to advertise the community attribute to their peers.

```
[R1]bgp 65100
[R1-bgp] peer IBGP advertise-community
[R1-bgp] peer 10.0.11.1 advertise-community

[R2]bgp 65100
[R2-bgp] peer IBGP advertise-community

[R3]bgp 65100
[R3-bgp] peer IBGP advertise-community
[R3-bgp] peer 10.0.35.5 advertise-community

[R4]bgp 65100
[R4-bgp] peer IBGP advertise-community

[R5]bgp 65100
```

```
[R5-bgp] peer IBGP advertise-community
[R5-bgp] peer 10.0.25.2 advertise-community
```

Configure S1, S2, and S5 to advertise the community attribute to their peers.

```
[S1]bgp 65001
[S1-bgp] peer 10.0.11.2 advertise-community

[S2]bgp 65001
[S2-bgp] peer 10.0.25.5 advertise-community

[S5]bgp 65001
[S5-bgp] peer 10.0.35.3 advertise-community
```

Configure routing policies on S1, S2, and S5 to add the community attribute to routes of Loopback0 interfaces.

Configure S1.

```
[S1]ip ip-prefix Com index 10 permit 10.0.1.1 32
[S1]route-policy Attr permit node 10
[S1-route-policy] if-match ip-prefix Com
[S1-route-policy] apply community 65001:1
[S1-route-policy] quit
[S1]route-policy Attr permit node 100
[S1-route-policy] quit
[S1]bgp 65001
[S1-bgp] peer 10.0.11.2 route-policy Attr export
```

Configure S2.

```
[S2]ip ip-prefix Com index 10 permit 10.0.2.1 32
[S2]route-policy Attr permit node 10
[S2-route-policy] if-match ip-prefix Com
[S2-route-policy] apply community 65002:2
[S2-route-policy] quit
[S2]route-policy Attr permit node 100
[S2-route-policy quit
[S2]bgp 65002
[S2-bgp] peer 10.0.25.5 route-policy Attr export
```

Configure S5.

```
[S5]ip ip-prefix Com index 10 permit 10.0.3.1 32
[S5]route-policy Attr permit node 10
[S5-route-policy] if-match ip-prefix Com
[S5-route-policy] apply community 65003:2
[S5-route-policy] quit
[S5]route-policy Attr permit node 100
[S5-route-policy] quit
[S5]bgp 65003
[S5-bgp] peer 10.0.35.3 route-policy Attr export
```

Check the configuration.

Check the configurations on S1, S2, and S5. S1 is used as an example. The configurations of S2 and S5 are similar to that of S1.

```
[S1]display bgp routing-table 10.0.3.1
```

```
BGP local router ID : 10.0.11.1
Local AS number : 65001
Paths:   1 available, 1 best, 1 select
BGP routing table entry information of 10.0.3.1/32:
From: 10.0.11.2 (10.10.10.1)
Route Duration: 00h03m50s
Direct Out-interface: Vlanif1
Original nexthop: 10.0.11.2
Qos information : 0x0
Community:<65003:1>
AS-path 65100 65003, origin igp, pref-val 0, valid, external, best, select, active, pre 255
Not advertised to any peer yet
```

14 Configure a routing policy to control financial services.

Traffic of confidential financial services can be forwarded only between S1 and S5 and between S2 and S5. If no VPN is deployed, you can only control route sending and receiving.

To simplify filtering configuration, you can use the AS_Path filter and routing policy to filter routes on R1 and R2.

For route control, do not filter the routes of Loopback0 interfaces. You can use the community attribute to allow routes of Loopback0 interfaces in advance and then filter routes of Loopback1 interfaces.

Configure R1.

```
[R1]ip community-filter basic OA permit 65002:1
[R1]ip as-path-filter Finance permit 65002$
[R1]route-policy Finance permit node 10
[R1-route-policy] if-match community-filter OA
[R1-route-policy] quit
[R1]route-policy Finance deny node 20
[R1-route-policy] if-match as-path-filter Finance
[R1-route-policy] quit
[R1]route-policy Finance permit node 100
[R1-route-policy] quit
[R1]bgp 65100
[R1-bgp] peer 10.0.11.1
[R1-bgp] route-policy Finance export
```

Configure R5.

```
[R5]ip community-filter basic OA permit 65001:1
[R5]ip as-path-filter Finance permit 65001$
[R5]route-policy Finance permit node 10
[R5-route-policy] if-match community-filter OA
[R5-route-policy] quit
[R5]route-policy Finance deny node 20
[R5-route-policy] if-match as-path-filter Finance
```

```
[R5-route-policy] quit
[R5]route-policy Finance permit node 100
[R5-route-policy] quit
[R5]bgp 65100
[R5-bgp] peer 10.0.25.2
[R5-bgp] route-policy Finance export
```

Check the configuration.

Check the configurations on S1, S2, and S5.

```
<S1>display bgp routing-table

 BGP Local router ID is 10.0.11.1
 Status codes: * - valid, > - best, d - damped,
               h - history,  i - internal, s - suppressed, S - Stale
               Origin : i - IGP, e - EGP, ? - incomplete


 Total Number of Routes: 7
      Network          NextHop        MED        LocPrf    PrefVal Path/Ogn

 *>   10.0.1.1/32      0.0.0.0        0                    0       i
 *>   10.0.2.1/32      10.0.11.2                            0       65100 65002i
 *>   10.0.3.1/32      10.0.11.2                            0       65100 65003i
 *>   10.1.1.1/32      0.0.0.0        0                    0       i
 *>   10.1.3.1/32      10.0.11.2                            0       65100 65003i

<S2>display bgp routing-table

 BGP Local router ID is 10.0.25.2
 Status codes: * - valid, > - best, d - damped,
               h - history,  i - internal, s - suppressed, S - Stale
               Origin : i - IGP, e - EGP, ? - incomplete


 Total Number of Routes: 7
      Network          NextHop        MED        LocPrf    PrefVal Path/Ogn

 *>   10.0.1.1/32      10.0.25.5                            0       65100 65001i
 *>   10.0.2.1/32      0.0.0.0        0                    0       i
 *>   10.0.3.1/32      10.0.25.5                            0       65100 65003i
 *>   10.1.2.1/32      0.0.0.0        0                    0       i
 *>   10.1.3.1/32      10.0.25.5                            0       65100 65003i

<S5>display bgp routing-table

 BGP Local router ID is 10.0.35.5
 Status codes: * - valid, > - best, d - damped,
               h - history,  i - internal, s - suppressed, S - Stale
               Origin : i - IGP, e - EGP, ? - incomplete


 Total Number of Routes: 9
      Network          NextHop        MED        LocPrf    PrefVal Path/Ogn
```

```
*>  10.0.1.1/32      10.0.35.3                  0      65100 65001i
*>  10.0.2.1/32      10.0.35.3                  0      65100 65002i
*>  10.0.3.1/32      0.0.0.0        0           0      i
*>  10.1.1.1/32      10.0.35.3                  0      65100 65001i
*>  10.1.2.1/32      10.0.35.3                  0      65100 65002i
*>  10.1.3.1/32      0.0.0.0        0           0      i
```

The command output shows that the route 10.1.2.1/32 of Loopback1 on S2 has been filtered out in the BGP routing table of S1, the route 10.1.1.1/32 of Loopback1 on S1 has been filtered out in the BGP routing table of S2, and the BGP routing table of S5 remains unchanged.

2.1.3 Quiz

How can I prevent a BGP peer from receiving a large number of unnecessary routes?

2.1.4 Configuration Reference

Configuration on R1

```
#
 sysname R1
#
router id 10.10.10.1
#
interface GigabitEthernet0/0/2
 ip address 10.0.12.1 255.255.255.0
#
interface GigabitEthernet0/0/3
 ip address 10.0.11.2 255.255.255.0
#
interface LoopBack0
 ip address 10.10.10.1 255.255.255.255
#
bgp 65100
 peer 10.0.11.1 as-number 65001
 peer 10.0.11.1 password cipher Huawei@123
 peer 10.0.11.1 valid-ttl-hops 255
 group IBGP internal
 peer IBGP connect-interface LoopBack0
 peer 10.10.10.2 as-number 65100
 peer 10.10.10.2 group IBGP
 peer 10.10.10.4 as-number 65100
 peer 10.10.10.4 group IBGP
 #
 ipv4-family unicast
  undo synchronization
  peer 10.0.11.1 enable
  peer 10.0.11.1 route-policy Finance export
  peer 10.0.11.1 advertise-community
  peer IBGP enable
  peer IBGP next-hop-local
  peer IBGP advertise-community
  peer 10.10.10.2 enable
  peer 10.10.10.2 group IBGP
```

```
  peer 10.10.10.4 enable
  peer 10.10.10.4 group IBGP
#
ospf 1
 area 0.0.0.0
  network 10.0.12.1 0.0.0.0
  network 10.10.10.1 0.0.0.0
#
route-policy Finance permit node 10
 if-match community-filter OA
#
route-policy Finance deny node 20
 if-match as-path-filter Finance
#
route-policy Finance permit node 100
#
ip as-path-filter Finance permit 65002$
#
ip community-filter basic OA permit 65002:1
#
return
```

Configuration on R2

```
#
 sysname R2
#
router id 10.10.10.2
#
interface GigabitEthernet0/0/2
 ip address 10.0.23.2 255.255.255.0
#
interface GigabitEthernet0/0/3
 ip address 10.0.12.2 255.255.255.0
#
interface LoopBack0
 ip address 10.10.10.2 255.255.255.255
#
bgp 65100
 group IBGP internal
 peer IBGP connect-interface LoopBack0
 peer 10.10.10.1 as-number 65100
 peer 10.10.10.1 group IBGP
 peer 10.10.10.3 as-number 65100
 peer 10.10.10.3 group IBGP
 peer 10.10.10.4 as-number 65100
 peer 10.10.10.4 group IBGP
 peer 10.10.10.5 as-number 65100
 peer 10.10.10.5 group IBGP
 #
 ipv4-family unicast
  undo synchronization
  reflector cluster-id 24.24.24.24
  peer IBGP enable
  peer IBGP reflect-client
  peer IBGP advertise-community
```

```
 peer 10.10.10.1 enable
 peer 10.10.10.1 group IBGP
 peer 10.10.10.3 enable
 peer 10.10.10.3 group IBGP
 peer 10.10.10.4 enable
 peer 10.10.10.4 group IBGP
 peer 10.10.10.5 enable
 peer 10.10.10.5 group IBGP
#
ospf 1
 area 0.0.0.0
  network 10.0.12.2 0.0.0.0
  network 10.0.23.2 0.0.0.0
  network 10.10.10.2 0.0.0.0
#
return
```

Configuration on R3

```
#
 sysname R3
#
router id 10.10.10.3
#
interface GigabitEthernet0/0/1
 ip address 10.0.35.3 255.255.255.0
#
interface GigabitEthernet0/0/2
 ip address 10.0.34.3 255.255.255.0
#
interface GigabitEthernet0/0/3
 ip address 10.0.23.3 255.255.255.0
#
interface LoopBack0
 ip address 10.10.10.3 255.255.255.255
#
bgp 65100
 peer 10.0.35.5 as-number 65003
 peer 10.0.35.5 password cipher Huawei@123
 peer 10.0.35.5 valid-ttl-hops 255
 group IBGP internal
 peer IBGP connect-interface LoopBack0
 peer 10.10.10.2 as-number 65100
 peer 10.10.10.2 group IBGP
 peer 10.10.10.4 as-number 65100
 peer 10.10.10.4 group IBGP
 #
 ipv4-family unicast
  undo synchronization
  peer 10.0.35.5 enable
  peer 10.0.35.5 advertise-community
  peer IBGP enable
  peer IBGP reflect-client
  peer IBGP next-hop-local
  peer IBGP advertise-community
  peer 10.10.10.2 enable
```

```
  peer 10.10.10.2 group IBGP
  peer 10.10.10.4 enable
  peer 10.10.10.4 group IBGP
#
ospf 1
 area 0.0.0.0
  network 10.0.23.3 0.0.0.0
  network 10.0.34.3 0.0.0.0
  network 10.0.35.3 0.0.0.0
  network 10.10.10.3 0.0.0.0
#
return
```

Configuration on R4

```
#
 sysname R4
#
router id 10.10.10.4
#
interface GigabitEthernet0/0/2
 ip address 10.0.45.4 255.255.255.0
#
interface GigabitEthernet0/0/3
 ip address 10.0.34.4 255.255.255.0
#
interface LoopBack0
 ip address 10.10.10.4 255.255.255.255
#
bgp 65100
 group IBGP internal
 peer IBGP connect-interface LoopBack0
 peer 10.10.10.1 as-number 65100
 peer 10.10.10.1 group IBGP
 peer 10.10.10.2 as-number 65100
 peer 10.10.10.2 group IBGP
 peer 10.10.10.3 as-number 65100
 peer 10.10.10.3 group IBGP
 peer 10.10.10.5 as-number 65100
 peer 10.10.10.5 group IBGP
 #
 ipv4-family unicast
  undo synchronization
  reflector cluster-id 24.24.24.24
  peer IBGP enable
  peer IBGP reflect-client
  peer IBGP advertise-community
  peer 10.10.10.1 enable
  peer 10.10.10.1 group IBGP
  peer 10.10.10.2 enable
  peer 10.10.10.2 group IBGP
  peer 10.10.10.3 enable
  peer 10.10.10.3 group IBGP
  peer 10.10.10.5 enable
  peer 10.10.10.5 group IBGP
#
```

```
ospf 1
 area 0.0.0.0
  network 10.0.34.4 0.0.0.0
  network 10.0.45.4 0.0.0.0
  network 10.10.10.4 0.0.0.0
#
return
```

Configuration on R5

```
#
 sysname R5
#
router id 10.10.10.5
#
firewall zone Local
 priority 15
#
interface GigabitEthernet0/0/3
 ip address 10.0.45.5 255.255.255.0
#
interface GigabitEthernet0/0/4
 ip address 10.0.25.5 255.255.255.0
#
interface LoopBack0
 ip address 10.10.10.5 255.255.255.255
#
bgp 65100
 peer 10.0.25.2 as-number 65002
 peer 10.0.25.2 password cipher Huawei@123
 peer 10.0.25.2 valid-ttl-hops 255
 group IBGP internal
 peer IBGP connect-interface LoopBack0
 peer 10.10.10.2 as-number 65100
 peer 10.10.10.2 group IBGP
 peer 10.10.10.4 as-number 65100
 peer 10.10.10.4 group IBGP
 #
 ipv4-family unicast
  undo synchronization
  peer 10.0.25.2 enable
  peer 10.0.25.2 route-policy Finance export
  peer 10.0.25.2 advertise-community
  peer IBGP enable
  peer IBGP next-hop-local
  peer IBGP advertise-community
  peer 10.10.10.2 enable
  peer 10.10.10.2 group IBGP
  peer 10.10.10.4 enable
  peer 10.10.10.4 group IBGP
#
ospf 1
 area 0.0.0.0
  network 10.0.45.5 0.0.0.0
  network 10.10.10.5 0.0.0.0
#
```

```
route-policy Finance permit node 10
 if-match community-filter OA
#
route-policy Finance deny node 20
 if-match as-path-filter Finance
#
route-policy Finance permit node 100
#
ip as-path-filter Finance permit 65001$
#
ip community-filter basic OA permit 65001:1
#
return
```

Configuration on S1

```
#
sysname S1
#
interface Vlanif1
 ip address 10.0.11.1 255.255.255.0
#
interface GigabitEthernet0/0/1
 port link-type access
#
interface LoopBack0
 ip address 10.0.1.1 255.255.255.255
#
interface LoopBack1
 ip address 10.1.1.1 255.255.255.255
#
bgp 65001
 peer 10.0.11.2 as-number 65100
 peer 10.0.11.2 password cipher Huawei@123
 #
 ipv4-family unicast
  undo synchronization
  network 10.0.1.1 255.255.255.255
  network 10.1.1.1 255.255.255.255
  peer 10.0.11.2 enable
  peer 10.0.11.2 route-policy Attr export
  peer 10.0.11.2 advertise-community
#
route-policy Attr permit node 10
 if-match ip-prefix Com
 apply community 65001:1
#
route-policy Attr permit node 100
#
ip ip-prefix Com index 10 permit 10.0.1.1 32
#
return
```

Configuration on S2

```
#
sysname S2
```

```
#
interface Vlanif1
 ip address 10.0.25.2 255.255.255.0
#
interface GigabitEthernet0/0/5
 port link-type access
interface LoopBack0
 ip address 10.0.2.1 255.255.255.255
#
interface LoopBack1
 ip address 10.1.2.1 255.255.255.255
#
bgp 65002
 peer 10.0.25.5 as-number 65100
 peer 10.0.25.5 password cipher Huawei@123
 #
 ipv4-family unicast
  undo synchronization
  network 10.0.2.1 255.255.255.255
  network 10.1.2.1 255.255.255.255
  peer 10.0.25.5 enable
  peer 10.0.25.5 route-policy Attr export
  peer 10.0.25.5 advertise-community
#
route-policy Attr permit node 10
 if-match ip-prefix Com
 apply community 65002:1
#
route-policy Attr permit node 100
#
ip ip-prefix Com index 10 permit 10.0.2.1 32
#
return
```

Configuration on S5

```
#
sysname S5
#
interface Vlanif1
 ip address 10.0.35.5 255.255.255.0
#
interface GigabitEthernet0/0/1
 port link-type access
#
interface LoopBack0
 ip address 10.0.3.1 255.255.255.255
#
interface LoopBack1
 ip address 10.1.3.1 255.255.255.255
#
bgp 65003
 peer 10.0.35.3 as-number 65100
 peer 10.0.35.3 password cipher Huawei@123
 #
 ipv4-family unicast
```

```
 undo synchronization
 network 10.0.3.1 255.255.255.255
 network 10.1.3.1 255.255.255.255
 network 10.2.3.1 255.255.255.255
 peer 10.0.35.3 enable
 peer 10.0.35.3 route-policy Attr export
 peer 10.0.35.3 advertise-community
#
route-policy Attr permit node 10
 if-match ip-prefix Com
 apply community 65003:1
#
route-policy Attr permit node 100
#
ip ip-prefix Com index 10 permit 10.0.3.1 32
#
return
```

3 IPv6 Routing

3.1 IPv4/IPv6 Dual Stack Experiment

3.1.1 Introduction

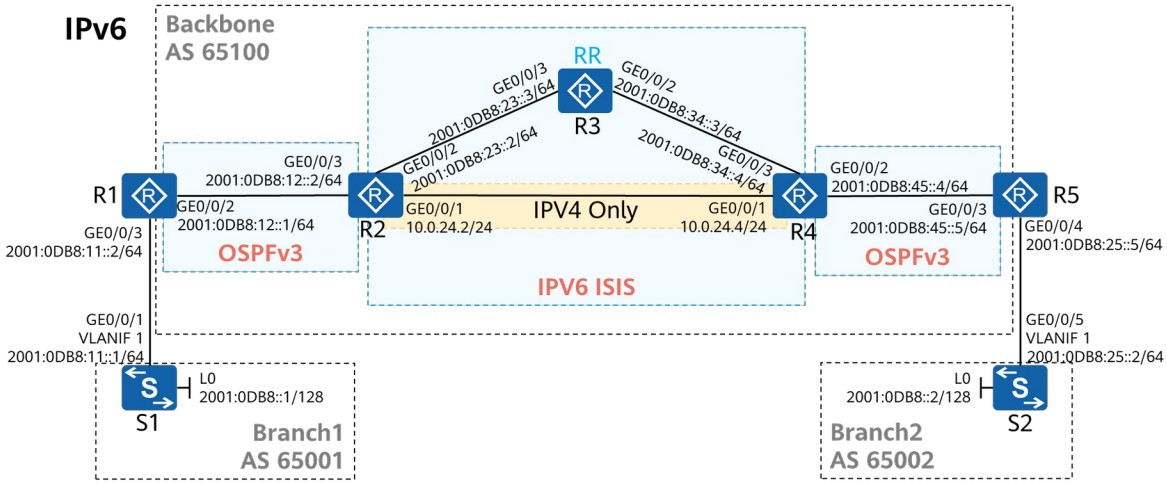
3.1.1.1 Objectives

Upon completion of this task, you will be able to:

- Configure MP-BGP.
- Configure IS-IS dual-stack.
- Configure OSPF dual-stack.

3.1.1.2 Networking Topology

Networking of IPv4/IPv6 dual-stack



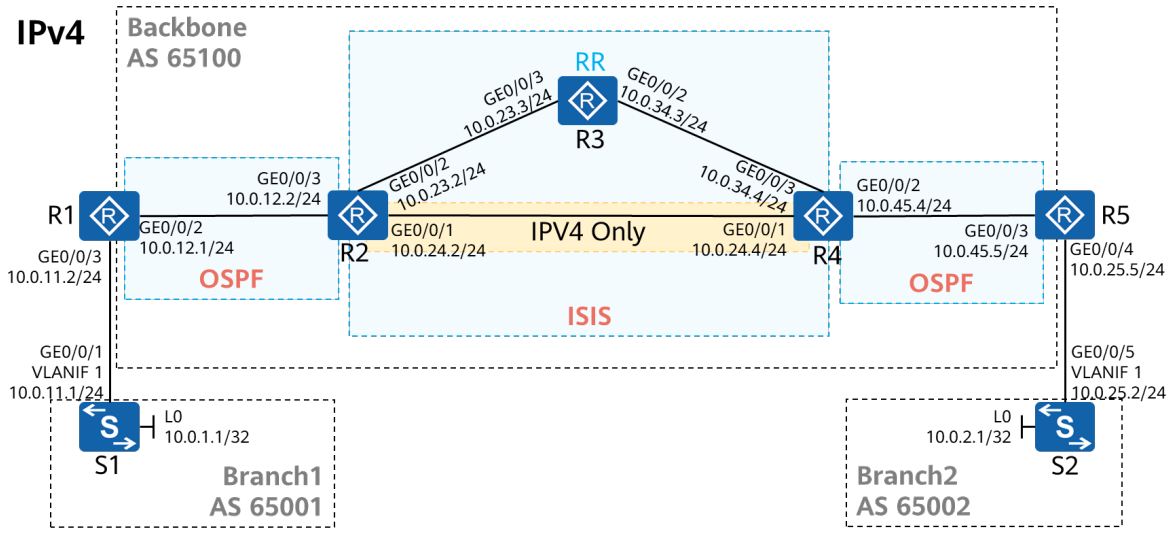


Figure 3-1 shows IP addresses of interconnection interfaces and BGP AS numbers. R3 is an RR, and R1, R2, R4, and R5 are clients of R3. In BGP AS 65100, OSPF and IS-IS are used to build the underlying network.

3.1.1.3 Lab Background

The IPv4+IPv6 service needs to be deployed on an enterprise network so that devices on IPv4 and IPv6 network segments between branch 1 and branch 2 can communicate with each other.

The enterprise has a large-scale backbone network. The access layer of the backbone network uses OSPF or Open Shortest Path First Version 3 (OSPFv3), and the core layer uses IPv4 or IPv6 IS-IS.

Some pure IPv4 networks exist at the core layer of the backbone network, and IPv6 cannot be deployed.

3.1.2 Lab Configuration

3.1.2.1 Configuration Roadmap

1. Configure IPv4 or IPv6 addresses for devices.
2. Deploy OSPF or OSPFv3 at the access layer of the backbone network.
3. Deploy IPv4 or IPv6 IS-IS at the core layer of the backbone network.
4. Configure IPv4 IS-IS and OSPF to import routes from each other, and IPv6 IS-IS and OSPFv3 to import routes from each other.
5. Configure R1, R2, R4, and R5 to establish External BGP (EBGP) peer relationships with the RR, and configure R3 as the RR.
6. Configure S1 and S2 to establish EBGP peer relationships with R1 and R5 and advertise IPv4 and IPv6 routes.

3.1.2.2 Configuration Procedure

Step 1 Configure IPv4 and IPv6 addresses for interconnection interfaces and loopback interfaces.

Name the devices.

The configuration details are not provided.

Disable the interfaces that are not used in this experiment.

The configuration details are not provided.

Configure IPv4 addresses for GE0/0/2, GE0/0/3, and Loopback0 on R1.

```
[R1]interface LoopBack0
[R1-LoopBack0] ip address 10.10.10.1 255.255.255.255
[R1-LoopBack0] quit
[R1]interface GigabitEthernet0/0/2
[R1-GigabitEthernet0/0/2] ip address 10.0.12.1 255.255.255.0
[R1-GigabitEthernet0/0/2] quit
[R1]interface GigabitEthernet0/0/3
[R1-GigabitEthernet0/0/3] ip address 10.0.11.2 255.255.255.0
[R1-GigabitEthernet0/0/3] quit
```

Configure IPv6 addresses for GE0/0/2, GE0/0/3, and Loopback0 on R1.

```
[R1]ipv6
[R1]interface LoopBack0
[R1-LoopBack0] ipv6 enable
[R1-LoopBack0] ipv6 address 1::1/128
[R1-LoopBack0] quit
[R1]interface GigabitEthernet0/0/2
[R1-GigabitEthernet0/0/2] ipv6 enable
[R1-GigabitEthernet0/0/2] ipv6 address 2001:DB8:12::1/64
[R1-GigabitEthernet0/0/2] quit
[R1]interface GigabitEthernet0/0/3
[R1-GigabitEthernet0/0/3] ipv6 enable
[R1-GigabitEthernet0/0/3] ipv6 address 2001:DB8:11::2/64
[R1-GigabitEthernet0/0/3] quit
```

Configure IPv4 addresses for GE0/0/1, GE0/0/2, GE0/0/3, and Loopback0 on R2.

```
[R2]interface LoopBack0
[R2-LoopBack0] ip address 10.10.10.2 255.255.255.255
[R2-LoopBack0] quit
[R2]interface GigabitEthernet0/0/1
[R2-GigabitEthernet0/0/1] ip address 10.0.24.2 255.255.255.0
[R2-GigabitEthernet0/0/1] quit
[R2]interface GigabitEthernet0/0/2
[R2-GigabitEthernet0/0/2] ip address 10.0.23.2 255.255.255.0
[R2-GigabitEthernet0/0/2] quit
[R2]interface GigabitEthernet0/0/3
[R2-GigabitEthernet0/0/3] ip address 10.0.12.2 255.255.255.0
[R2-GigabitEthernet0/0/3] quit
```

Configure IPv6 addresses for GE0/0/2, GE0/0/3, and Loopback0 on R2. GE0/0/1 supports only IPv4 and does not need to be configured with an IPv6 address.

```
[R2]ipv6
[R2]interface LoopBack0
[R2-LoopBack0] ipv6 enable
[R2-LoopBack0] ipv6 address 2::2/128
[R2-LoopBack0] quit
```

```
[R2]interface GigabitEthernet0/0/2
[R2-GigabitEthernet0/0/2] ipv6 enable
[R2-GigabitEthernet0/0/2] ipv6 address 2001:DB8:23::2/64
[R2-GigabitEthernet0/0/2] quit
[R2]interface GigabitEthernet0/0/3
[R2-GigabitEthernet0/0/3] ipv6 enable
[R2-GigabitEthernet0/0/3] ipv6 address 2001:DB8:12::2/64
[R2-GigabitEthernet0/0/3] quit
```

Configure IPv4 addresses for GE0/0/2, GE0/0/3, and Loopback0 on R3.

```
[R3]interface LoopBack0
[R3-LoopBack0] ip address 10.10.10.3 255.255.255.255
[R3-LoopBack0] quit
[R3]interface GigabitEthernet0/0/2
[R3-GigabitEthernet0/0/2] ip address 10.0.34.3 255.255.255.0
[R3-GigabitEthernet0/0/2] quit
[R3]interface GigabitEthernet0/0/3
[R3-GigabitEthernet0/0/3] ip address 10.0.23.3 255.255.255.0
[R3-GigabitEthernet0/0/3] quit
```

Configure IPv6 addresses for GE0/0/2, GE0/0/3, and Loopback0 on R3.

```
[R3]ipv6
[R3]interface LoopBack0
[R3-LoopBack0] ipv6 enable
[R3-LoopBack0] ipv6 address 3::3/128
[R3-LoopBack0] quit
[R3]interface GigabitEthernet0/0/2
[R3-GigabitEthernet0/0/2] ipv6 enable
[R3-GigabitEthernet0/0/2] ipv6 address 2001:DB8:23::3/64
[R3-GigabitEthernet0/0/2] quit
[R3]interface GigabitEthernet0/0/3
[R3-GigabitEthernet0/0/3] ipv6 enable
[R3-GigabitEthernet0/0/3] ipv6 address 2001:DB8:34::3/64
[R3-GigabitEthernet0/0/3] quit
```

Configure IPv4 addresses for GE0/0/1, GE0/0/2, GE0/0/3, and Loopback0 on R4.

```
[R4]interface LoopBack0
[R4-LoopBack0] ip address 10.10.10.4 255.255.255.255
[R4-LoopBack0] quit
[R4]interface GigabitEthernet0/0/1
[R4-GigabitEthernet0/0/1] ip address 10.0.24.4 255.255.255.0
[R4-GigabitEthernet0/0/1] quit
[R4]interface GigabitEthernet0/0/2
[R4-GigabitEthernet0/0/2] ip address 10.0.45.4 255.255.255.0
[R4-GigabitEthernet0/0/2] quit
[R4]interface GigabitEthernet0/0/3
[R4-GigabitEthernet0/0/3] ip address 10.0.34.4 255.255.255.0
[R4-GigabitEthernet0/0/3] quit
```

Configure IPv6 addresses for GE0/0/2, GE0/0/3, and Loopback0 on R4. GE0/0/1 supports only IPv4 and does not need to be configured with an IPv6 address.

```
[R4]ipv6
[R4]interface LoopBack0
[R4-LoopBack0] ipv6 enable
```

```
[R4-LoopBack0] ipv6 address 4::4/128
[R4-LoopBack0] quit
[R4]interface GigabitEthernet0/0/2
[R4-GigabitEthernet0/0/2] ipv6 enable
[R4-GigabitEthernet0/0/2] ipv6 address 2001:DB8:45::4/64
[R4-GigabitEthernet0/0/2] quit
[R4]interface GigabitEthernet0/0/3
[R4-GigabitEthernet0/0/3] ipv6 enable
[R4-GigabitEthernet0/0/3] ipv6 address 2001:DB8:34::4/64
[R4-GigabitEthernet0/0/3] quit
```

Configure IPv4 addresses for GE0/0/3, GE0/0/4, and Loopback0 on R5.

```
[R5]interface LoopBack0
[R5-LoopBack0] ip address 10.10.10.5 255.255.255.255
[R5-LoopBack0] quit
[R5]interface GigabitEthernet0/0/3
[R5-GigabitEthernet0/0/3] ip address 10.0.45.5 255.255.255.0
[R5-GigabitEthernet0/0/3] quit
[R5]interface GigabitEthernet0/0/4
[R5-GigabitEthernet0/0/4] ip address 10.0.25.5 255.255.255.0
[R5-GigabitEthernet0/0/4] quit
```

Configure IPv6 addresses for GE0/0/2, GE0/0/3, and Loopback0 on R5.

```
[R5]ipv6
[R5]interface LoopBack0
[R5-LoopBack0] ipv6 enable
[R5-LoopBack0] ipv6 address 5::5/128
[R5-LoopBack0] quit
[R5]interface GigabitEthernet0/0/3
[R5-GigabitEthernet0/0/2] ipv6 enable
[R5-GigabitEthernet0/0/2] ipv6 address 2001:DB8:45::5/64
[R5-GigabitEthernet0/0/2] quit
[R5]interface GigabitEthernet0/0/4
[R5-GigabitEthernet0/0/3] ipv6 enable
[R5-GigabitEthernet0/0/3] ipv6 address 2001:DB8:25::2/64
[R5-GigabitEthernet0/0/3] quit
```

Configure the VLAN to which GE0/0/1 of S1 belongs and IPv4 addresses of VLANIF 1 and Loopback0.

```
[S1]interface LoopBack0
[S1-LoopBack0] ip address 10.0.1.1 255.255.255.255
[S1-LoopBack0] quit
[S1]interface GigabitEthernet0/0/1
[S1-GigabitEthernet0/0/1] port link-type access
[S1-GigabitEthernet0/0/1] port default vlan 1
[S1-GigabitEthernet0/0/1] quit
[S1]interface Vlanif 1
[S1-Vlanif1] ip address 10.1.11.1 24
[S1-Vlanif1] quit
```

Configure the VLAN to which GE0/0/1 of S1 belongs and IPv6 addresses for VLANIF 1 and Loopback0.

```
[S1]ipv6
[S1]interface LoopBack0
```

```
[S1-LoopBack0] ipv6 enable
[S1-LoopBack0] ipv6 address 2001:DB8::1/128
[S1-LoopBack0] quit
[S1]interface GigabitEthernet0/0/1
[S1-GigabitEthernet0/0/1] ipv6 enable
[S1-GigabitEthernet0/0/1] ipv6 address 2001:DB8:11::1/64
[S1-GigabitEthernet0/0/1] quit
```

Configure the VLAN to which GE0/0/5 of S2 belongs and IPv4 addresses of VLANIF 1 and Loopback0.

```
[S2]interface LoopBack0
[S2-LoopBack0] ip address 10.0.2.1 255.255.255.255
[S2-LoopBack0] quit
[S2]interface GigabitEthernet0/0/5
[S2-GigabitEthernet0/0/5] port link-type access
[S2-GigabitEthernet0/0/5] port default vlan 1
[S2-GigabitEthernet0/0/5] quit
[S2]interface Vlanif 1
[S2-Vlanif1] ip address 10.0.25.2 24
[S2-Vlanif1] quit
```

Configure the VLAN to which GE0/0/5 of S2 belongs and IPv6 addresses for VLANIF 1 and Loopback0.

```
[S2]ipv6
[S2]interface LoopBack0
[S2-LoopBack0] ipv6 enable
[S2-LoopBack0] ipv6 address 2001:DB8::2/128
[S2-LoopBack0] quit
[S2]interface GigabitEthernet0/0/1
[S2-GigabitEthernet0/0/5] ipv6 enable
[S2-GigabitEthernet0/0/5] ipv6 address 2001:DB8:25::2/64
[S2-GigabitEthernet0/0/5] quit
```

Check the connectivity of IP addresses of interconnection interfaces on R1, R3, and R5. (IPv6 addresses are used as an example.)

```
<R1>ping ipv6 2001:0db8:11::1
  PING 2001:0db8:11::1 : 56  data bytes, press CTRL_C to break
    Reply from 2001:DB8:11::1
    bytes=56 Sequence=1 hop limit=64  time = 20 ms

  --- 2001:0db8:11::1 ping statistics ---
    1 packet(s) transmitted
    1 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 20/30/40 ms

<R1>ping ipv6 2001:0db8:12::2
  PING 2001:0db8:12::2 : 56  data bytes, press CTRL_C to break
    Reply from 2001:DB8:12::2
    bytes=56 Sequence=1 hop limit=64  time = 20 ms

  --- 2001:0db8:12::2 ping statistics ---
    1 packet(s) transmitted
    1 packet(s) received
```

```
  0.00% packet loss
  round-trip min/avg/max = 20/30/40 ms

<R3>ping ipv6 2001:0db8:23::2
 PING 2001:0db8:23::2 : 56  data bytes, press CTRL_C to break
  Reply from 2001:DB8:23::2
  bytes=56 Sequence=1 hop limit=64  time = 20 ms

 --- 2001:0db8:23::2 ping statistics ---
  1 packet(s) transmitted
  1 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 20/30/40 ms

<R3>ping ipv6 2001:0db8:34::4
 PING 2001:0db8:34::4 : 56  data bytes, press CTRL_C to break
  Reply from 2001:DB8:34::4
  bytes=56 Sequence=1 hop limit=64  time = 20 ms

 --- 2001:0db8:34::4 ping statistics ---
  1 packet(s) transmitted
  1 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 20/30/40 ms

<R5>ping ipv6 2001:0db8:45::4
 PING 2001:0db8:45::4 : 56  data bytes, press CTRL_C to break
  Reply from 2001:DB8:45::4
  bytes=56 Sequence=1 hop limit=64  time = 20 ms

 --- 2001:0db8:45::4 ping statistics ---
  1 packet(s) transmitted
  1 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 20/30/40 ms

<R5>ping ipv6 2001:0db8:25::5
 PING 2001:0db8: 25::5 : 56  data bytes, press CTRL_C to break
  Reply from 2001:DB8: 25::5
  bytes=56 Sequence=1 hop limit=64  time = 20 ms

 --- 2001:0db8: 25::5 ping statistics ---
  1 packet(s) transmitted
  1 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 20/30/40 ms
```

15 Configure OSPF and OSPFv3 at the access layer of the backbone network to build the underlying network.

Deploy OSPF processes on R1, R2, R4, and R5 according to the topology design. Set the process ID to 1, area ID to 0, and router ID to the loopback interface's IP address of the router. R1 with the router ID of 10.10.10.1 is used as an example.

Configure routes of OSPF interfaces in AS 65100 to be accurately advertised.

The configuration of OSPFv3 is similar to the configuration of OSPF. Set the process ID to 1, area ID to 0, instance ID to 1, and router ID to the loopback interface's address of the router. R1 with the router ID of 10.10.10.1 is used as an example.

Configure OSPF on R1.

```
[R1]router id 10.10.10.1
[R1]ospf 1
[R1-ospf-1] area 0
[R1-ospf-1-area-0.0.0.0] network 10.10.10.1 0.0.0.0
[R1-ospf-1-area-0.0.0.0] network 10. 0.12.1 0.0.0.0
```

Configure OSPFv3 on R1.

```
[R1]ospfv3 1
[R1-ospfv3-1] router-id 10.10.10.1
[R1-ospfv3-1] quit
[R1]interface GigabitEthernet0/0/2
[R1-GigabitEthernet0/0/2] ospfv3 1 area 0 instance 1
[R1-GigabitEthernet0/0/2] quit
[R1]interface LoopBack0
[R1-LoopBack0] ospfv3 1 area 0 instance 1
```

Configure OSPF on R2.

```
[R2]router id 10.10.10.2
[R2]ospf 1
[R2-ospf-1] area 0
[R2-ospf-1-area-0.0.0.0] network 10.0.12.2 0.0.0.0
```

Configure OSPFv3 on R2.

```
[R2]ospfv3 1
[R2-ospfv3-1] router-id 10.10.10.2
[R2-ospfv3-1] quit
[R2]interface GigabitEthernet0/0/3
[R2-GigabitEthernet0/0/3] ospfv3 1 area 0 instance 1
[R2-GigabitEthernet0/0/3] quit
```

Configure OSPF on R4.

```
[R4]router id 10.10.10.4
[R4]ospf 1
[R4-ospf-1] area 0
[R4-ospf-1-area-0.0.0.0] network 10.0.45.4 0.0.0.0
```

Configure OSPFv3 on R4.

```
[R4]ospfv3 1
[R4-ospfv3-1] router-id 10.10.10.4
[R4-ospfv3-1] quit
[R4]interface GigabitEthernet0/0/2
[R4-GigabitEthernet0/0/2] ospfv3 1 area 0 instance 1
```

```
[R4-GigabitEthernet0/0/2] quit
```

Configure OSPF on R5.

```
[R5]router id 10.10.10.5
[R5]ospf 1
[R5-ospf-1] area 0
[R5-ospf-1-area-0.0.0.0] network 10.10.10.5 0.0.0.0
[R5-ospf-1-area-0.0.0.0] network 10.0.45.5 0.0.0.0
```

Configure OSPFv3 on R5.

```
[R5]ospfv3 1
[R5-ospfv3-1] router-id 10.10.10.5
[R5-ospfv3-1] quit
[R5]interface GigabitEthernet0/0/3
[R5-GigabitEthernet0/0/3] ospfv3 1 area 0 instance 1
[R5-GigabitEthernet0/0/3] quit
[R5]interface LoopBack0
[R5-LoopBack0] ospfv3 1 area 0 instance 1
```

Check the OSPF/OSPFv3 configuration.

Check OSPF neighbor relationship information on R2 and R4.

```
[R2]display ospf peer brief

  OSPF Process 1 with Router ID 10.10.10.2
    Peer Statistic Information
 ----------------------------------------------------------------------------
 Area Id          Interface                    Neighbor id      State
 0.0.0.0          GigabitEthernet0/0/3             10.10.10.1       Full
----------------------------------------------------------------------------

[R4]display ospf peer brief

  OSPF Process 1 with Router ID 10.10.10.4
    Peer Statistic Information
 ----------------------------------------------------------------------------
 Area Id          Interface                    Neighbor id      State
0.0.0.0          GigabitEthernet0/0/2             10.10.10.5       Full
 ----------------------------------------------------------------------------
# Check OSPFv3 neighbor relationship information on R1 and R5.
[R1]display ospfv3 peer
OSPFv3 Process (1)
OSPFv3 Area (0.0.0.0)
Neighbor ID     Pri  State           Dead Time Interface          Instance ID
10.10.10.2        1  Full/Backup      00:00:35  GE0/0/2                1

[R5]display ospfv3 peer
OSPFv3 Process (1)
OSPFv3 Area (0.0.0.0)
Neighbor ID     Pri  State           Dead Time Interface          Instance ID
10.10.10.4        1  Full/DR          00:00:36  GE0/0/3                1
# Check the OSPF and OSPFv3 routing tables on routers. R1 is used as an example.
[R1]display ospf routing

  OSPF Process 1 with Router ID 10.10.10.1
```

```
  Routing Tables

 Routing for Network
 Destination        Cost  Type      NextHop      AdvRouter      Area
 10.0.12.0/24        1    Transit    10.0.12.1     10.10.10.1     0.0.0.0
 10.10.10.1/32       0    Stub      10.10.10.1     10.10.10.1     0.0.0.0

 Total Nets: 2
 Intra Area: 2  Inter Area: 0  ASE: 0  NSSA: 0

[R1]display ospfv3 routing

Codes : E2 - Type 2 External, E1 - Type 1 External, IA - Inter-Area,
        N - NSSA, U - Uninstalled

OSPFv3 Process (1)
    Destination                                    Metric
      Next-hop
    1::1/128                                        0
      directly connected, LoopBack0
    2001:DB8:12::/64                                     1
      directly connected, GigabitEthernet0/0/1
```

The command output shows that IPv4 and IPv6 routes have been learned by the router.

16 Deploy dual-stack IS-IS at the core layer of the backbone network to build the underlying network.

Deploy IS-IS processes on R2, R3, and R4 based on the topology design. Set the process ID to 1, configure R2, R3, and R4 as Level-2 routers, set the cost type to wide, set the area ID of the NET to 49.0001, set the system ID based on the loopback interface, and set the IS-IS host name to be the same as the device name.

Because a pure IPv4 network exists between R2 and R4, the multi-topology function needs to be deployed when dual-stack IS-IS is deployed.

Deploy dual-stack IS-IS on R2. GE0/0/1 supports only IPv4, so the route of GE0/0/1 does not need to be advertised to IPv6 IS-IS.

```
[R2]isis 1
[R2-isis-1] is-level level-2
[R2-isis-1] cost-style wide
[R2-isis-1] network-entity 49.0001.0100.1001.0002.00
[R2-isis-1] is-name R2
[R2-isis-1] ipv6 enable topology ipv6
[R2-isis-1] quit
[R2]interface LoopBack0
[R2-LoopBack0] isis enable 1
[R2-LoopBack0] isis ipv6 enable 1
```

```
[R2-LoopBack0] quit
[R2]interface GigabitEthernet0/0/1
[R2-GigabitEthernet0/0/1] isis enable 1
[R2-GigabitEthernet0/0/1] quit
[R2]interface GigabitEthernet0/0/2
[R2-GigabitEthernet0/0/2] isis enable 1
[R2-GigabitEthernet0/0/2] isis ipv6 enable 1
[R2-GigabitEthernet0/0/2] quit
```

Deploy dual-stack IS-IS on R3.

```
[R3]isis 1
[R3-isis-1] is-level level-2
[R3-isis-1] cost-style wide
[R3-isis-1] network-entity 49.0001.0100.1001.0003.00
[R3-isis-1] is-name R3
[R3-isis-1] ipv6 enable topology ipv6
[R3-isis-1] quit
[R3]interface LoopBack0
[R3-LoopBack0] isis enable 1
[R3-LoopBack0] isis ipv6 enable 1
[R3-LoopBack0] quit
[R3]interface GigabitEthernet0/0/2
[R3-GigabitEthernet0/0/2] isis enable 1
[R3-GigabitEthernet0/0/2] isis ipv6 enable 1
[R3-GigabitEthernet0/0/2] quit
[R3]interface GigabitEthernet0/0/3
[R3-GigabitEthernet0/0/3] isis enable 1
[R3-GigabitEthernet0/0/3] isis ipv6 enable 1
[R3-GigabitEthernet0/0/3] quit
```

Deploy dual-stack IS-IS on R4. GE0/0/1 supports only IPv4, so the route of GE0/0/1 does not need to be advertised to IPv6 IS-IS.

```
[R4]isis 1
[R4-isis-1] is-level level-2
[R4-isis-1] cost-style wide
[R4-isis-1] network-entity 49.0001.0100.1001.0004.00
[R4-isis-1] is-name R4
[R4-isis-1] ipv6 enable topology ipv6
[R4-isis-1] quit
[R4]interface LoopBack0
[R4-LoopBack0] isis enable 1
[R4-LoopBack0] isis ipv6 enable 1
[R4-LoopBack0] quit
[R4]interface GigabitEthernet0/0/1
[R4-GigabitEthernet0/0/1] isis enable 1
[R4-GigabitEthernet0/0/1] quit
[R4]interface GigabitEthernet0/0/3
[R4-GigabitEthernet0/0/3] isis enable 1
[R4-GigabitEthernet0/0/3] isis ipv6 enable 1
[R4-GigabitEthernet0/0/3] quit
```

Check the dual-stack IS-IS configuration.

Check the IS-IS neighbor relationship on R3.

```
[R3]display isis peer
```

```
                    Peer information for ISIS(1)

 System Id      Interface          Circuit Id     State HoldTime  Type     PRI
-----------------------------------------------------------------------------
R4*             GE0/0/2            R4.02          Up    8s        L2       64
R2*             GE0/0/3            R2.02          Up    9s        L2       64

Total Peer(s): 2

[R3]display isis interface

                    Interface information for ISIS(1)
                    ---------------------------------
 Interface      Id      IPV4.State          IPV6.State        MTU   Type   DIS
 GE0/0/2       001        Up                  Up            1497   L1/L2   No/No
 GE0/0/3       002        Up                  Up            1497   L1/L2   No/No
 Loop0          001        Up                  Up            1500   L1/L2   --
```

Check IPv4 and IPv6 IS-IS routing tables on each router. R3 is used as an example.

```
[R3]display isis route

                    Route information for ISIS(1)
                    -----------------------------

                    ISIS(1) Level-2 Forwarding Table
                    --------------------------------

IPV4 Destination      IntCost     ExtCost ExitInterface   NextHop          Flags
-----------------------------------------------------------------------------
10.0.24.0/24            20        NULL    GE0/0/2       10.0.34.4          A/-/-/-
                                        GE0/0/3       10.0.23.2
10.10.10.3/32           0         NULL    Loop0          Direct            D/-/L/-
10.10.10.2/32           10        NULL    GE0/0/3       10.0.23.2          A/-/-/-
10.0.23.0/24            10        NULL    GE0/0/3       Direct             D/-/L/-
10.0.34.0/24            10        NULL    GE0/0/2       Direct             D/-/L/-
10.10.10.4/32           10        NULL    GE0/0/2       10.0.34.4          A/-/-/-

 IPV6 Dest.       ExitInterface   NextHop                        Cost       Flags
-----------------------------------------------------------------------------
 4::4/128          GE0/0/2        FE80::2E0:FCFF:FE92:2B1C        10          A/-/-
 3::3/128          Loop0           Direct                          0          D/L/-
 2::2/128          GE0/0/3        FE80::2E0:FCFF:FE1C:24DE        10          A/-/-
 2001:DB8:23::/64
                  GE0/0/3        Direct                          10          D/L/-
 2001:DB8:34::/64
                  GE0/0/2        Direct                          10          D/L/-

      Flags: D-Direct, A-Added to URT, L-Advertised in LSPs, S-IGP Shortcut,
                                U-Up/Down Bit Set
```

The command output shows that IPv4 and IPv6 routes have been learned by the router.

17 Configure IS-IS and OSPF to import routes from each other.

Configure IS-IS and OSPF to import routes from each other on R2 and R4, implementing connectivity of the underlying network of the backbone network and preparing for BGP deployment.

R2 and R4 need to import routes from both IPv4 and IPv6 networks.

Configure IS-IS and OSPF to import routes from each other on R2.

```
[R2]isis 1
[R2-isis-1] import-route ospf 1
[R2-isis-1] ipv6 import-route ospfv3 1
[R2-isis-1] quit
[R2]ospf 1
[R2-ospf-1] import-route isis 1
[R2-ospf-1] quit
[R2]ospfv3 1
[R2-ospfv3-1] import-route isis 1
```

Configure IS-IS and OSPF to import routes from each other on R4.

```
[R4]isis 1
[R4-isis-1] import-route ospf 1
[R4-isis-1] ipv6 import-route ospfv3 1
[R4-isis-1] quit
[R4]ospf 1
[R4-ospf-1] import-route isis 1
[R4-ospf-1] quit
[R4]ospfv3 1
[R4-ospfv3-1] import-route isis 1
```

Check the bidirectional route import configuration.

Check IPv4 and IPv6 routing tables on routers. R1 is used as an example.

```
[R1]display ip routing-table protocol ospf
Route Flags: R - relay, D - download to fib
------------------------------------------------------------------------------
Public routing table : OSPF
         Destinations : 8        Routes : 8

OSPF routing table status : <Active>
         Destinations : 8        Routes : 8

Destination/Mask    Proto    Pre  Cost      Flags NextHop         Interface

      10.0.23.0/24  O_ASE   150  1           D   10.0.12.2       GigabitEthernet0/0/2
      10.0.24.0/24  O_ASE   150  1           D   10.0.12.2       GigabitEthernet0/0/2
      10.0.34.0/24  O_ASE   150  1           D   10.0.12.2       GigabitEthernet0/0/2
      10.0.45.0/24  O_ASE   150  1           D   10.0.12.2       GigabitEthernet0/0/2
     10.10.10.2/32  O_ASE   150  1           D   10.0.12.2       GigabitEthernet0/0/2
```

```
   10.10.10.3/32  O_ASE   150  1          D   10.0.12.2      GigabitEthernet0/0/2
   10.10.10.4/32  O_ASE   150  1          D   10.0.12.2      GigabitEthernet0/0/2
   10.10.10.5/32  O_ASE   150  1          D   10.0.12.2      GigabitEthernet0/0/2

OSPF routing table status : <Inactive>
       Destinations : 0        Routes : 0

[R1]display ipv6 routing-table protocol ospfv3
Public Routing Table : OSPFv3
Summary Count : 9

OSPFv3 Routing Table's Status : < Active >
Summary Count : 7

 Destination    : 2::2                        PrefixLength : 128
 NextHop      : FE80::2E0:FCFF:FE1C:24DF       Preference   : 150
 Cost         : 1                        Protocol     : OSPFv3ASE
 RelayNextHop  : ::                         TunnelID    : 0x0
 Interface      : GigabitEthernet0/0/2             Flags      : D

 Destination    : 3::3                        PrefixLength : 128
 NextHop      : FE80::2E0:FCFF:FE1C:24DF       Preference   : 150
 Cost         : 1                        Protocol     : OSPFv3ASE
 RelayNextHop  : ::                         TunnelID    : 0x0
 Interface      : GigabitEthernet0/0/2             Flags      : D

 Destination    : 4::4                        PrefixLength : 128
 NextHop      : FE80::2E0:FCFF:FE1C:24DF       Preference   : 150
 Cost         : 1                        Protocol     : OSPFv3ASE
 RelayNextHop  : ::                         TunnelID    : 0x0
 Interface      : GigabitEthernet0/0/2             Flags      : D

 Destination    : 5::5                        PrefixLength : 128
 NextHop      : FE80::2E0:FCFF:FE1C:24DF       Preference   : 150
 Cost         : 1                        Protocol     : OSPFv3ASE
 RelayNextHop  : ::                         TunnelID    : 0x0
 Interface      : GigabitEthernet0/0/2             Flags      : D

 Destination     : 2001:DB8:23::                PrefixLength : 64
 NextHop      : FE80::2E0:FCFF:FE1C:24DF       Preference   : 150
 Cost         : 1                        Protocol     : OSPFv3ASE
 RelayNextHop  : ::                         TunnelID    : 0x0
 Interface      : GigabitEthernet0/0/2             Flags      : D

 Destination    : 2001:DB8:34::                PrefixLength : 64
 NextHop      : FE80::2E0:FCFF:FE1C:24DF       Preference   : 150
 Cost         : 1                        Protocol     : OSPFv3ASE
 RelayNextHop  : ::                         TunnelID    : 0x0
 Interface      : GigabitEthernet0/0/2             Flags      : D

 Destination    : 2001:DB8:45::                PrefixLength : 64
 NextHop      : FE80::2E0:FCFF:FE1C:24DF       Preference   : 150
 Cost         : 1                        Protocol     : OSPFv3ASE
 RelayNextHop  : ::                         TunnelID    : 0x0
 Interface      : GigabitEthernet0/0/2             Flags      : D
```

```
OSPFv3 Routing Table's Status : < Inactive >
Summary Count : 2

 Destination    : 1::1                        PrefixLength : 128
 NextHop      : ::                          Preference   : 10
 Cost         : 0                         Protocol     : OSPFv3
 RelayNextHop  : ::                           TunnelID    : 0x0
 Interface      : LoopBack0                    Flags       :

 Destination    : 2001:DB8:12::                  PrefixLength : 64
 NextHop      : ::                          Preference   : 10
 Cost         : 1                         Protocol     : OSPFv3
 RelayNextHop  : ::                           TunnelID    : 0x0
 Interface      : GigabitEthernet0/0/2               Flags       :
```

The routing table of R1 shows that the underlying network in AS 65100 has been established.

18 Deploy IBGP peers on the backbone network.

Establish an IBGP peer relationship in AS 65100. Configure R3 as the RR, and R1, R2, R4, and R5 as clients of R3.

You can configure a peer group to reduce the configuration workload.

Configure R1 to establish IPv4 and IPv6 IBGP peer relationships.

```
[R1]bgp 65100
[R1-bgp] peer 10.10.10.3 as-number 65100
[R1-bgp] peer 10.10.10.3 connect-interface LoopBack0
[R1-bgp] peer 3::3 as-number 65100
[R1-bgp] peer 3::3 connect-interface LoopBack0
[R1-bgp] ipv4-family unicast
[R1-bgp-af-ipv4] peer 10.10.10.3 enable
[R1-bgp-af-ipv4] peer 10.10.10.3 next-hop-local
[R1-bgp-af-ipv4] quit
[R1-bgp] ipv6-family unicast
[R1-bgp-af-ipv6] peer 3::3 enable
[R1-bgp-af-ipv6] peer 3::3 next-hop-local
[R1-bgp-af-ipv6] quit
```

Configure R2 to establish IPv4 and IPv6 IBGP peer relationships.

```
[R2]bgp 65100
[R2-bgp] peer 10.10.10.3 as-number 65100
[R2-bgp] peer 10.10.10.3 connect-interface LoopBack0
[R2-bgp] peer 3::3 as-number 65100
[R2-bgp] peer 3::3 connect-interface LoopBack0
[R2-bgp] ipv4-family unicast
[R2-bgp-af-ipv4] peer 10.10.10.3 enable
[R2-bgp-af-ipv4] quit
```

```
[R2-bgp] ipv6-family unicast
[R2-bgp-af-ipv6] peer 3::3 enable
[R2-bgp-af-ipv6] quit
```

Configure R3 to establish IPv4 and IPv6 IBGP peer relationships. R3 functions as an RR and needs to connect to many clients. The peer group is recommended.

```
[R3]bgp 65100
[R3-bgp] group IBGP internal
[R3-bgp] peer IBGP connect-interface LoopBack0
[R3-bgp] peer 10.10.10.1 group IBGP
[R3-bgp] peer 10.10.10.2 group IBGP
[R3-bgp] peer 10.10.10.4 group IBGP
[R3-bgp] peer 10.10.10.5 group IBGP
[R3-bgp] peer 1::1group IBGP
[R3-bgp] peer 2::2group IBGP
[R3-bgp] peer 4::4group IBGP
[R3-bgp] peer 5::5group IBGP
[R3-bgp] ipv4-family unicast
[R3-bgp-af-ipv4] peer IBGP enable
[R3-bgp-af-ipv4] peer IBGP reflect-client
[R3-bgp-af-ipv4] peer 10.10.10.1 enable
[R3-bgp-af-ipv4] peer 10.10.10.2 enable
[R3-bgp-af-ipv4] peer 10.10.10.4 enable
[R3-bgp-af-ipv4] peer 10.10.10.5 enable
[R3-bgp-af-ipv4] quit
[R3-bgp] ipv6-family unicast
[R3-bgp-af-ipv6] peer 3::3 enable
[R3-bgp-af-ipv6] peer IBGP enable
[R3-bgp-af-ipv6] peer IBGP reflect-client
[R3-bgp-af-ipv6] peer 1::1 enable
[R3-bgp-af-ipv6] peer 2::2 enable
[R3-bgp-af-ipv6] peer 4::4 enable
[R3-bgp-af-ipv6] peer 5::5 enable
[R3-bgp-af-ipv6] quit
```

Configure R4 to establish IPv4 and IPv6 IBGP peer relationships.

```
[R4]bgp 65100
[R4-bgp] peer 10.10.10.3 as-number 65100
[R4-bgp] peer 10.10.10.3 connect-interface LoopBack0
[R4-bgp] peer 3::3 as-number 65100
[R4-bgp] peer 3::3 connect-interface LoopBack0
[R4-bgp] ipv4-family unicast
[R4-bgp-af-ipv4] peer 10.10.10.3 enable
[R4-bgp-af-ipv4] quit
[R4-bgp] ipv6-family unicast
[R4-bgp-af-ipv6] peer 3::3 enable
[R4-bgp-af-ipv6] quit
```

Configure R5 to establish IPv4 and IPv6 IBGP peer relationships.

```
[R5]bgp 65100
[R5-bgp] peer 10.10.10.3 as-number 65100
[R5-bgp] peer 10.10.10.3 connect-interface LoopBack0
[R5-bgp] peer 3::3 as-number 65100
[R5-bgp] peer 3::3 connect-interface LoopBack0
```

```
[R5-bgp] ipv4-family unicast
[R5-bgp-af-ipv4] peer 10.10.10.3 enable
[R5-bgp-af-ipv4] peer 10.10.10.3 next-hop-local
[R5-bgp-af-ipv4] quit
[R5-bgp] ipv6-family unicast
[R5-bgp-af-ipv6] peer 3::3 enable
[R5-bgp-af-ipv6] peer 3::3 next-hop-local
[R5-bgp-af-ipv6] quit
```

Check the BGP configuration.

Check IPv4 and IPv6 IBGP peer relationships on R3.

```
[R3]display bgp peer

 BGP local router ID : 10.10.10.3
 Local AS number : 65100
 Total number of peers : 4    Peers in established state : 4

  Peer            V          AS  MsgRcvd  MsgSent  OutQ  Up/Down       State PrefRcv

  10.10.10.1      4       65100     200      201     0   03:17:18  Established      0
  10.10.10.2      4       65100     344      352     0   05:42:08  Established      0
  10.10.10.4      4       65100     343      352     0   05:41:49  Established      0
  10.10.10.5      4       65100     199      200     0   03:16:45  Established      0

[R3]display BGP IPV6 peer

 BGP local router ID : 10.10.10.3
 Local AS number : 65100
 Total number of peers : 4    Peers in established state : 4

  Peer            V          AS  MsgRcvd  MsgSent  OutQ  Up/Down       State PrefRcv

  1::1            4       65100     201      202     0  03:18:39   Established      0
  2::2            4       65100     345      355     0  05:43:56   Established      0
  4::4            4       65100     345      356     0  05:43:37   Established      0
  5::5            4       65100     201      202     0  03:18:26   Established      0
```

The command output shows that both IPv4 and IPv6 IBGP peer relationships have been established.

19 EBGP peer relationships have been established between the branch and backbone network, and routes are advertised.

IPv4 and IPv6 EBGP peer relationships have been established between the branch and backbone network, and routes of loopback interfaces on S1 and S2 are imported to BGP.

Establish IPv4 and IPv6 EBGP peer relationships between R1 and S1 and advertise routes.

```
[R1]bgp 65100
[R1-bgp] peer 10.0.11.1 as-number 65001
[R1-bgp] peer 2001:DB8:11::1 as-number 65001
[R1-bgp] ipv4-family unicast
[R1-bgp-af-ipv4] peer 10.0.11.1 enable
[R1-bgp-af-ipv4] quit
[R1-bgp] ipv6-family unicast
[R1-bgp-af-ipv6] peer 2001:DB8:11::1 enable
[R1-bgp-af-ipv6] quit

[S1]bgp 65001
[S1-bgp] peer 10.0.11.2 as-number 65100
[S1-bgp] peer 2001:DB8:11::2 as-number 65100
[S1-bgp] ipv4-family unicast
[S1-bgp-af-ipv4] peer 10.0.11.2 enable
[S1-bgp-af-ipv4] network 10.0.1.1 255.255.255.255
[S1-bgp-af-ipv4] quit
[S1-bgp] ipv6-family unicast
[S1-bgp-af-ipv6] peer 2001:DB8:11::2 enable
[S1-bgp-af-ipv6] network 2001:DB8::1 128
[S1-bgp-af-ipv6] quit
```

Establish IPv4 and IPv6 EBGP peer relationships between R5 and S2 and advertise routes.

```
[R5]bgp 65100
[R5-bgp] peer 10.0.25.2 as-number 65002
[R5-bgp] peer 2001:DB8:25::2 as-number 65002
[R5-bgp] ipv4-family unicast
[R5-bgp-af-ipv4]p eer 10.0.25.2 enable
[R5-bgp-af-ipv4] quit
[R5-bgp] ipv6-family unicast
[R5-bgp-af-ipv6] peer 2001:DB8:25::2 enable
[R5-bgp-af-ipv6] quit

[S2]bgp 65002
[S2-bgp] peer 10.0.25.5 as-number 65100
[S2-bgp] peer 2001:DB8:25::5 as-number 65100
[S2-bgp] ipv4-family unicast
[S2-bgp-af-ipv4] peer 10.0.25.5 enable
[S2-bgp-af-ipv4] network 10.0.2.1 255.255.255.255
[S2-bgp-af-ipv4] quit
[S2-bgp] ipv6-family unicast
[S2-bgp-af-ipv6] peer 2001:DB8:25::5 enable
[S2-bgp-af-ipv6] network 2001:DB8::2 128
[S2-bgp-af-ipv6] quit
```

Check the configuration.

Check the IPv4 and IPv6 EBGP peer relationships on S1 and S2. S1 is used as an example.

```
[S1]display bgp peer

 BGP local router ID : 10.0.11.1
 Local AS number : 65001
 Total number of peers : 1    Peers in established state : 1
```

```
 Peer          V        AS  MsgRcvd  MsgSent  OutQ  Up/Down      State PrefRcv

 10.0.11.2      4       65100     377     371     0   06:08:10  Established     1

[S1]display bgp ipv6 peer

 BGP local router ID : 10.0.11.1
 Local AS number : 65001
 Total number of peers : 1    Peers in established state : 1

 Peer          V        AS  MsgRcvd  MsgSent  OutQ  Up/Down      State PrefRcv

 2001:DB8:11::2   4       65100     375     373     0   06:08:05  Established     1
```

The preceding command output shows that IPv4 and IPv6 BGP peer relationships have been established.

Check the route transmission result. Check IPv4 and IPv6 routing tables on S1 and S2. S1 is used as an example.

```
[S1]display ip routing-table protocol bgp
Route Flags: R - relay, D - download to fib
------------------------------------------------------------------------------
Public routing table : BGP
        Destinations : 1       Routes : 1

BGP routing table status : <Active>
        Destinations : 1       Routes : 1

Destination/Mask    Proto     Pre  Cost       Flags  NextHop       Interface

      10.0.2.1/32  EBGP    255  0          D   10.0.11.2       Vlanif1

BGP routing table status : <Inactive>
        Destinations : 0       Routes : 0

[S1]display ipv6 routing-table protocol bgp
Public Routing Table : BGP
Summary Count : 1

BGP Routing Table's Status : < Active >
Summary Count : 1

 Destination    : 2001:DB8::2                    PrefixLength : 128
 NextHop      : 2001:DB8:11::2                 Preference   : 255
 Cost         : 0                          Protocol     : EBGP
 RelayNextHop  : ::                            TunnelID    : 0x0
 Interface      : Vlanif1                      Flags       : D

BGP Routing Table's Status : < Inactive >
Summary Count : 0
```

The command output shows that S1 has received the route of Loopback0 on S2.

Test the connectivity between Loopback0 interfaces of S1 and S2.

```
[S1]ping -a 10.0.1.1 10.0.2.1
  PING 10.0.2.1: 56  data bytes, press CTRL_C to break
    Reply from 10.0.2.1: bytes=56 Sequence=1 ttl=251 time=60 ms

  --- 10.0.2.1 ping statistics ---
    1 packet(s) transmitted
    1 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 40/66/100 ms

[S1]ping ipv6 -a 2001:0db8::1 2001:0db8::2
  PING 2001:0db8::2 : 56  data bytes, press CTRL_C to break
    Reply from 2001:DB8::2
    bytes=56 Sequence=1 hop limit=59  time = 100 ms

  --- 2001:0db8::2 ping statistics ---
    1 packet(s) transmitted
    1 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 30/76/100 ms
```

The ping operation is successful, indicating that the lab configuration is successful.

3.1.3 Quiz

If IS-IS multi-topology is not deployed, can IPv4 and IPv6 networks communicate with each other?

3.1.4 Configuration Reference

Configuration on R1

```
#
sysname R1
#
ipv6
#
router id 10.10.10.1
#
ospfv3 1
 router-id 10.10.10.1
#
interface GigabitEthernet0/0/2
 ipv6 enable
 ip address 10.0.12.1 255.255.255.0
 ipv6 address 2001:DB8:12::1/64
 ospfv3 1 area 0.0.0.0
#
interface GigabitEthernet0/0/3
 ipv6 enable
 ip address 10.0.11.2 255.255.255.0
 ipv6 address 2001:DB8:11::2/64
#
interface LoopBack0
 ipv6 enable
```

```
 ip address 10.10.10.1 255.255.255.255
 ipv6 address 1::1/128
 ospfv3 1 area 0.0.0.0
#
bgp 65100
 peer 10.0.11.1 as-number 65001
 peer 10.10.10.3 as-number 65100
 peer 10.10.10.3 connect-interface LoopBack0
 peer 3::3 as-number 65100
 peer 3::3 connect-interface LoopBack0
 peer 2001:DB8:11::1 as-number 65001
 #
 ipv4-family unicast
  undo synchronization
  peer 10.0.11.1 enable
  peer 10.10.10.3 enable
  peer 10.10.10.3 next-hop-local
 #
 ipv6-family unicast
  undo synchronization
  peer 3::3 enable
  peer 3::3 next-hop-local
  peer 2001:DB8:11::1 enable
#
ospf 1
 area 0.0.0.0
  network 10.0.12.1 0.0.0.0
  network 10.10.10.1 0.0.0.0
#
return
```

Configuration on R2

```
#
 sysname R2
#
ipv6
#
router id 10.10.10.2
#
isis 1
 is-level level-2
 cost-style wide
 network-entity 49.0001.0100.1001.0002.00
 is-name R2
 import-route ospf 1
 #
 ipv6 enable topology ipv6
 ipv6 import-route ospfv3 1
 #
#
ospfv3 1
 router-id 10.10.10.2
 import-route isis 1
#
interface GigabitEthernet0/0/1
```

```
 ip address 10.0.24.2 255.255.255.0
 isis enable 1
#
interface GigabitEthernet0/0/2
 ipv6 enable
 ip address 10.0.23.2 255.255.255.0
 ipv6 address 2001:DB8:23::2/64
 isis enable 1
 isis ipv6 enable 1
#
interface GigabitEthernet0/0/3
 ipv6 enable
 ip address 10.0.12.2 255.255.255.0
 ipv6 address 2001:DB8:12::2/64
 ospfv3 1 area 0.0.0.0
#
interface LoopBack0
 ipv6 enable
 ip address 10.10.10.2 255.255.255.255
 ipv6 address 2::2/128
 isis enable 1
 isis ipv6 enable 1
#
bgp 65100
 peer 10.10.10.3 as-number 65100
 peer 10.10.10.3 connect-interface LoopBack0
 peer 3::3 as-number 65100
 peer 3::3 connect-interface LoopBack0
 #
 ipv4-family unicast
  undo synchronization
  peer 10.10.10.3 enable
 #
 ipv6-family unicast
  undo synchronization
  peer 3::3 enable
#
ospf 1
 import-route isis 1
 area 0.0.0.0
  network 10.0.12.2 0.0.0.0
#
return
```

Configuration on R3

```
#
 sysname R3
#
ipv6
#
router id 10.10.10.3
#
isis 1
 is-level level-2
 cost-style wide
```

```
 network-entity 49.0001.0100.1001.0003.00
 is-name R3
 #
 ipv6 enable topology ipv6
 #
#
interface GigabitEthernet0/0/2
 ipv6 enable
 ip address 10.0.34.3 255.255.255.0
 ipv6 address 2001:DB8:34::3/64
 isis enable 1
 isis ipv6 enable 1
#
interface GigabitEthernet0/0/3
 ipv6 enable
 ip address 10.0.23.3 255.255.255.0
 ipv6 address 2001:DB8:23::3/64
 isis enable 1
 isis ipv6 enable 1
#
interface LoopBack0
 ipv6 enable
 ip address 10.10.10.3 255.255.255.255
 ipv6 address 3::3/128
 isis enable 1
 isis ipv6 enable 1
#
bgp 65100
 group IBGP internal
 peer IBGP connect-interface LoopBack0
 peer 10.10.10.1 as-number 65100
 peer 10.10.10.1 group IBGP
 peer 10.10.10.2 as-number 65100
 peer 10.10.10.2 group IBGP
 peer 10.10.10.4 as-number 65100
 peer 10.10.10.4 group IBGP
 peer 10.10.10.5 as-number 65100
 peer 10.10.10.5 group IBGP
 peer 1::1 as-number 65100
 peer 1::1 group IBGP
 peer 2::2 as-number 65100
 peer 2::2 group IBGP
 peer 4::4 as-number 65100
 peer 4::4 group IBGP
 peer 5::5 as-number 65100
 peer 5::5 group IBGP
 #
 ipv4-family unicast
  undo synchronization
  peer IBGP enable
  peer IBGP reflect-client
  peer 10.10.10.1 enable
  peer 10.10.10.1 group IBGP
  peer 10.10.10.2 enable
  peer 10.10.10.2 group IBGP
```

```
 peer 10.10.10.4 enable
 peer 10.10.10.4 group IBGP
 peer 10.10.10.5 enable
 peer 10.10.10.5 group IBGP
#
ipv6-family unicast
 undo synchronization
 peer IBGP enable
 peer IBGP reflect-client
 peer 1::1 enable
 peer 1::1 group IBGP
 peer 2::2 enable
 peer 2::2 group IBGP
 peer 4::4 enable
 peer 4::4 group IBGP
 peer 5::5 enable
 peer 5::5 group IBGP
#
return
```

Configuration on R4

```
#
 sysname R4
#
ipv6
#
router id 10.10.10.4
#
isis 1
 is-level level-2
 cost-style wide
 network-entity 49.0001.0100.1001.0004.00
 is-name R4
 import-route ospf 1
 #
 ipv6 enable topology ipv6
 ipv6 import-route ospfv3 1
 #
#
ospfv3 1
 router-id 10.10.10.4
 import-route isis 1
#
interface GigabitEthernet0/0/1
 ip address 10.0.24.4 255.255.255.0
 isis enable 1
#
interface GigabitEthernet0/0/2
 ipv6 enable
 ip address 10.0.45.4 255.255.255.0
 ipv6 address 2001:DB8:45::4/64
 ospfv3 1 area 0.0.0.0 instance 1
#
interface GigabitEthernet0/0/3
 ipv6 enable
```

```
 ip address 10.0.34.4 255.255.255.0
 ipv6 address 2001:DB8:34::4/64
 isis enable 1
 isis ipv6 enable 1
#
interface LoopBack0
 ipv6 enable
 ip address 10.10.10.4 255.255.255.255
 ipv6 address 4::4/128
 isis enable 1
 isis ipv6 enable 1
#
bgp 65100
 peer 10.10.10.3 as-number 65100
 peer 10.10.10.3 connect-interface LoopBack0
 peer 3::3 as-number 65100
 peer 3::3 connect-interface LoopBack0
 #
 ipv4-family unicast
  undo synchronization
  peer 10.10.10.3 enable
 #
 ipv6-family unicast
  undo synchronization
  peer 3::3 enable
#
ospf 1
 import-route isis 1
 area 0.0.0.0
  network 10.0.45.4 0.0.0.0
#
return
```

Configuration on R5

```
#
 sysname R5
#
ipv6
#
router id 10.10.10.5
#
ospfv3 1
 router-id 10.10.10.5
#
interface GigabitEthernet0/0/3
 ipv6 enable
 ip address 10.0.45.5 255.255.255.0
 ipv6 address 2001:DB8:45::5/64
 ospfv3 1 area 0.0.0.0 instance 1
#
interface GigabitEthernet0/0/4
 ipv6 enable
 ip address 10.0.25.5 255.255.255.0
 ipv6 address 2001:DB8:25::5/64
#
```

```
interface LoopBack0
 ipv6 enable
 ip address 10.10.10.5 255.255.255.255
 ipv6 address 5::5/128
 ospfv3 1 area 0.0.0.0 instance 1
#
bgp 65100
 peer 10.0.25.2 as-number 65002
 peer 10.10.10.3 as-number 65100
 peer 10.10.10.3 connect-interface LoopBack0
 peer 3::3 as-number 65100
 peer 3::3 connect-interface LoopBack0
 peer 2001:DB8:25::2 as-number 65002
 #
 ipv4-family unicast
  undo synchronization
  peer 10.0.25.2 enable
  peer 10.10.10.3 enable
  peer 10.10.10.3 next-hop-local
 #
 ipv6-family unicast
  undo synchronization
  peer 3::3 enable
  peer 3::3 next-hop-local
  peer 2001:DB8:25::2 enable
#
ospf 1
 area 0.0.0.0
  network 10.0.45.5 0.0.0.0
  network 10.10.10.5 0.0.0.0
#
return
```

Configuration on S1

```
#
sysname S1
#
ipv6
#
interface Vlanif1
 ipv6 enable
 ip address 10.0.11.1 255.255.255.0
 ipv6 address 2001:DB8:11::1/64
#
interface GigabitEthernet0/0/1
 port link-type access
#
interface LoopBack0
 ipv6 enable
 ip address 10.0.1.1 255.255.255.255
 ipv6 address 2001:DB8::1/128
#
bgp 65001
 peer 10.0.11.2 as-number 65100
 peer 2001:DB8:11::2 as-number 65100
```

```
 #
 ipv4-family unicast
  undo synchronization
  network 10.0.1.1 255.255.255.255
  peer 10.0.11.2 enable
 #
 ipv6-family unicast
  undo synchronization
  network 2001:DB8::1 128
  peer 2001:DB8:11::2 enable
#
return
```

Configuration on S2

```
#
sysname SW2
#
ipv6
#
interface Vlanif1
 ipv6 enable
 ip address 10.0.25.2 255.255.255.0
 ipv6 address 2001:DB8:25::2/64
#
interface GigabitEthernet0/0/1
 port link-type access
#
interface LoopBack0
 ipv6 enable
 ip address 10.0.2.1 255.255.255.255
 ipv6 address 2001:DB8::2/128
#
bgp 65002
 peer 10.0.25.5 as-number 65100
 peer 2001:DB8:25::5 as-number 65100
 #
 ipv4-family unicast
  undo synchronization
  network 10.0.2.1 255.255.255.255
  peer 10.0.25.5 enable
 #
 ipv6-family unicast
  undo synchronization
  network 2001:DB8::2 128
  peer 2001:DB8:25::5 enable
#
return
```

4 Advanced VLAN Technologies

4.1 Advanced VLAN Technologies

4.1.1 Introduction

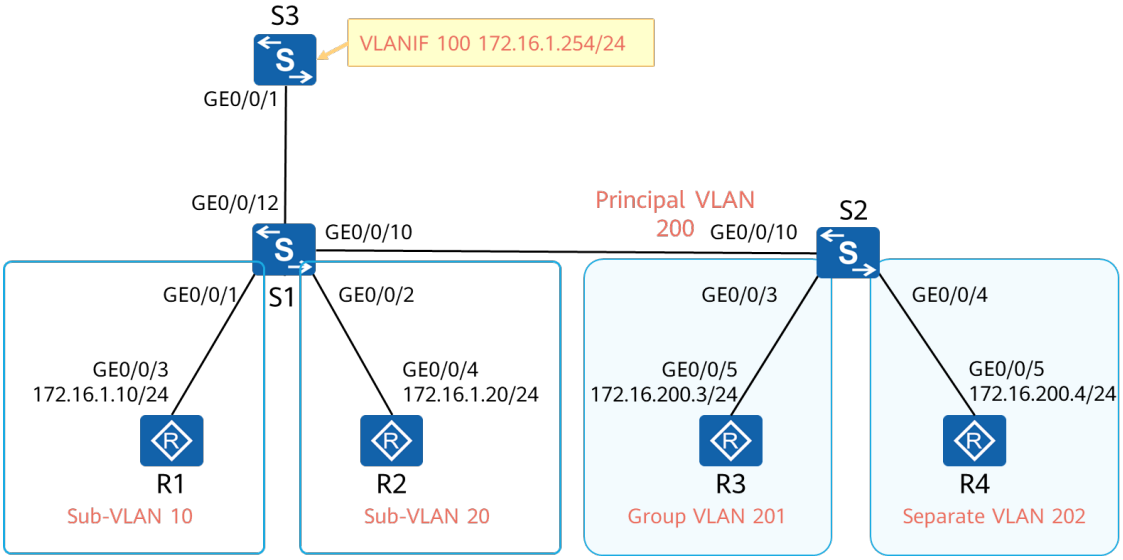
4.1.1.1 Objectives

Upon completion of this task, you will be able to:

- Configure VLAN aggregation.
- Configure MUX VLAN.
- Configure proxy ARP to implement communication between sub-VLANs.

4.1.1.2 Networking Topology

Networking of advanced VLAN technologies



VLAN aggregation is configured on S1 and S2, VLAN 100 is configured as the super-VLAN, the VLANIF interface on S3 is used as the gateway of R1 and R2, and VLAN 10 and VLAN 20 are configured as sub-VLANs.

MUX VLAN is configured on S2, and VLAN 200 is configured as the principal VLAN, VLAN 201 as the group VLAN, and VLAN 202 as the separate VLAN.

4.1.1.3 Lab Background

Assume that you are the network administrator of a company. To save the IP address space of the intranet, you need to deploy VLAN aggregation on the intranet and deploy MUX VLAN to restrict access between different services.

4.1.2 Lab Configuration

4.1.2.1 Configuration Roadmap

1. Complete basic VLAN aggregation configurations on S1 and S3, and configure IP addresses for interfaces on R1 and R2 to simulate terminal users.
2. Enable proxy ARP on VLANIF 100 of the super-VLAN on S3 and observe the communication between R1 and R2.
3. Configure VLAN 200 on S1, S2, and S3 to simulate the communication between the external network and sub-VLANs.
4. Configure MUX VLAN on S2 and assign IP addresses to interfaces of R3 and R4 to verify isolation through MUX VLAN.

4.1.2.2 Configuration Procedure

Step 1 Configure VLAN aggregation.

Create sub-VLANs 10 and 20 on S1, add the interfaces connected to R1 and R2 to sub-VLAN 10 and sub-VLAN 20, create super-VLAN 100 on S3, and create VLANIF 100 as the gateway. Configure the interconnection interfaces between S1 and S3 as trunk interfaces and configure the interfaces to allow packets from VLAN 10 and VLAN 20 to pass through. Configure IP addresses for interfaces on R1 and R2 to simulate terminal users.

Name the devices.

The configuration details are not provided.

Disable the interfaces that are not used in this experiment.

The configuration details are not provided.

Create sub-VLANs on S1.

```
[S1]vlan batch 10 20
```

Add S1's interfaces connected to R1 and R2 to sub-VLANs 10 and 20, respectively.

```
[S1]interface GigabitEthernet0/0/1
[S1-GigabitEthernet0/0/1] port link-type access
[S1-GigabitEthernet0/0/1] port default vlan 10
[S1-GigabitEthernet0/0/1] quit

[S1]interface GigabitEthernet0/0/2
[S1-GigabitEthernet0/0/2] port link-type access
```

```
[S1-GigabitEthernet0/0/2] port default vlan 20
[S1-GigabitEthernet0/0/2] quit
```

Create super-VLAN 100 on S3 and associate it with sub-VLANs 10 and 20. (VLANs 10 and 20 must exist on S3.)

```
[S3]vlan batch 10 20 100

[S3]vlan 100
[S3-vlan100] aggregate-vlan
[S3-vlan100] access-vlan 10 20
[S3-vlan100] quit
```

Create VLANIF 100 on S3 as the gateway.

```
[S3]interface Vlanif 100
[S3-Vlanif100] ip address 172.16.1.254 24
[S3-Vlanif100] quit
```

Configure interconnection interfaces between S1 and S3 to allow packets from VLANs 10 and 20.

```
[S1]interface GigabitEthernet0/0/12
[S1-GigabitEthernet0/0/12] port link-type trunk
[S1-GigabitEthernet0/0/12] port trunk allow-pass vlan 10 20
[S1-GigabitEthernet0/0/12] quit

[S3]interface GigabitEthernet0/0/1
[S3-GigabitEthernet0/0/1] port link-type trunk
[S3-GigabitEthernet0/0/1] port trunk allow-pass vlan 10 20
[S3-GigabitEthernet0/0/1] quit
```

Configure IP addresses for interfaces on R1 and R2.

```
[R1]interface GigabitEthernet0/0/3
[R1-GigabitEthernet0/0/3] ip address 172.16.1.10 255.255.255.0
[R1-GigabitEthernet0/0/3] quit

[R2]interface GigabitEthernet0/0/4
[R2-GigabitEthernet0/0/4] ip address 172.16.1.20 255.255.255.0
[R2-GigabitEthernet0/0/4] quit
```

Check the connectivity between R1 and the gateway and between R2 and the gateway.

```
<S3>ping -c 1 172.16.1.10
  PING 172.16.1.10: 56  data bytes, press CTRL_C to break
    Reply from 172.16.1.10: bytes=56 Sequence=1 ttl=255 time=110 ms

  --- 172.16.1.10 ping statistics ---
    1 packet(s) transmitted
    1 packet(s) received
    0.00% packet loss
round-trip min/avg/max = 110/110/110 ms

<S3>ping -c 1 172.16.1.20
  PING 172.16.1.20: 56  data bytes, press CTRL_C to break
    Reply from 172.16.1.20: bytes=56 Sequence=1 ttl=255 time=80 ms
```

```
 --- 172.16.1.20 ping statistics ---
   1 packet(s) transmitted
   1 packet(s) received
   0.00% packet loss
   round-trip min/avg/max = 80/80/80 ms
```

R1 and R2 can communicate with the gateway.

20 Observe the communication between sub-VLANs.

Enable proxy ARP on VLANIF 100 of S3 and observe the communication between sub-VLANs.

On R1, test the connectivity with R2.

```
<R1>ping -c 1 172.16.1.20
  PING 172.16.1.20: 56  data bytes, press CTRL_C to break
    Request time out

  --- 172.16.1.20 ping statistics ---
    1 packet(s) transmitted
    0 packet(s) received
    100.00% packet loss
```

R1 cannot communicate with R2.

Check ARP entries on R1 and S3.

```
<R1>display arp all
IP ADDRESS  MAC ADDRESS              EXPIRE(M) TYPE            INTERFACE    VPN-INSTANCE
                                         VLAN/CEVLAN     PVC
-------------------------------------------------------------------------------
172.16.1.10      5489-98bf-6225                    I -              GE0/0/3
172.16.1.254     4c1f-ccea-53ea          18        D-0               GE0/0/3
-------------------------------------------------------------------------------
Total:2          Dynamic:1        Static:0      Interface:1
```

In this case, R1 can learn only the ARP entry of VLANIF 100.

```
<S3>display arp all
IP ADDRESS  MAC ADDRESS              EXPIRE(M) TYPE            INTERFACE    VPN-INSTANCE
                                         VLAN/CEVLAN     PVC
-------------------------------------------------------------------------------
172.16.1.254     4c1f-ccea-53ea                    I -              Vlanif100
172.16.1.10      5489-98bf-6225          15        D-0               GE0/0/1
                                             10
172.16.1.20      5489-982a-0ad0          13        D-0               GE0/0/1
                                             20
-------------------------------------------------------------------------------
Total:3          Dynamic:2        Static:0      Interface:1
```

S3 has ARP entries of R1 and R2. The MAC address of VLANIF 100 is 4c1f-ccea-53ea.

On S3, enable inter-VLAN proxy ARP on VLANIF 100.

```
[S3]interface Vlanif 100
[S3-Vlanif100] arp-proxy inter-sub-vlan-proxy enable
```

Configure debugging arp process, debugging arp packet, and debugging arp-proxy on S3 to check the ARP proxy process.

```
<S3>terminal debugging
<S3>terminal monitor

<S3>debugging arp process
<S3>debugging arp packet
<S3>debugging arp-proxy
```

On R1, test the connectivity with R2 again.

```
<R1>ping -c 1 172.16.1.20
  PING 172.16.1.20: 56  data bytes, press CTRL_C to break
    Reply from 172.16.1.20: bytes=56 Sequence=1 ttl=254 time=190 ms

  --- 172.16.1.20 ping statistics ---
    1 packet(s) transmitted
    1 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 190/190/190 ms
```

R1 and R2 can communicate with each other.

Display debugging information on S3.

```
<S3>
Jul 13 2020 14:44:04.730.1-08:00 S3 ARP/7/arp_rcv:Receive an ARP Packet, operation : 1, sender_eth_addr : 5489-98bf-6225, sender_ip_addr : 172.16.1.10, target_eth_addr : 0000-0000-0000, target_ip_addr : 172.16.1.20
Jul 13 2020 14:44:04.730.2-08:00 S3 ARP/7/arp_info:ETHARP_ArpInput() Update an ARP entry, IP = 0xa0110ac.
Jul 13 2020 14:44:04.730.3-08:00 S3 ARP/7/eth_proxy_reply:The interface Vlanif100 outputs a proxy reply, origin VLAN ID is 10, type of proxy is inter-sub-vlan-proxy
Jul 13 2020 14:44:04.730.4-08:00 S3 ARP/7/arp_proxy:Send an ARP Proxy Packet, operation : 2, sender_eth_addr : 4c1f-ccea-53ea,sender_ip_addr : 172.16.1.20, target_eth_addr : 5489-98bf-6225, target_ip_addr : 172.16.1.10


Jul 13 2020 14:44:04.840.1-08:00 S3 ARP/7/arp_rcv:Receive an ARP Packet, operation : 1, sender_eth_addr : 5489-982a-0ad0, sender_ip_addr : 172.16.1.20, target_eth_addr : 0000-0000-0000, target_ip_addr : 172.16.1.10
Jul 13 2020 14:44:04.840.2-08:00 S3 ARP/7/arp_info:ETHARP_ArpInput() Update an ARP entry, IP = 0x140110ac.
Jul 13 2020 14:44:04.840.3-08:00 S3 ARP/7/eth_proxy_reply:The interface Vlanif100 outputs a proxy reply, origin VLAN ID is 20, type of proxy is inter-sub-vlan-proxy
Jul 13 2020 14:44:04.840.4-08:00 S3 ARP/7/arp_proxy:Send an ARP Proxy Packet, operation : 2, sender_eth_addr : 4c1f-ccea-53ea,sender_ip_addr : 172.16.1.10, target_eth_addr : 5489-982a-0ad0, target_ip_addr : 172.16.1.20
```

Analyze debugging information on S3 step by step.

```
Jul 13 2020 14:44:04.730.1-08:00 S3 ARP/7/arp_rcv:Receive an ARP Packet, operation : 1,
sender_eth_addr : 5489-98bf-6225, sender_ip_addr : 172.16.1.10, target_eth_addr : 0000-
0000-0000, target_ip_addr : 172.16.1.20
Jul 13 2020 14:44:04.730.2-08:00 S3 ARP/7/arp_info:ETHARP_ArpInput() Update an ARP entry,
IP = 0xa0110ac.
Jul 13 2020 14:44:04.730.3-08:00 S3 ARP/7/eth_proxy_reply:The interface Vlanif100 outputs a
proxy reply, origin VLAN ID is 10, type of proxy is inter-sub-vlan-proxy
Jul 13 2020 14:44:04.730.4-08:00 S3 ARP/7/arp_proxy:Send an ARP Proxy Packet, operation :
2, sender_eth_addr : 4c1f-ccea-53ea,sender_ip_addr : 172.16.1.20, target_eth_addr : 5489-
98bf-6225, target_ip_addr : 172.16.1.10
```

S3 receives the ARP Request packet from R1 at 172.16.1.10 and requests to resolve the MAC address of R2 at 172.16.1.20. Proxy ARP allows VLANIF 100 on S3 to respond to the ARP Request packet from R1. The source MAC address in the ARP Reply packet is 4c1f-ccea-53ea (MAC address of VLANIF 100), and the source IP address is 172.16.1.20.

After R1 receives the ARP Reply packet, it sends an ICMP packet with destination IP address 172.16.1.20 and destination MAC address 4c1f-ccea-53ea (MAC address of VLANIF 100). After receiving the ICMP packet, S3 forwards the packet.

```
Jul 13 2020 14:44:04.840.1-08:00 S3 ARP/7/arp_rcv:Receive an ARP Packet, operation : 1,
sender_eth_addr : 5489-982a-0ad0, sender_ip_addr : 172.16.1.20, target_eth_addr : 0000-
0000-0000, target_ip_addr : 172.16.1.10
Jul 13 2020 14:44:04.840.2-08:00 S3 ARP/7/arp_info:ETHARP_ArpInput() Update an ARP entry,
IP = 0x140110ac.
Jul 13 2020 14:44:04.840.3-08:00 S3 ARP/7/eth_proxy_reply:The interface Vlanif100 outputs a
proxy reply, origin VLAN ID is 20, type of proxy is inter-sub-vlan-proxy
Jul 13 2020 14:44:04.840.4-08:00 S3 ARP/7/arp_proxy:Send an ARP Proxy Packet, operation :
2, sender_eth_addr : 4c1f-ccea-53ea,sender_ip_addr : 172.16.1.10, target_eth_addr : 5489-
982a-0ad0, target_ip_addr : 172.16.1.20
```

After receiving the ICMP packet from R1, R2 sends an ARP request to R1 at 172.16.1.20. After S3 receives the ARP request, proxy ARP enables VLANIF 100 on S3 to respond to the ARP Request packet from R2. In this case, the source MAC address of the ARP packet is 4c1f-ccea-53ea (MAC address of VLANIF 100), and the source IP address is 172.16.1.10.

R1 and R2 have learned MAC addresses of VLANIF 100 on S3.

Check ARP entries on R1 and R2.

```
<R1>display arp all
IP ADDRESS  MAC ADDRESS     EXPIRE(M) TYPE              INTERFACE   VPN-INSTANCE
                                  VLAN/CEVLAN     PVC
--------------------------------------------------------------------------------
172.16.1.10     5489-98bf-6225              I -             GE0/0/3
172.16.1.254    4c1f-ccea-53ea       3        D-0            GE0/0/3
172.16.1.20     4c1f-ccea-53ea       3        D-0            GE0/0/3
--------------------------------------------------------------------------------
Total:3         Dynamic:2       Static:0     Interface:1

<R2>display arp all
IP ADDRESS  MAC ADDRESS     EXPIRE(M) TYPE              INTERFACE   VPN-INSTANCE
                                  VLAN/CEVLAN     PVC
--------------------------------------------------------------------------------
172.16.1.20     5489-982a-0ad0              I -             GE0/0/4
172.16.1.254    4c1f-ccea-53ea       2        D-0            GE0/0/4
```

```
172.16.1.10     4c1f-ccea-53ea      2       D-0          GE0/0/4
--------------------------------------------------------------------------------
Total:3        Dynamic:2      Static:0     Interface:1
```

The peer MAC addresses in the ARP entries learned by R1 and R2 are the MAC address of VLANIF 100 on S3.

21 Configure Layer 3 communication between sub-VLANs and the external network.

Create VLAN 200 on S1, S2, and S3 and create VLANIF 200 on S3 to simulate Layer 3 communication between sub-VLANs and the external network.

Create VLAN 200.

```
[S1]vlan 200

[S2]vlan 200

[S3]vlan 200
```

Configure interconnection interfaces between S1 and S3 to allow packets from VLAN 200 to pass through.

```
[S3]interface GigabitEthernet0/0/1
[S3-GigabitEthernet0/0/1] port link-type trunk
[S3-GigabitEthernet0/0/1] port trunk allow-pass vlan 200
[S3-GigabitEthernet0/0/1] quit

[S1]interface GigabitEthernet0/0/12
[S1-GigabitEthernet0/0/12] port link-type trunk
[S1-GigabitEthernet0/0/12] port trunk allow-pass vlan 200
[S1-GigabitEthernet0/0/12] quit
[S1]interface GigabitEthernet0/0/10
[S1-GigabitEthernet0/0/10] port link-type access
[S1-GigabitEthernet0/0/10] port default vlan 200
[S1-GigabitEthernet0/0/10] quit
```

Note that GE0/0/10 on S1 is configured as the access interface.

Create VLANIF 200 on S3 and set its IP address to 172.16.200.xx/24 (x indicates the device number).

```
[S3]interface Vlanif200
[S3-Vlanif200] ip address 172.16.200.33 255.255.255.0
[S3-Vlanif200] quit
```

Configure static routes to the external network on R1 and R2.

```
[R1]ip route-static 172.16.200.0 24 172.16.1.254

[R2]ip route-static 172.16.200.0 24 172.16.1.254
```

22 Configure MUX VLAN.

Configure MUX VLAN on S2, configure VLAN 200 as the principal VLAN, configure VLAN 201 as the group VLAN, configure VLAN 202 as the separate VLAN, and configure IP addresses for interfaces on R3 and R4 to verify isolation through MUX VLAN.

Create VLANs 201 and 202.

```
[S2]vlan batch 201 202
```

Configure VLAN 200 as the principal VLAN, VLAN 201 as the group VLAN, and VLAN 202 as the separate VLAN.

```
[S2]vlan 200
[S2-vlan200] mux-vlan
[S2-vlan200] subordinate separate 202
[S2-vlan200] subordinate group 201
[S2-vlan200] quit
```

Add interfaces to the VLANs and enable the MUX VLAN function.

```
[S2]interface GigabitEthernet0/0/10
[S2-GigabitEthernet0/0/10] port link-type access
[S2-GigabitEthernet0/0/10] port default vlan 200
[S2-GigabitEthernet0/0/10] port mux-vlan enable
[S2-GigabitEthernet0/0/10] quit
[S2]interface GigabitEthernet0/0/3
[S2-GigabitEthernet0/0/3] port link-type access
[S2-GigabitEthernet0/0/3] port default vlan 201
[S2-GigabitEthernet0/0/3] port mux-vlan enable
[S2-GigabitEthernet0/0/3]quit
[S2]interface GigabitEthernet0/0/4
[S2-GigabitEthernet0/0/4] port link-type access
[S2-GigabitEthernet0/0/4] port default vlan 202
[S2-GigabitEthernet0/0/4] port mux-vlan enable
[S2-GigabitEthernet0/0/4]quit
```

Check the MUX VLAN configuration.

```
[S2]display mux-vlan
Principal Subordinate     Type        Interface
-------------------------------------------------------------------------
200       -           principal     GigabitEthernet0/0/10
200       202           separate      GigabitEthernet0/0/4
200       201           group        GigabitEthernet0/0/3
-------------------------------------------------------------------------
```

Configure IP addresses for interfaces on R3 and R4.

```
[R3]interface GigabitEthernet0/0/5
[R3-GigabitEthernet0/0/5] ip address 172.16.200.3 255.255.255.0
[R3-GigabitEthernet0/0/5] quit
```

```
[R4]interface GigabitEthernet0/0/5
[R4-GigabitEthernet0/0/5] ip address 172.16.200.4 255.255.255.0
[R4-GigabitEthernet0/0/5] quit
```

Configure static routes on R3 and R4 to communicate with the network segment 172.16.1.0/24.

```
[R3]ip route-static 172.16.1.0 24 172.16.200.33

[R4]ip route-static 172.16.1.0 24 172.16.200.33
```

Check the connectivity.

```
<R3>ping -c 1 172.16.1.10
  PING 172.16.1.10: 56  data bytes, press CTRL_C to break
    Reply from 172.16.1.10: bytes=56 Sequence=1 ttl=254 time=190 ms

  --- 172.16.1.10 ping statistics ---
    1 packet(s) transmitted
    1 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 190/190/190 ms

<R3>ping -c 1 172.16.200.4
  PING 172.16.200.4: 56  data bytes, press CTRL_C to break
    Request time out

  --- 172.16.200.4 ping statistics ---
    1 packet(s) transmitted
    0 packet(s) received
    100.00% packet loss
```

R3 in the group VLAN can communicate with R1 through the principal VLAN, but cannot communicate with R4 in the separate VLAN.

```
<R4>ping -c 1 172.16.1.10
  PING 172.16.1.10: 56  data bytes, press CTRL_C to break
    Reply from 172.16.1.10: bytes=56 Sequence=1 ttl=254 time=190 ms

  --- 172.16.1.10 ping statistics ---
    1 packet(s) transmitted
    1 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 190/190/190 ms

<R4>ping -c 1 172.16.200.3
  PING 172.16.200.3: 56  data bytes, press CTRL_C to break
    Request time out

  --- 172.16.200.3 ping statistics ---
    1 packet(s) transmitted
    0 packet(s) received
    100.00% packet loss
```

R4 in the separate VLAN can communicate with R1 through the principal VLAN, but cannot communicate with R3 in the group VLAN.

```
<R1>ping -c 1 172.16.200.3
  PING 172.16.200.3: 56  data bytes, press CTRL_C to break
    Reply from 172.16.200.3: bytes=56 Sequence=1 ttl=254 time=110 ms

  --- 172.16.200.3 ping statistics ---
    1 packet(s) transmitted
    1 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 110/110/110 ms

<R1>ping -c 1 172.16.200.4
  PING 172.16.200.4: 56  data bytes, press CTRL_C to break
    Reply from 172.16.200.4: bytes=56 Sequence=1 ttl=254 time=120 ms

  --- 172.16.200.4 ping statistics ---
    1 packet(s) transmitted
    1 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 120/120/120 ms
```

R1 can communicate with R3 and R4 in the group VLAN and separate VLAN through the principal VLAN.

4.1.3 Quiz

After inter-VLAN proxy ARP is enabled on a VLANIF interface in a super-VLAN, in which situation will the VLANIF interface respond to ARP Request packets?

4.1.4 Configuration Reference

Configuration on R1

```
#
sysname R1
#
interface GigabitEthernet0/0/3
 ip address 172.16.1.10 255.255.255.0
#
ip route-static 172.16.200.0 255.255.255.0 172.16.1.254
```

Configuration on R2

```
#
sysname R2
#
interface GigabitEthernet0/0/4
 ip address 172.16.1.20 255.255.255.0
#
ip route-static 172.16.200.0 255.255.255.0 172.16.1.254
#
return
```

Configuration on R3

```
#
sysname R3
#
```

```
interface GigabitEthernet0/0/5
 ip address 172.16.200.3 255.255.255.0
#
ip route-static 172.16.1.0 255.255.255.0 172.16.200.33
#
return
```

Configuration on R4

```
#
sysname R4
#
interface GigabitEthernet0/0/5
 ip address 172.16.200.4 255.255.255.0
#
ip route-static 172.16.1.0 255.255.255.0 172.16.200.33
#
return
```

Configuration on S1

```
#
sysname S1
#
vlan batch 10 20 200
#
interface GigabitEthernet0/0/1
 port link-type access
 port default vlan 10
#
interface GigabitEthernet0/0/2
 port link-type access
 port default vlan 20
#
interface GigabitEthernet0/0/10
 port link-type access
 port default vlan 200
#
interface GigabitEthernet0/0/12
 port link-type trunk
 port trunk allow-pass vlan 10 20 200
#
return
```

Configuration on S2

```
#
sysname S2
#
vlan batch 200 to 202
#
vlan 200
 mux-vlan
 subordinate separate 202
 subordinate group 201
#
interface GigabitEthernet0/0/3
```

```
 port link-type access
 port default vlan 201
 port mux-vlan enable
#
interface GigabitEthernet0/0/4
 port link-type access
 port default vlan 202
 port mux-vlan enable
#
interface GigabitEthernet0/0/10
 port link-type access
 port default vlan 200
 port mux-vlan enable
#
ip route-static 172.16.1.0 255.255.255.0 172.16.200.3
#
return
```

Configuration on S3

```
#
sysname S3
#
vlan batch 10 20 100 200
#
vlan 100
 aggregate-vlan
 access-vlan 10 20
#
interface Vlanif100
 ip address 172.16.1.254 255.255.255.0
 arp-proxy inter-sub-vlan-proxy enable
#
interface Vlanif200
 ip address 172.16.200.33 255.255.255.0
#
interface GigabitEthernet0/0/1
 port link-type trunk
 port trunk allow-pass vlan 10 20 200
#
return
```

5 Ethernet Switching Security

5.1 Ethernet Switching Security

5.1.1 Introduction

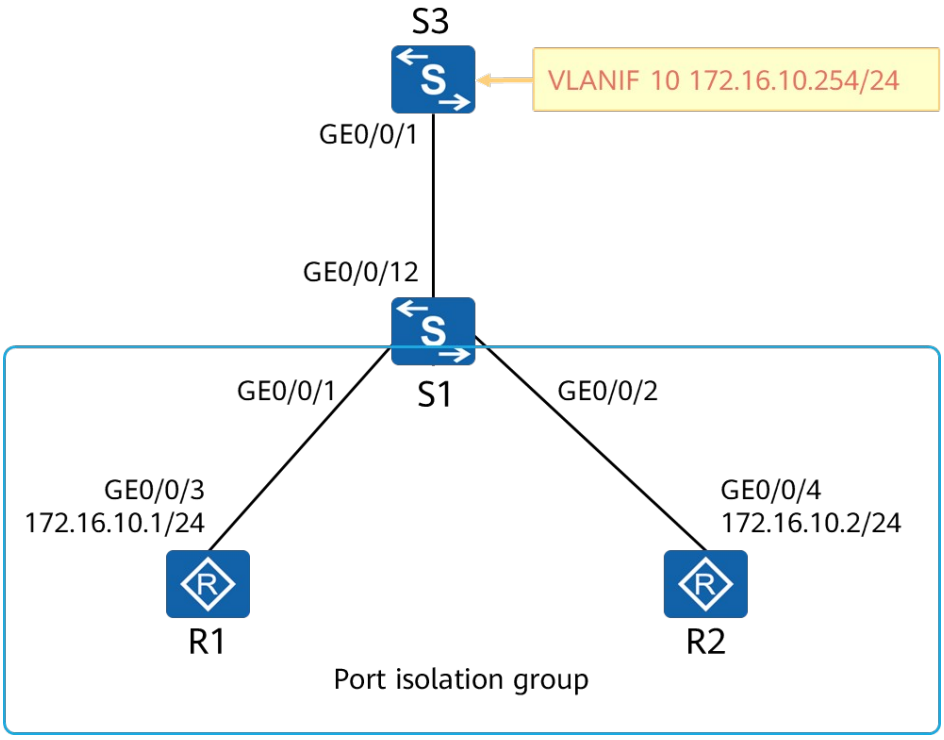
5.1.1.1 Objectives

Upon completion of this task, you will be able to:

- Configure basic functions of DHCP snooping.
- Configure IP Source Guard (IPSG) to prevent users from changing terminal IP addresses without permission.
- Configure port security to limit the number of access terminals.
- Configure port isolation to restrict Layer 2 communication between users on the same network segment.

5.1.1.2 Networking Topology

Networking of Ethernet switching security



R1 and R2 are connected to GE0/0/1 and GE0/0/2 of S1 and are assigned to VLAN 10. The interfaces connecting S1 and S3 are configured as trunk interfaces and configured to allow packets from VLAN 10 to pass through.

The DHCP server is configured on S3 to statically assign fixed IP addresses to R1 and R2, and DHCP snooping, IPSG, port security, and port isolation are configured on S1.

5.1.1.3 Lab Background

You are a network administrator of a company. To improve network security, you need to deploy the following security technologies on access switches: DHCP snooping, IPSG, port security, and port isolation.

5.1.2 Lab Configuration

5.1.2.1 Configuration Roadmap

1. Complete basic Layer 2 configurations on S1 and S2, and enable the DHCP service on S3 to assign fixed IP addresses to R1 and R2.
2. Enable DHCP snooping on S1 and configure the interface connected to S3 as the trusted interface.
3. Enable IPSG on S1 to check packets against dynamic binding entries generated by DHCP snooping.

4. Configure port security on S1 to limit the number of MAC addresses learned by the interface.
5. Configure a port isolation group on S1 to restrict Layer 2 communication between R1 and R2. Configure proxy ARP on VLANIF 10 of S3 to implement mutual access between R1 and R2.

5.1.2.2 Configuration Procedure

Step 1 Configure DHCP.

Create VLAN 10 on S1 and add the interfaces connected to R1, R2, and S3 to VLAN 10. Configure the interfaces connected to R1 and R2 as access interfaces and the interface connected to S3 as the trunk interface.

Enable the DHCP service on S3 to allocate fixed IP addresses to R1 and R2.

Name the devices.

The configuration details are not provided.

Disable the interfaces that are not used in this experiment.

The configuration details are not provided.

Create VLAN 10 on S1 and S3.

```
[S1]vlan 10

[S3]vlan 10
```

On S1, configure GE0/0/1 and GE0/0/2 as access interfaces and set the PVID to VLAN 10.

```
[S1]interface GigabitEthernet0/0/1
[S1-GigabitEthernet0/0/1] port link-type access
[S1-GigabitEthernet0/0/1] port default vlan 10
[S1-GigabitEthernet0/0/1] quit

[S1]interface GigabitEthernet0/0/2
[S1-GigabitEthernet0/0/2] port link-type access
[S1-GigabitEthernet0/0/2] port default vlan 10
[S1-GigabitEthernet0/0/2] quit
```

Configure GE0/0/12 of S1 as the trunk interface and configure it to allow packets from VLAN 10 to pass through.

```
[S1]interface GigabitEthernet0/0/12
[S1-GigabitEthernet0/0/12] port link-type trunk
[S1-GigabitEthernet0/0/12] port trunk allow-pass vlan 10
[S1-GigabitEthernet0/0/12] quit
```

Configure GE0/0/1 of S3 as the trunk interface and configure it to allow packets from VLAN 10 to pass through.

```
[S3]interface GigabitEthernet0/0/1
[S3-GigabitEthernet0/0/1] port link-type trunk
[S3-GigabitEthernet0/0/1] port trunk allow-pass vlan 10
[S3-GigabitEthernet0/0/1] quit
```

Create VLANIF 10 on S3 as the gateway of R1 and R2.

```
[S3]interface Vlanif10
[S3-Vlanif10] ip address 172.16.10.254 255.255.255.0
[S3-Vlanif10] quit
```

Enable the DHCP service on S3 and configure a global address pool to assign fixed IP addresses to R1 and R2.

```
[S3]dhcp enable

[S3]ip pool vlan10
[S3-ip-pool-vlan10] gateway-list 172.16.10.254
[S3-ip-pool-vlan10] network 172.16.10.0 mask 255.255.255.0
[S3-ip-pool-vlan10] dns-list 172.16.10.254
[S3-ip-pool-vlan10] quit
```

Enable the DHCP server function on VLANIF 10 of S3.

```
[S3]interface Vlanif 10
[S3-Vlanif10] dhcp select global
```

Check MAC addresses of interfaces on R1 and R2.

```
<R1>display interface GigabitEthernet 0/0/3 | in Hardware
IP Sending Frames' Format is PKTFMT_ETHNT_2, Hardware address is 5489-98bf-6225
Hardware address is 5489-98bf-6225

<R2>display interface GigabitEthernet 0/0/4 | in Hardware
IP Sending Frames' Format is PKTFMT_ETHNT_2, Hardware address is 5489-982a-0ad0
```

Hardware address is 5489-982a-0ad0

Configure S3 to assign fixed IP addresses to R1 and R2.

```
[S3]ip pool vlan10
[S3-ip-pool-vlan10] static-bind ip-address 172.16.10.1 mac-address 5489-98bf-6225
[S3-ip-pool-vlan10] static-bind ip-address 172.16.10.2 mac-address 5489-982a-0ad0
```

Configure R1 and R2 to obtain IP addresses through DHCP.

```
[R1]dhcp enable
[R1]interface GigabitEthernet0/0/3
[R1-GigabitEthernet0/0/3] ip address dhcp-alloc
[R1-GigabitEthernet0/0/3] quit

[R2]dhcp enable
[R2]interface GigabitEthernet0/0/4
[R2-GigabitEthernet0/0/4] ip address dhcp-alloc
[R2-GigabitEthernet0/0/4] quit
```

Check the IP address allocation result on R1 and R2.

```
<R1>display ip interface  brief | in GigabitEthernet0/0/3
*down: administratively down
!down: FIB overload down
^down: standby
(l): loopback
(s): spoofing
(d): Dampening Suppressed
The number of interface that is UP in Physical is 2
The number of interface that is DOWN in Physical is 9
```

```
The number of interface that is UP in Protocol is 2
The number of interface that is DOWN in Protocol is 9

Interface                     IP Address/Mask      Physical   Protocol
GigabitEthernet0/0/3            172.16.10.1/24       up         up

<R2>display ip interface brief | in GigabitEthernet0/0/4
*down: administratively down
!down: FIB overload down
^down: standby
(l): loopback
(s): spoofing
(d): Dampening Suppressed
The number of interface that is UP in Physical is 2
The number of interface that is DOWN in Physical is 9
The number of interface that is UP in Protocol is 2
The number of interface that is DOWN in Protocol is 9

Interface                     IP Address/Mask      Physical   Protocol
GigabitEthernet0/0/4            172.16.10.2/24       up         up
```

R1 and R2 have successfully obtained fixed IP addresses through DHCP.

23 Configure DHCP snooping.

To prevent unauthorized DHCP servers from assigning IP addresses, enable DHCP snooping on S1, configure the interface connected to the DHCP server as the trusted interface, and enable DHCP snooping on the interface connected to terminals.

Enable DHCP snooping globally on S1 and configure the interface connected to S3 as the trusted interface.

```
[S1]dhcp enable
[S1]dhcp snooping enable ipv4

[S1]interface GigabitEthernet 0/0/12
[S1-GigabitEthernet0/0/12] dhcp snooping trusted
[S1-GigabitEthernet0/0/12] quit
```

Note that the DHCP function must be enabled first.

Enable DHCP snooping on S1's interfaces connected to R1 and R2.

```
[S1]interface GigabitEthernet 0/0/1
[S1-GigabitEthernet0/0/1] dhcp snooping enable
[S1-GigabitEthernet0/0/1] quit
[S1]interface GigabitEthernet 0/0/2
[S1-GigabitEthernet0/0/2] dhcp snooping enable
[S1-GigabitEthernet0/0/2] quit
```

Shut down interfaces on R1 and R2, wait for a period of time, and enable the interfaces again to obtain IP addresses through DHCP.

```
[R1]interface GigabitEthernet0/0/3
[R1-GigabitEthernet0/0/3] shutdown

[R2]interface GigabitEthernet0/0/4
[R2-GigabitEthernet0/0/4] shutdown

[R1]interface GigabitEthernet0/0/3
[R1-GigabitEthernet0/0/3] undo shutdown

[R2]interface GigabitEthernet0/0/4
[R2-GigabitEthernet0/0/4] undo shutdown
```

Check dynamic DHCP snooping entries on S1.

```
[S1]display dhcp snooping user-bind all
DHCP Dynamic Bind-table:
Flags:O - outer vlan ,I - inner vlan ,P - map vlan
IP Address      MAC Address      VSI/VLAN(O/I/P)    Interface      Lease

--------------------------------------------------------------------------------
172.16.10.1      5489-98bf-6225    10  /--  /--      GE0/0/1       2020.07.14-22:42
172.16.10.2      5489-982a-0ad0     10  /--  /--       GE0/0/2       2020.07.14-22:46
--------------------------------------------------------------------------------
print count:          2         total count:          2
```

In this case, dynamic entries are generated.

24 Configure IPSG.

Enable IPSG on S1 to prevent terminal users from configuring static IP addresses without permission, allow hosts to use only the IP addresses allocated by the DHCP server, and prevent access of unauthorized users.

Enable IPSG on GE0/0/1 and GE0/0/2 of S1.

```
[S1]interface GigabitEthernet 0/0/1
[S1-GigabitEthernet0/0/1] ip source check user-bind enable
[S1-GigabitEthernet0/0/1] ip source check user-bind alarm enable
[S1-GigabitEthernet0/0/1] ip source check user-bind alarm threshold 3
[S1-GigabitEthernet0/0/1] quit
[S1]interface GigabitEthernet 0/0/2
[S1-GigabitEthernet0/0/2] ip source check user-bind enable
[S1-GigabitEthernet0/0/2] ip source check user-bind alarm enable
[S1-GigabitEthernet0/0/2] ip source check user-bind alarm threshold 3
```

Enable the alarm function for IP packet check and set the alarm threshold for the number of discarded IP packets to 3.

On R1, test the connectivity of the link between R1 and R5.

```
<R1>ping -c 1 172.16.10.254
  PING 172.16.10.254: 56  data bytes, press CTRL_C to break
    Reply from 172.16.10.254: bytes=56 Sequence=1 ttl=255 time=110 ms

  --- 172.16.10.254 ping statistics ---
    1 packet(s) transmitted
    1 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 110/110/110 ms
```

R1 can communicate with the gateway.

Change the IP address of R1 to 172.16.10.11/24 so that the packets sent by R1 cannot match dynamic DHCP snooping entries. That is, the packets cannot pass the IPSG check.

```
[R1]interface GigabitEthernet0/0/3
[R1-GigabitEthernet0/0/3] ip address 172.16.10.11 24
```

On R1, test the connectivity with the gateway again.

```
[R1]ping -c 4 172.16.10.254
  PING 172.16.10.254: 56  data bytes, press CTRL_C to break
    Request time out
    Request time out
    Request time out
    Request time out

  --- 172.16.10.254 ping statistics ---
    4 packet(s) transmitted
    0 packet(s) received
    100.00% packet loss
```

R1 cannot communicate with the gateway.

Check the log buffer of S1.

```
[S1]display logbuffer
Logging buffer configuration and contents : enabled
Allowed max buffer size : 1024
Actual buffer size : 512
Channel number : 4 , Channel name : logbuffer
Dropped messages : 0
Overwritten messages : 0
Current messages : 5

Jul 14 2020 11:53:43 S1 %%01SECE/4/IPSG_DROP_PACKET(l)[0]:IP packets dropped by IPSG.
(SourceInterface=GigabitEthernet0/0/1,DropPacketNumber=4,DropTime=2020/07/14
11:53:43, The most dropped packet is: VLAN=10, SourceMAC=5489-98bf-6225,
SourceIP=172.16.10.11.)
```

The packets from R1 are discarded by IPSG, and the number of discarded packets is 4, which exceeds the alarm threshold. An alarm log is generated and can be viewed in the log buffer.

Create a static binding entry.

```
[S1]user-bind static ip-address 172.16.10.11 mac-address 5489-98bf-6225 interface
GigabitEthernet 0/0/1 vlan 10
```

Check the static binding entry.

```
[S1]display dhcp  static user-bind all
DHCP static Bind-table:
Flags:O - outer vlan ,I - inner vlan ,P - Vlan-mapping
IP Address                    MAC Address     VSI/VLAN(O/I/P)        Interface
--------------------------------------------------------------------------------
172.16.10.11                   5489-98bf-6225    10  /--  /--         GE0/0/1
--------------------------------------------------------------------------------
Print count:          1         Total count:
```

On R1, test the connectivity with the gateway again.

```
<R1>ping -c 4 172.16.10.254
  PING 172.16.10.254: 56  data bytes, press CTRL_C to break
    Reply from 172.16.10.254: bytes=56 Sequence=1 ttl=255 time=1 ms
    Reply from 172.16.10.254: bytes=56 Sequence=2 ttl=255 time=1 ms
    Reply from 172.16.10.254: bytes=56 Sequence=3 ttl=255 time=1 ms
    Reply from 172.16.10.254: bytes=56 Sequence=4 ttl=255 time=1 ms

  --- 172.16.10.254 ping statistics ---
    4 packet(s) transmitted
    4 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 1/1/1 ms
```

Because the static binding entry exists, R1 can communicate with the gateway.

25 Configure port security.

To allow only one terminal to connect to an interface of an access switch, enable port security on the interface of S1 connected to only one terminal. When the number of terminals connected to the interface exceeds the limit, the packets whose source MAC addresses are not in the secure MAC address list are discarded and an alarm is generated.

Enable port security on S1's interface connected to R2.

```
[S1]interface GigabitEthernet 0/0/2
[S1-GigabitEthernet0/0/2] port-security enable
[S1-GigabitEthernet0/0/2] port-security max-mac-num 1
[S1-GigabitEthernet0/0/2] port-security mac-address sticky
[S1-GigabitEthernet0/0/2] port-security protect-action restrict
[S1-GigabitEthernet0/0/2] quit
```

R2 communicates with the gateway to trigger the generation of a sticky MAC address entry.

```
<R2>ping -c 1 172.16.10.254
  PING 172.16.10.254: 56  data bytes, press CTRL_C to break
    Reply from 172.16.10.254: bytes=56 Sequence=1 ttl=255 time=60 ms

  --- 172.16.10.254 ping statistics ---
```

```
  1 packet(s) transmitted
  1 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 60/60/60 ms
```

Check the sticky MAC address entry generated on S1.

```
<S1>display mac-address sticky
MAC address table of slot 0:
-------------------------------------------------------------------------------
MAC Address    VLAN/      PEVLAN  CEVLAN  Port          Type      LSP/LSR-ID
               VSI/SI                                   MAC-Tunnel
-------------------------------------------------------------------------------
5489-982a-0ad0  10            -     -    GE0/0/2       sticky        -
-------------------------------------------------------------------------------
Total matching items on slot 0 displayed = 1
```

Change the MAC address of the interface on R2.

```
[R2]interface GigabitEthernet0/0/4
[R2-GigabitEthernet0/0/4] mac-address 0000-0000-0002
[R2-GigabitEthernet0/0/4] quit
```

On R2, test the connectivity with the gateway.

```
<R2>ping -c 1 172.16.10.254
 PING 172.16.10.254: 56  data bytes, press CTRL_C to break
   Request time out

 --- 172.16.10.254 ping statistics ---
   1 packet(s) transmitted
   0 packet(s) received
   100.00% packet loss
```

S2 cannot communicate with the gateway.

Check MAC address entries on GE0/0/2 of S1.

```
<S1>display mac-address sticky
MAC address table of slot 0:
-------------------------------------------------------------------------------
MAC Address    VLAN/      PEVLAN  CEVLAN  Port          Type      LSP/LSR-ID
               VSI/SI                                   MAC-Tunnel
-------------------------------------------------------------------------------
5489-982a-0ad0  10            -     -    GE0/0/2       sticky        -
-------------------------------------------------------------------------------
Total matching items on slot 0 displayed = 1
```

There is still only one sticky MAC address entry, and the MAC address manually changed on the interface of R2 is not learned.

Change the maximum number of MAC addresses that can be learned by GE0/0/2 of S1 to 2.

```
[S1]interface GigabitEthernet 0/0/2
[S1-GigabitEthernet0/0/2] port-security max-mac-num 2
[S1-GigabitEthernet0/0/2] quit
```

On R2, test the connectivity with the gateway.

```
<R2>ping -c 1 172.16.10.254
  PING 172.16.10.254: 56  data bytes, press CTRL_C to break
    Request time out

  --- 172.16.10.254 ping statistics ---
    1 packet(s) transmitted
    0 packet(s) received
    100.00% packet loss
```

In this case, R2 cannot communicate with the gateway due to IPSG configuration.

Check MAC address entries on GE0/0/2 of S1.

```
<S1>display mac-address sticky
MAC address table of slot 0:
--------------------------------------------------------------------------------
MAC Address    VLAN/      PEVLAN  CEVLAN  Port              Type      LSP/LSR-ID
               VSI/SI                                     MAC-Tunnel
--------------------------------------------------------------------------------
5489-982a-0ad0  10            -     -      GE0/0/2        sticky         -
0000-0000-0002  10            -     -      GE0/0/2        sticky         -
--------------------------------------------------------------------------------
Total matching items on slot 0 displayed = 2
```

After the MAC address entry is manually changed on the interface of R2, the interface learns the sticky MAC address entry.

Create a static binding entry.

```
[S1]user-bind static ip-address 172.16.10.2 mac-address 0000-0000-0002 interface
GigabitEthernet 0/0/2 vlan 10
```

On R2, test the connectivity with the gateway again.

```
<R2>ping -c 1 172.16.10.254
  PING 172.16.10.254: 56  data bytes, press CTRL_C to break
    Reply from 172.16.10.254: bytes=56 Sequence=1 ttl=255 time=80 ms

  --- 172.16.10.254 ping statistics ---
    1 packet(s) transmitted
    1 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 80/80/80 ms
```

R2 can communicate with the gateway.

26 Configure a port isolation group.

Enable port isolation on S1 to restrict mutual access between R1 and R2.

Test the connectivity between R1 and R2 before restricting mutual access between R1 and R2.

```
<R1>ping -c 1 172.16.10.2
```

```
 PING 172.16.10.2: 56  data bytes, press CTRL_C to break
   Reply from 172.16.10.2: bytes=56 Sequence=1 ttl=255 time=70 ms

 --- 172.16.10.2 ping statistics ---
   1 packet(s) transmitted
   1 packet(s) received
   0.00% packet loss
   round-trip min/avg/max = 70/70/70 ms
```

Enable port isolation on S1 and set the isolation mode to Layer 2 isolation and Layer 3 interconnection.

```
[S1]port-isolate mode l2
[S1]interface GigabitEthernet 0/0/1
[S1-GigabitEthernet0/0/1] port-isolate enable group 1
[S1-GigabitEthernet0/0/1] quit
[S1]interface GigabitEthernet 0/0/2
[S1-GigabitEthernet0/0/2] port-isolate enable group 1
[S1-GigabitEthernet0/0/2] quit
```

Test the connectivity between R1 and R2.

```
<R1>ping -c 1 172.16.10.2
 PING 172.16.10.2: 56  data bytes, press CTRL_C to break
   Request time out

 --- 172.16.10.2 ping statistics ---
   1 packet(s) transmitted
   0 packet(s) received
   100.00% packet loss
```

R1 and R2 cannot communicate with each other.

Enable intra-VLAN proxy ARP on VLANIF 10 of S3.

```
[S3]interface Vlanif 10
[S3-Vlanif10] arp-proxy inner-sub-vlan-proxy enable
[S3-Vlanif10] quit
```

Clear ARP entries on R1 and R2.

```
<R1>reset arp all
Warning: This operation will reset all static and dynamic ARP entries, and clear the
configurations of all static ARP, continue?[Y/N]:y

<R2>reset arp all
Warning: This operation will reset all static and dynamic ARP entries, and clear the
configurations of all static ARP, continue?[Y/N]:y
```

In this case, R1 and R2 learn ARP entries again. Because proxy ARP is enabled on the gateway, R1 and R2 can communicate with each other through the gateway.

Test the connectivity between R1 and R2.

```
<R1>ping -c 1 172.16.10.2
 PING 172.16.10.2: 56  data bytes, press CTRL_C to break
   Reply from 172.16.10.2: bytes=56 Sequence=1 ttl=254 time=190 ms

 --- 172.16.10.2 ping statistics ---
```

```
  1 packet(s) transmitted
  1 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 190/190/190 ms
```

R1 and R2 can communicate with each other.

Check ARP entries on R1.

```
<R1>display arp all
IP ADDRESS       MAC ADDRESS      EXPIRE(M)     TYPE              INTERFACE    VPN-
INSTANCE
                                                VLAN/CEVLAN   PVC
--------------------------------------------------------------------------------
172.16.10.11         5489-98bf-6225                    I -             GE0/0/3
172.16.10.254        4c1f-ccea-53ea        14          D-0              GE0/0/3
172.16.10.2          4c1f-ccea-53ea        14          D-0              GE0/0/3
--------------------------------------------------------------------------------
Total:3          Dynamic:2        Static:0      Interface:1
```

The command output shows that MAC addresses mapping 172.16.10.2 and 172.16.10.254 in the ARP entries are the same.

5.1.3 Quiz

After port isolation is configured, the gateway functions as a proxy to implement communication on the same network segment. What are the advantages and disadvantages of this mode?

5.1.4 Configuration Reference

Configuration on R1

```
#
sysname R1
#
interface GigabitEthernet0/0/3
 ip address 172.16.10.11 255.255.255.0
#
return
```

Configuration on R2

```
#
sysname R2
#
interface GigabitEthernet0/0/4
 mac-address 0000-0000-0002
 ip address dhcp-alloc
#
return
```

Configuration on S1

```
#
sysname S1
#
vlan batch 10
```

```
#
port-isolate mode all
#
dhcp enable
#
dhcp snooping enable ipv4
user-bind static ip-address 172.16.10.11 mac-address 5489-98bf-6225 interface
GigabitEthernet0/0/1 vlan 10
user-bind static ip-address 172.16.10.2 mac-address 0000-0000-0002 interface
GigabitEthernet0/0/2 vlan 10
#
interface GigabitEthernet0/0/1
 port link-type access
 port default vlan 10
 port-isolate enable group 1
 ip source check user-bind enable
 ip source check user-bind alarm enable
 ip source check user-bind alarm threshold 3
 dhcp snooping enable
#
interface GigabitEthernet0/0/2
 port link-type access
 port default vlan 10
 port-security enable
 port-security max-mac-num 2
 port-security mac-address sticky
 port-isolate enable group 1
 ip source check user-bind enable
 ip source check user-bind alarm enable
 ip source check user-bind alarm threshold 3
 dhcp snooping enable
#
interface GigabitEthernet0/0/12
 port link-type trunk
 port trunk allow-pass vlan 10
 dhcp snooping trusted
#
return
```

Configuration on S3

```
#
sysname S3
#
vlan batch 10
#
dhcp enable
#
ip pool vlan10
 gateway-list 172.16.10.254
 network 172.16.10.0 mask 255.255.255.0
 static-bind ip-address 172.16.10.1 mac-address 5489-98bf-6225
 static-bind ip-address 172.16.10.2 mac-address 5489-982a-0ad0
 dns-list 172.16.10.254
#
interface Vlanif10
```

```
 ip address 172.16.10.254 255.255.255.0
 arp-proxy inner-sub-vlan-proxy enable
 dhcp select global
#
interface GigabitEthernet0/0/1
 port link-type trunk
 port trunk allow-pass vlan 10
#
return
```

6 BGP/MPLS IP VPN

6.1 BGP/MPLS IP VPN

6.1.1 Introduction

6.1.1.1 Objectives

Upon completion of this task, you will be able to:

- Configure MPLS.
- Configure MPLS LDP.
- Configure MPLS VPN.

6.1.1.2 Networking Topology

Networking of BGP/MPLS IP VPN

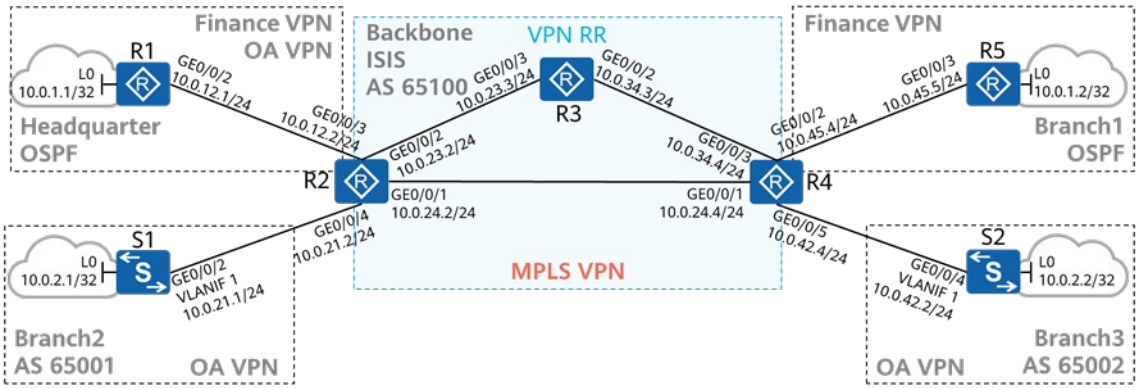


Figure 6-1 shows IP addresses of interconnection interfaces and BGP AS numbers. R3 is a VPN RR, and R2 and R4 are clients of R3. In BGP AS 65100, IS-IS is used to build the underlying network.

R2 and R4 are PEs of the MPLS VPN network, R1 and R5 are CEs of the finance VPN, and R1, S1, and S2 are CEs of the OA VPN.

6.1.1.3 Lab Background

An enterprise network has one headquarters and three branches. Branch1 mainly provides financial services, and Branch2 and Branch3 mainly provide enterprise office services.

The enterprise headquarters needs to receive routing entries of Branch1, Branch2, and Branch3. Branch1 is not allowed to receive routing entries of Branch2 and Branch3.

The enterprise backbone network is large, so the VPN RR needs to be used to simplify the configuration.

6.1.2 Lab Configuration

6.1.2.1 Configuration Roadmap

1. Configure IP addresses for devices.
2. Deploy IS-IS on the backbone network to build the underlying network.
3. Properly plan the RT and RD of the VPN instance and bind the VPN to the corresponding interface on the PE.
4. Deploy MPLS and MPLS LDP on the backbone network.
5. Create VPNv4 IBGP peers on the backbone network.
6. Advertise routing entries in the finance VPN.
7. Advertise routing entries in the OA VPN.

6.1.2.2 Configuration Procedure

Step 1 Configure IP addresses for interconnection interfaces and loopback interfaces.

Name the devices.

The configuration details are not provided.

Disable the interfaces that are not used in this experiment.

The configuration details are not provided.

Configure IP addresses for GE0/0/2 and Loopback0 on R1.

```
[R1]interface LoopBack0
[R1-LoopBack0] ip address 10.0.1.1 255.255.255.255
[R1-LoopBack0] quit
[R1]interface GigabitEthernet0/0/2
[R1-GigabitEthernet0/0/2] ip address 10.0.12.1 255.255.255.0
[R1-GigabitEthernet0/0/2] quit
```

Configure IP addresses for GE0/0/1, GE0/0/2, and Loopback0 on R2. GE0/0/3 and GE0/0/4 need to be bound to a VPN instance, so no IP address is assigned to them.

```
[R2]interface LoopBack0
[R2-LoopBack0] ip address 10.10.10.2 255.255.255.255
[R2-LoopBack0] quit
[R2]interface GigabitEthernet0/0/1
```

```
[R2-GigabitEthernet0/0/1] ip address 10.0.24.2 255.255.255.0
[R2-GigabitEthernet0/0/1] quit
[R2]interface GigabitEthernet0/0/2
[R2-GigabitEthernet0/0/2] ip address 10.0.23.2 255.255.255.0
[R2-GigabitEthernet0/0/2] quit
```

Configure IP addresses for GE0/0/2, GE0/0/3, and Loopback0 on R3.

```
[R3]interface LoopBack0
[R3-LoopBack0] ip address 10.10.10.3 255.255.255.255
[R3-LoopBack0]quit
[R3]interface GigabitEthernet0/0/2
[R3-GigabitEthernet0/0/2] ip address 10.0.34.3 255.255.255.0
[R3-GigabitEthernet0/0/2] quit
[R3]interface GigabitEthernet0/0/3
[R3-GigabitEthernet0/0/3]ip address 10.0.23.3 255.255.255.0
[R3-GigabitEthernet0/0/3]quit
```

Assign IP addresses to GE0/0/1, GE0/0/3, and Loopback0 on R4. GE0/0/2 and GE0/0/5 need to be bound to a VPN instance, so no IP address is assigned to them.

```
[R4]interface LoopBack0
[R4-LoopBack0] ip address 10.10.10.4 255.255.255.255
[R4-LoopBack0] quit
[R4]interface GigabitEthernet0/0/1
[R4-GigabitEthernet0/0/1] ip address 10.0.24.4 255.255.255.0
[R4-GigabitEthernet0/0/1] quit
[R4]interface GigabitEthernet0/0/3
[R4-GigabitEthernet0/0/3] ip address 10.0.34.4 255.255.255.0
[R4-GigabitEthernet0/0/3] quit
```

Configure IP addresses for GE0/0/3 and Loopback0 on R5.

```
[R5]interface LoopBack0
[R5-LoopBack0] ip address 10.0.1.2 255.255.255.255
[R5-LoopBack0] quit
[R5]interface GigabitEthernet0/0/3
[R5-GigabitEthernet0/0/3] ip address 10.0.45.5 255.255.255.0
[R5-GigabitEthernet0/0/3] quit
```

Configure the VLAN to which GE0/0/2 of S1 belongs and IP addresses for VLANIF 1 and Loopback0.

```
[S1]interface LoopBack0
[S1-LoopBack0] ip address 10.0.2.1 255.255.255.255
[S1-LoopBack0] quit
[S1]interface GigabitEthernet0/0/2
[S1-GigabitEthernet0/0/2] port link-type access
[S1-GigabitEthernet0/0/2] port default vlan 1
[S1-GigabitEthernet0/0/2] quit
[S1]interface Vlanif 1
[S1-Vlanif1] ip address 10.1.21.1 24
[S1-Vlanif1] quit
```

Configure the VLAN to which GE0/0/4 of S2 belongs and IP addresses for VLANIF 1 and Loopback0.

```
[S2]interface LoopBack0
```

```
[S2-LoopBack0] ip address 10.0.2.2 255.255.255.255
[S2-LoopBack0] quit
[S2]interface GigabitEthernet0/0/4
[S2-GigabitEthernet0/0/4] port link-type access
[S2-GigabitEthernet0/0/4] port default vlan 1
[S2-GigabitEthernet0/0/4] quit
[S2]interface Vlanif 1
[S2-Vlanif1] ip address 10.0.42.2 24
[S2-Vlanif1] quit
```

27 Deploy IS-IS on the backbone network to build the underlying network.

Deploy IS-IS processes on R2, R3, and R4 based on the topology design. Set the process ID to 1, configure R2, R3, and R4 as Level-2 routers, set the cost type to wide, set the area ID of the NET to 49.0001, set the system ID based on the loopback interface, and set the IS-IS host name to be the same as the device name.

Configure IS-IS on R2.

```
[R2]isis 1
[R2-isis-1] is-level level-2
[R2-isis-1] cost-style wide
[R2-isis-1] network-entity 49.0001.0100.1001.0002.00
[R2-isis-1] is-name R2
[R2-isis-1] quit
[R2]interface LoopBack0
[R2-LoopBack0] isis enable 1
[R2-LoopBack0] quit
[R2]interface GigabitEthernet0/0/1
[R2-GigabitEthernet0/0/1] isis enable 1
[R2-GigabitEthernet0/0/1] quit
[R2]interface GigabitEthernet0/0/2
[R2-GigabitEthernet0/0/2] isis enable 1
[R2-GigabitEthernet0/0/2] quit
```

Configure IS-IS on R3.

```
[R3]isis 1
[R3-isis-1] is-level level-2
[R3-isis-1] cost-style wide
[R3-isis-1] network-entity 49.0001.0100.1001.0003.00
[R3-isis-1] is-name R3
[R3-isis-1] quit
[R3]interface LoopBack0
[R3-LoopBack0] isis enable 1
[R3-LoopBack0] quit
[R3]interface GigabitEthernet0/0/2
[R3-GigabitEthernet0/0/2] isis enable 1
[R3-GigabitEthernet0/0/2] quit
```

```
[R3]interface GigabitEthernet0/0/3
[R3-GigabitEthernet0/0/3] isis enable 1
[R3-GigabitEthernet0/0/3] quit
```

Configure IS-IS on R4.

```
[R4]isis 1
[R4-isis-1] is-level level-2
[R4-isis-1] cost-style wide
[R4-isis-1] network-entity 49.0001.0100.1001.0004.00
[R4-isis-1] is-name R4
[R4-isis-1] quit
[R4]interface LoopBack0
[R4-LoopBack0] isis enable 1
[R4-LoopBack0] quit
[R4]interface GigabitEthernet0/0/1
[R4-GigabitEthernet0/0/1] isis enable 1
[R4-GigabitEthernet0/0/1] quit
[R4]interface GigabitEthernet0/0/3
[R4-GigabitEthernet0/0/3] isis enable 1
[R4-GigabitEthernet0/0/3] quit
```

Check the IS-IS configuration.

Check the IS-IS neighbor relationship on R3.

```
[R3]display isis peer

                    Peer information for ISIS(1)

  System Id      Interface          Circuit Id     State  HoldTime Type      PRI1
-------------------------------------------------------------------------------
R4*              GE0/0/2            R4.02          Up   8s        L2        64
R2*              GE0/0/3            R2.02          Up   9s        L2        64

Total Peer(s): 2

[R3]display isis interface

                    Interface information for ISIS(1)
                    ---------------------------------
 Interface       Id       IPV4.State          IPV6.State        MTU    Type    DIS
 GE0/0/2       001          Up                  Up            1497    L1/L2  No/No
 GE0/0/3       002          Up                  Up            1497    L1/L2  No/No
 Loop0         001          Up                  Up            1500    L1/L2  –
```

Check the IPv4 IS-IS routing table on each router. R3 is used as the example.

```
[R3]display isis route

                    Route information for ISIS(1)
                    -----------------------------

                    ISIS(1) Level-2 Forwarding Table
                    --------------------------------

IPV4 Destination      IntCost     ExtCost ExitInterface   NextHop          Flags
-------------------------------------------------------------------------------
```

```
10.0.24.0/24          20        NULL     GE0/0/2      10.0.34.4          A/-/-/-
                                       GE0/0/3      10.0.23.2
10.10.10.3/32         0         NULL     Loop0         Direct             D/-/L/-
10.10.10.2/32         10        NULL     GE0/0/3      10.0.23.2          A/-/-/-
10.0.23.0/24          10        NULL     GE0/0/3      Direct             D/-/L/-
10.0.34.0/24          10        NULL     GE0/0/2      Direct             D/-/L/-
10.10.10.4/32         10        NULL     GE0/0/2      10.0.34.4          A/-/-/-

      Flags:D-Direct,A-Added to URT,L-Advertised in LSPs, S-IGP Shortcut,
                                U-Up/Down Bit Set
```

The command output shows that IPv4 and IPv6 routes have been learned by the router.

28

Configure a VPN instance and bind an interface of each device to the VPN instance.

RDs and RTs need to be planned for VPN instances.

RDs can be used to identify duplicate routes and identify the areas from which routes are originated.

The RD of the route from the headquarters is 65100:12.

The RD of the route from Branch1 is 65100:1.

The RD of the route from Branch2 is 65001:2.

The RD of the route from Branch3 is 65002:2.

RTs need to be planned based on service requirements. The headquarters needs to communicate with Branch1, the headquarters, Branch2, and Branch3 need to communicate with each other, and Branch1 cannot communicate with Branch2 or Branch3.

The export RTs of the routes from the headquarters are 65100:12 and 65001:65002, and import RTs are 65100:12 and 65001:65002.

The export RT and IRT of the routes from Branch1 are 65100:12 and 65100:12, respectively.

The export RT and import RT of the routes from Branch2 are 65001:65002 and 65001:65002, respectively.

The export RT and import RT of the routes from Branch3 are 65001:65002 and 65001:65002, respectively.

On R2, configure VPN instances, bind interfaces to VPN instances, and configure IP addresses for the interfaces.

```
[R2]ip vpn-instance Finance&OA
[R2-vpn-instance-Finance&OA] route-distinguisher 65100:12
[R2-vpn-instance-Finance&OA-af-ipv4] vpn-target 65100:12 65001:65002
[R2-vpn-instance-Finance&OA-af-ipv4] quit
```

```
[R2]ip vpn-instance OA
[R2-vpn-instance-OA]
[R2-vpn-instance-OA] route-distinguisher 65001:2
[R2-vpn-instance-OA-af-ipv4] vpn-target 65001:65002
[R2-vpn-instance-OA-af-ipv4] quit
[R2]interface GigabitEthernet0/0/3
[R2-GigabitEthernet0/0/3] ip binding vpn-instance Finance&OA
[R2-GigabitEthernet0/0/3] ip address 10.0.12.2 255.255.255.0
[R2-GigabitEthernet0/0/3] quit
[R2]interface GigabitEthernet0/0/4
[R2-GigabitEthernet0/0/4] ip binding vpn-instance OA
[R2-GigabitEthernet0/0/4] ip address 10.0.21.2 255.255.255.0
[R2-GigabitEthernet0/0/4] quit
```

On R4, configure VPN instances, bind interfaces to VPN instances, and configure IP addresses for the interfaces.

```
[R4]ip vpn-instance Finance
[R4-vpn-instance-Finance] route-distinguisher 65100:1
[R4-vpn-instance-Finance -af-ipv4] vpn-target 65100:12
[R4-vpn-instance-Finance -af-ipv4] quit
[R4]ip vpn-instance OA
[R4-vpn-instance-OA] route-distinguisher 65002:2
[R4-vpn-instance-OA-af-ipv4] vpn-target 65001:65002
[R4-vpn-instance-OA-af-ipv4] quit
[R4]interface GigabitEthernet0/0/2
[R4-GigabitEthernet0/0/2] ip binding vpn-instance Finance
[R4-GigabitEthernet0/0/2] ip address 10.0.45.4 255.255.255.0
[R4-GigabitEthernet0/0/2] quit
[R4]interface GigabitEthernet0/0/5
[R4-GigabitEthernet0/0/5] ip binding vpn-instance OA
[R4-GigabitEthernet0/0/5] ip address 10.0.42.4 255.255.255.0
[R4-GigabitEthernet0/0/5] quit
```

Check the VPN instance configuration.

Check the VPN instance configuration on each PE. R2 is used as an example.

```
[R2]display current-configuration section vpn-instance
#
ip vpn-instance Finance&OA
 ipv4-family
  route-distinguisher 65100:1
  vpn-target 65100:12 65001:65002 export-extcommunity
  vpn-target 65100:12 65001:65002 import-extcommunity
#
ip vpn-instance OA
 ipv4-family
  route-distinguisher 65001:2
  vpn-target 65001:65002 export-extcommunity
  vpn-target 65001:65002 import-extcommunity
```

Check VPN instance binding information on the PE. R2 is used as an example.

```
[R2]display ip vpn-instance interface
 Total VPN-Instances configured      : 2

 VPN-Instance Name and ID : Finance&OA, 1
```

```
 Interface Number : 1
 Interface list : GigabitEthernet0/0/3

VPN-Instance Name and ID : OA, 2
 Interface Number : 1
 Interface list : GigabitEthernet0/0/4
```

The preceding command output helps you can check whether VPN instances are correctly configured and whether interfaces are correctly bound to VPN instances.

29 Deploy MPLS and MPLS LDP.

MPLS and MPLS LDP need to be configured on the BGP/MPLS IP VPN data forwarding path.

In this experiment, the path R2 -> R4 is the active path, and the path R2 -> R3 -> R4 is the standby path. Therefore, MPLS and MPLS LDP must be enabled on R2, R3, R4, and interfaces connecting R2, R3, and R4.

The IP address of the loopback interface is used as the MPLS LSR ID.

Configure MPLS on R2.

```
[R2]mpls lsr-id 10.10.10.2
[R2]mpls
[R2-mpls] quit
[R2]mpls ldp
[R2-mpls-ldp] quit
[R2]interface GigabitEthernet0/0/1
[R2-GigabitEthernet0/0/1] mpls
[R2-GigabitEthernet0/0/1] mpls ldp
[R2-GigabitEthernet0/0/1] quit
[R2]interface GigabitEthernet0/0/2
[R2-GigabitEthernet0/0/2] mpls
[R2-GigabitEthernet0/0/2] mpls ldp
[R2-GigabitEthernet0/0/2] quit
```

Configure MPLS on R3.

```
[R3]mpls lsr-id 10.10.10.3
[R3]mpls
[R3-mpls] quit
[R3]mpls ldp
[R3-mpls-ldp] quit
[R3]interface GigabitEthernet0/0/2
[R3-GigabitEthernet0/0/2] mpls
[R3-GigabitEthernet0/0/2] mpls ldp
[R3-GigabitEthernet0/0/2] quit
[R3]interface GigabitEthernet0/0/3
[R3-GigabitEthernet0/0/3] mpls
[R3-GigabitEthernet0/0/3] mpls ldp
```

```
[R3-GigabitEthernet0/0/3] quit
```

Configure MPLS on R4.

```
[R4]mpls lsr-id 10.10.10.4
[R4]mpls
[R4-mpls] quit
[R4]mpls ldp
[R4-mpls-ldp] quit
[R4]interface GigabitEthernet0/0/1
[R4-GigabitEthernet0/0/1] mpls
[R4-GigabitEthernet0/0/1] mpls ldp
[R4-GigabitEthernet0/0/1] quit
[R4]interface GigabitEthernet0/0/3
[R4-GigabitEthernet0/0/3] mpls
[R4-GigabitEthernet0/0/3] mpls ldp
[R4-GigabitEthernet0/0/3] quit
```

Check the MPLS configuration.

Check information about LDP peers on R3.

```
[R3]display mpls ldp peer

 LDP Peer Information in Public network
 A '*' before a peer means the peer is being deleted.
 ------------------------------------------------------------------------------
 PeerID            TransportAddress    DiscoverySource
 ------------------------------------------------------------------------------
 10.10.10.2:0          10.10.10.2        GigabitEthernet0/0/3
 10.10.10.4:0          10.10.10.4        GigabitEthernet0/0/2
 ------------------------------------------------------------------------------
 TOTAL: 2 Peer(s) Found.
```

Check information about LSPs on R3.

```
[R3]display mpls lsp
-------------------------------------------------------------------------------------------------------
                      LSP Information: LDP LSP
-------------------------------------------------------------------------------------------------------
FEC                 In/Out Label     In/Out IF                          VrfName
10.10.10.2/32       NULL/3           -/GE0/0/3
10.10.10.2/32       1024/3           -/GE0/0/3
10.10.10.3/32       3/NULL           -/-
10.10.10.4/32       NULL/3           -/GE0/0/2
10.10.10.4/32       1025/3           -/GE0/0/2
```

The command output shows that an LDP LSP has been established.

30 Establish VPNv4 IBGP peer relationships on the backbone network.

Establish IBGP VPNv4 peer relationships in AS 65100, configure R3 as the VPN RR, and configure R2 and R4 as clients of R3.

As AS 65100 only needs to transmit VPNv4 routes, so IPv4 IBGP peer relationships can be not established.

When R3 functions as a VPN RR, the RT filtering function needs to be disabled because no VPN instance is configured.

Configure a VPNv4 IBGP peer on R2.

```
[R2]bgp 65100
[R2-bgp] undo default ipv4-unicast
[R2-bgp] peer 10.10.10.3 as-number 65100
[R2-bgp] peer 10.10.10.3 connect-interface LoopBack0
[R2-bgp] ipv4-family vpnv4
[R2-bgp-af-vpnv4] peer 10.10.10.3 enable
[R2-bgp-af-vpnv4] quit
```

Configure a VPNv4 IBGP peer on R3. The RT filtering function needs to be disabled on R3 that functions as the VPN RR.

```
[R3]bgp 65100
[R3-bgp] undo default ipv4-unicast
[R3-bgp] peer 10.10.10.2 as-number 65100
[R3-bgp] peer 10.10.10.2 connect-interface LoopBack0
[R3-bgp] peer 10.10.10.4 as-number 65100
[R3-bgp] peer 10.10.10.4 connect-interface LoopBack0
[R3-bgp] ipv4-family vpnv4
[R3-bgp-af-vpnv4] undo policy vpn-target
[R3-bgp-af-vpnv4] peer 10.10.10.2 enable
[R3-bgp-af-vpnv4] peer 10.10.10.2 reflect-client
[R3-bgp-af-vpnv4] peer 10.10.10.4 enable
[R3-bgp-af-vpnv4] peer 10.10.10.4 reflect-client
[R3-bgp-af-vpnv4] quit
```

Configure a VPNv4 IBGP peer on R4.

```
[R4]bgp 65100
[R4-bgp] undo default ipv4-unicast
[R4-bgp] peer 10.10.10.3 as-number 65100
[R4-bgp] peer 10.10.10.3 connect-interface LoopBack0
[R4-bgp] ipv4-family vpnv4
[R4-bgp-af-vpnv4] peer 10.10.10.3 enable
[R4-bgp-af-vpnv4] quit
```

Check the BGP configuration.

Check the BGP VPNv4 peer on R3.

```
[R3-bgp]display bgp vpnv4 all peer

 BGP local router ID : 10.10.10.3
 Local AS number : 65100
 Total number of peers : 2    Peers in established state : 2

  Peer            V          AS  MsgRcvd  MsgSent  OutQ  Up/Down       State PrefRcv

  10.10.10.2      4       65100     312      320     0   05:04:45  Established     0
```

```
10.10.10.4      4       65100     311     316     0   05:04:40  Established     0
```

The command output shows that an IBGP VPNv4 peer relationship has been established.

31 Advertise the routing entry of the finance VPN.

R1 and R5 function as CEs and use OSPF to send routes of Loopback0 to PEs. Then PEs exchange routes.

Set the process ID to 1, area ID to 0, and router ID to the loopback interface's IP address of the router. R2 with the router ID of 10.10.10.2 is used as an example.

Establish an OSPF neighbor relationship between R1 (CE) and R2 (PE).

```
[R1]router id 10.10.10.1
[R1]ospf 1
[R1-ospf-1] area 0
[R1-ospf-1-area-0.0.0.0] network 10.0.1.1 0.0.0.0
[R1-ospf-1-area-0.0.0.0] network 10.0.12.1 0.0.0.0

[R2]router id 10.10.10.2
[R2]ospf 1 vpn-instance Finance&OA
[R2-ospf-1] area 0
[R2-ospf-1-area-0.0.0.0] network 10.0.12.2 0.0.0.0
[R2-ospf-1-area-0.0.0.0] quit
```

Establish an OSPF neighbor relationship between R4 (CE) and R5 (PE).

```
[R5]router id 10.10.10.5
[R5]ospf 1
[R5-ospf-1] area 0
[R5-ospf-1-area-0.0.0.0] network 10.0.1.2 0.0.0.0
[R5-ospf-1-area-0.0.0.0] network 10.0.45.5 0.0.0.0

[R4]router id 10.10.10.4
[R4]ospf 1 vpn-instance Finance
[R4-ospf-1] area 0
[R4-ospf-1-area-0.0.0.0] network 10.0.45.4 0.0.0.0
[R4-ospf-1-area-0.0.0.0] quit
```

Configure R2 and R4 to import routes from each other.

```
[R2]ospf 1 vpn-instance Finance&OA
[R2-ospf-1] import-route bgp
[R2-ospf-1] quit
[R2]bgp 65100
[R2-bgp] ipv4-family vpn-instance Finance&OA
[R2-bgp-Finance&OA] network 10.0.1.1 32

[R4]ospf 1 vpn-instance Finance
[R4-ospf-1] import-route bgp
[R4-ospf-1] quit
```

```
[R4]bgp 65100
[R4-bgp] ipv4-family vpn-instance Finance
[R4-bgp-Finance&OA] network 10.0.1.2 32
```

Check the configuration.

Check routing tables on R1 and R5. R1 is used as an example.

```
[R1]display ip routing-table
Route Flags: R - relay, D - download to fib
------------------------------------------------------------------------------
Routing Tables: Public
         Destinations : 9        Routes : 9

Destination/Mask    Proto   Pre  Cost      Flags NextHop         Interface

        10.0.1.1/32  Direct  0    0           D   127.0.0.1       LoopBack0
        10.0.1.2/32  OSPF   10   3           D   10.0.12.2       GigabitEthernet0/0/2
       10.0.12.0/24  Direct  0    0           D   10.0.12.1       GigabitEthernet0/0/2
       10.0.12.1/32  Direct  0    0           D   127.0.0.1       GigabitEthernet0/0/2
    10.0.12.255/32   Direct  0    0           D   127.0.0.1       GigabitEthernet0/0/2
       127.0.0.0/8   Direct  0    0           D   127.0.0.1       InLoopBack0
       127.0.0.1/32  Direct  0    0           D   127.0.0.1       InLoopBack0
127.255.255.255/32   Direct  0    0           D   127.0.0.1       InLoopBack0
255.255.255.255/32   Direct  0    0           D   127.0.0.1       InLoopBack0
```

Test the connectivity between Loopback0 interfaces of R1 and R5.

```
[R1]ping -a 10.0.1.1 10.0.1.2
  PING 10.0.1.2: 56  data bytes, press CTRL_C to break
    Reply from 10.0.1.2: bytes=56 Sequence=1 ttl=253 time=60 ms

  --- 10.0.1.2 ping statistics ---
    1 packet(s) transmitted
    1 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 30/40/60 ms
```

The preceding information shows that connectivity on the network segment of the finance VPN is achieved.

32 Advertise routing entries in the OA VPN.

S1 and S2 function as CEs and use BGP to send routes of Loopback0 interfaces to PEs. The PEs then advertise routes to each other.

Establish a BGP peer relationship between S1 (CE) and R2 (PE) and advertise routes.

```
[S1]bgp 65001
[S1-bgp] peer 10.0.21.2 enable
[S1-bgp] network 10.0.2.1 32
[S1-bgp] quit
```

```
[R2]bgp 65100
[R2-bgp] ipv4 vpn-instance OA
[R2-bgp-OA] peer 10.0.21.1 as-number 65001
[R2-bgp-OA] quit
```

Establish a BGP peer relationship between S2 (CE) and R4 (PE) and advertise routes.

```
[S2]bgp 65002
[S2-bgp] peer 10.0.42.4 enable
[S2-bgp] network 10.0.2.2 32
[S2-bgp] quit

[R4]bgp 65100
[R4-bgp] ipv4 vpn-instance OA
[R4-bgp-OA] peer 10.0.42.2 as-number 65002
[R4-bgp-OA] quit
```

Check the configuration.

Check routing tables on R1, S1, and S2. R1 is used as an example.

```
[R1]display ip routing-table
Route Flags: R - relay, D - download to fib
------------------------------------------------------------------------------
Routing Tables: Public
         Destinations : 9        Routes : 9

Destination/Mask    Proto   Pre  Cost      Flags NextHop         Interface

        10.0.1.1/32  Direct  0    0           D   127.0.0.1       LoopBack0
        10.0.1.2/32  OSPF    10   3           D   10.0.12.2       GigabitEthernet0/0/2
        10.0.2.1/32  O_ASE   150  1           D   10.0.12.2       GigabitEthernet0/0/2
        10.0.2.2/32  O_ASE   150  1           D   10.0.12.2       GigabitEthernet0/0/2
       10.0.12.0/24  Direct  0    0           D   10.0.12.1       GigabitEthernet0/0/2
       10.0.12.1/32  Direct  0    0           D   127.0.0.1       GigabitEthernet0/0/2
     10.0.12.255/32  Direct  0    0           D   127.0.0.1       GigabitEthernet0/0/2
       127.0.0.0/8   Direct  0    0           D   127.0.0.1       InLoopBack0
       127.0.0.1/32  Direct  0    0           D   127.0.0.1       InLoopBack0
127.255.255.255/32   Direct  0    0           D   127.0.0.1       InLoopBack0
255.255.255.255/32   Direct  0    0           D   127.0.0.1       InLoopBack0
```

Test the connectivity between Loopback0 interfaces of R1 and S2.

```
<R1>ping -a 10.0.1.1 10.0.2.2
  PING 10.0.2.2: 56  data bytes, press CTRL_C to break
    Reply from 10.0.2.2: bytes=56 Sequence=1 ttl=253 time=50 ms

  --- 10.0.2.2 ping statistics ---
    1 packet(s) transmitted
    1 packet(s) received
    0.00% packet loss
round-trip min/avg/max = 30/42/50 ms
```

Test the connectivity between Loopback0 interfaces of S1 and S2.

```
[S1]ping -a 10.0.2.1 10.0.2.2
```

```
 PING 10.0.2.2: 56  data bytes, press CTRL_C to break
   Reply from 10.0.2.2: bytes=56 Sequence=1 ttl=253 time=50 ms

 --- 10.0.2.2 ping statistics ---
   1 packet(s) transmitted
   1 packet(s) received
   0.00% packet loss
   round-trip min/avg/max = 50/60/70 ms
```

Test the connectivity between Loopback0 interfaces of S1 and R5.

```
[S1]ping -a 10.0.2.1 10.0.1.2
 PING 10.0.1.2: 56  data bytes, press CTRL_C to break
   Request time out
   Request time out
   Request time out
   Request time out
   Request time out

 --- 10.0.1.2 ping statistics ---
   5 packet(s) transmitted
   0 packet(s) received
   100.00% packet loss
```

According to the preceding information, S1 and S2 can communicate with R1 in the OA VPN, but S1 and S2 cannot communicate with R5 in the finance VPN.

6.1.3 Quiz

If R5 needs to communicate with S1 and S2, how do we modify the VPN instance configuration?

6.1.4 Configuration Reference

Configuration on R1

```
#
 sysname R1
#
router id 10.10.10.1
#
interface GigabitEthernet0/0/2
 ip address 10.0.12.1 255.255.255.0
#
interface LoopBack0
 ip address 10.0.1.1 255.255.255.255
#
ospf 1
 area 0.0.0.0
  network 10.0.1.1 0.0.0.0
  network 10.0.12.1 0.0.0.0
#
return
```

Configuration on R2

```
#
```

```
 sysname R2
#
router id 10.10.10.2
#
ip vpn-instance Finance&OA
 ipv4-family
  route-distinguisher 65100:12
  vpn-target 65100:12 65001:65002 export-extcommunity
  vpn-target 65100:12 65001:65002 import-extcommunity
#
ip vpn-instance OA
 ipv4-family
  route-distinguisher 65001:2
  vpn-target 65001:65002 export-extcommunity
  vpn-target 65001:65002 import-extcommunity
#
mpls lsr-id 10.10.10.2
mpls
#
mpls ldp
#
isis 1
 is-level level-2
 cost-style wide
 network-entity 49.0001.0100.1001.0002.00
 is-name R2
#
interface GigabitEthernet0/0/1
 ip address 10.0.24.2 255.255.255.0
 isis enable 1
 mpls
 mpls ldp
#
interface GigabitEthernet0/0/2
 ip address 10.0.23.2 255.255.255.0
 isis enable 1
 mpls
 mpls ldp
#
interface GigabitEthernet0/0/3
 ip binding vpn-instance Finance&OA
 ip address 10.0.12.2 255.255.255.0
#
interface GigabitEthernet0/0/4
 ip binding vpn-instance OA
 ip address 10.0.21.2 255.255.255.0
#
interface LoopBack0
 ip address 10.10.10.2 255.255.255.255
 isis enable 1
#
bgp 65100
 undo default ipv4-unicast
 peer 10.10.10.3 as-number 65100
 peer 10.10.10.3 connect-interface LoopBack0
```

```
 #
 ipv4-family unicast
  undo synchronization
  undo peer 10.10.10.3 enable
 #
 ipv4-family vpnv4
  policy vpn-target
  peer 10.10.10.3 enable
 #
 ipv4-family vpn-instance Finance&OA
  network 10.0.1.1 255.255.255.255
 #
 ipv4-family vpn-instance OA
  peer 10.0.21.1 as-number 65001
#
ospf 1 vpn-instance Finance&OA
 import-route bgp
 area 0.0.0.0
  network 10.0.12.2 0.0.0.0
#
return
```

Configuration on R3

```
#
 sysname R3
#
router id 10.10.10.3
#
mpls lsr-id 10.10.10.3
mpls
#
mpls ldp
#
isis 1
 is-level level-2
 cost-style wide
 network-entity 49.0001.0100.1001.0003.00
 is-name R3
#
interface GigabitEthernet0/0/2
 ip address 10.0.34.3 255.255.255.0
 isis enable 1
 mpls
 mpls ldp
#
interface GigabitEthernet0/0/3
 ip address 10.0.23.3 255.255.255.0
 isis enable 1
 mpls
 mpls ldp
#
interface LoopBack0
 ip address 10.10.10.3 255.255.255.255
 isis enable 1
#
```

```
bgp 65100
 undo default ipv4-unicast
 peer 10.10.10.2 as-number 65100
 peer 10.10.10.2 connect-interface LoopBack0
 peer 10.10.10.4 as-number 65100
 peer 10.10.10.4 connect-interface LoopBack0
 #
 ipv4-family unicast
  undo synchronization
  undo peer 10.10.10.2 enable
  undo peer 10.10.10.4 enable
 #
 ipv4-family vpnv4
  undo policy vpn-target
  peer 10.10.10.2 enable
  peer 10.10.10.2 reflect-client
  peer 10.10.10.4 enable
  peer 10.10.10.4 reflect-client
#
return
```

Configuration on R4

```
#
 sysname R4
#
router id 10.10.10.4
#
ip vpn-instance Finance
 ipv4-family
  route-distinguisher 65100:1
  vpn-target 65100:12 export-extcommunity
  vpn-target 65100:12 import-extcommunity
#
ip vpn-instance OA
 ipv4-family
  route-distinguisher 65002:2
  vpn-target 65001:65002 export-extcommunity
  vpn-target 65001:65002 import-extcommunity
#
mpls lsr-id 10.10.10.4
mpls
#
mpls ldp
#
isis 1
 is-level level-2
 cost-style wide
 network-entity 49.0001.0100.1001.0004.00
 is-name R4
#
interface GigabitEthernet0/0/1
 ip address 10.0.24.4 255.255.255.0
 isis enable 1
 mpls
 mpls ldp
```

```
#
interface GigabitEthernet0/0/2
 ip binding vpn-instance Finance
 ip address 10.0.45.4 255.255.255.0
#
interface GigabitEthernet0/0/3
 ip address 10.0.34.4 255.255.255.0
 isis enable 1
 mpls
 mpls ldp
#
interface GigabitEthernet0/0/5
 ip binding vpn-instance OA
 ip address 10.0.42.4 255.255.255.0
#
interface LoopBack0
 ip address 10.10.10.4 255.255.255.255
 isis enable 1
#
bgp 65100
 undo default ipv4-unicast
 peer 10.10.10.3 as-number 65100
 peer 10.10.10.3 connect-interface LoopBack0
 #
 ipv4-family unicast
  undo synchronization
  undo peer 10.10.10.3 enable
 #
 ipv4-family vpnv4
  policy vpn-target
  peer 10.10.10.3 enable
 #
 ipv4-family vpn-instance Finance
  network 10.0.1.2 255.255.255.255
 #
 ipv4-family vpn-instance OA
  peer 10.0.42.2 as-number 65002
#
ospf 1 vpn-instance Finance
 import-route bgp
 area 0.0.0.0
  network 10.0.45.4 0.0.0.0
#
return
```

Configuration on R5

```
#
 sysname R5
#
router id 10.10.10.5
#
interface GigabitEthernet0/0/3
 ip address 10.0.45.5 255.255.255.0
#
interface LoopBack0
```

```
 ip address 10.0.1.2 255.255.255.255
#
ospf 1
 area 0.0.0.0
  network 10.0.1.2 0.0.0.0
  network 10.0.45.5 0.0.0.0
#
return
```

Configuration on S1

```
#
sysname S1
#
interface Vlanif1
 ip address 10.0.21.1 255.255.255.0
#
interface GigabitEthernet0/0/2
 port link-type access
#
interface LoopBack0
 ip address 10.0.2.1 255.255.255.255
#
bgp 65001
 peer 10.0.21.2 as-number 65100
 #
 ipv4-family unicast
  undo synchronization
  network 10.0.2.1 255.255.255.255
  peer 10.0.21.2 enable
#
return
```

Configuration on S2

```
#
sysname S2
#
interface Vlanif1
 ip address 10.0.42.2 255.255.255.0
#
interface GigabitEthernet0/0/4
 port link-type access
#
interface LoopBack0
 ip address 10.0.2.2 255.255.255.255
#
bgp 65002
 peer 10.0.42.4 as-number 65100
 #
 ipv4-family unicast
  undo synchronization
  network 10.0.2.2 255.255.255.255
  peer 10.0.42.4 enable
#
Return
```

Reference Answers

Answers to the advanced IGP

1. OSPF inter-area route advertisement depends on Type 3 LSAs. In OSPF, an ABR imports inter-area routes to other areas into the local area through Type 3 LSAs. You can run the **filter** command in the area view of the ABR to set filtering conditions for incoming and outgoing Type 3 LSAs in the area. Only the filtered information can be advertised and received.

   Note that the filter command in the OSPF area view can be executed only on ABRs and takes effect only for Type 3 LSAs. The internal route calculation in an OSPF area depends on Type 1 and Type 2 LSAs, so this command cannot be used to filter these LSAs.

Answers to the advanced BGP

1. When a BGP peer receives a route, it adds the route to the BGP Adj-IN routing table and processes the route based on the inbound BGP routing policy.

   If a BGP peer needs to receive only some routes, the peer does not need to send all routes. To solve this problem, use either of the following methods:

   a) Configure a routing policy on the peer to filter the routes to be sent to the BGP peer. This method requires the cooperation of the peer and has poor flexibility.

   b) With the outbound route filtering (ORF) function enabled, the BGP peer notifies the peer that only some routes are sent. This method is flexible, but both ends must support ORF.

   Use either of the two methods as required.

Answers to the IPv6 Routing

1. IS-IS uses LSPs in the type-length-value (TLV) structure. Therefore, no new protocol framework needs to be developed when IS-IS is used to deploy a dual-stack network. However, by default, IS-IS uses only the shortest path tree (SPT) constructed on an IPv4 network to calculate the topology.

   If a dual-stack IS-IS network needs to be deployed but some devices do not support IPv6, IS-IS still uses the SPT constructed based on the IPv4 network by default. As a result, IPv6 packets may be sent to the devices that do not support IPv6, causing packet loss. Therefore, the multi-topology feature is required.

   OSPFv3 has developed for IPv6 networks. OSPFv3 can form SPTs based on IPv6 networks, and therefore multi-topology is not required.

   Advanced VLAN Technologies

If the ARP entry of the requested host can be found in local ARP entries (dynamic learned or statically bound entries), the VLANIF interface of the super-VLAN responds to the ARP Request packet.

Answers to the Ethernet Switching Security

1. All terminals on the same network segment communicate with each other through the gateway. The gateway can restrict and monitor the communication between terminals. However, the traffic that should be forwarded by the switch at Layer 2 is forwarded by the gateway at Layer 3, which increases the load of the gateway.

Answers to the BGP/MPLS IP VPN

1. Multiple ERTs and IRTs can be configured in a VPN instance. After the peer receives a route, it accepts the route as long as one ERT in the route matches the locally configured IRT.

   To enable R5 to receive routes from S1 and S2, add Import RT 65001:65002 to the finance VPN on R4.